# Provider Management

Use the **VCF Automation** Provider Management UI as an IT administrator (or provider administrator) to set up organizations for Lines of Businesses (LOBs) and allocate resources.

**In This Chapter**


- Overview of VCF Automation Provider Management
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**Getting to Know VCF**

To apply this documentation, you must be acquainted with the VCF Product Overview, VCF Design, and VCF Release Notes.

**Extending Your VCF Private Cloud with Services in VCF Automation and vSphere Supervisor**

You can extend the capabilities of your VCF private cloud by installing services in VCF Automation or vSphere Supervisor. For example, you can install a Harbor OCI registry for storing containers, Cavel packages, and other types of deployment artifacts. For more information about installing services, see the VMware Cloud Foundation Service Administration and Development documentation.

**SDKs, APIs, and CLI for VCF Administration**

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See Administration SDKs, APIs, and CLI.

**Migration Tool**

In VCF Automation 9.1, you can use the **Migration Tool** to perform an in-place migration from VMware Cloud Director to VCF Automation. For more information, contact your Broadcom representative.

## Overview of VCF Automation Provider Management

With VCF Automation, as a provider administrator, you can build secure, multi-tenant clouds by pooling virtual infrastructure resources, which are an abstraction of their underlying vSphere and NSX resources, into regions and exposing them to users through web-based portals and programmatic interfaces.

The VCF Automation Provider Management Guide provides information about adding resources to the system, creating and provisioning organizations, and managing resources and organizations. This guide is intended for provider administrators responsible for managing the overall infrastructure, services, and tenancy model across multiple organizations or tenants. The provider administrator role can either be an Enterprise IT Administrator in an enterprise environment or a Provider Administrator in a service provider context. Within this guide, provider administrator and system administrator are used interchangeably.

The VCF Automation Provider Management Guide uses the following terms.

### vSphere and NSX Resources

VCF Automation relies on vSphere resources to provide CPU and memory, and on NSX to support the VCF Automation networking capabilities. Adding storage is provided through vSphere, whether vSAN or external storage.

You can use the underlying vSphere and NSX resources to create cloud resources.

### Infrastructure Resources

Infrastructure resources consist of Supervisors, regions, external connections, and region quotas. Infrastructure resources provide an abstraction of the underlying vCenter and NSX resources that supply compute, storage, memory, and networking to VCF Automation. Using the VCF Automation Provider Management UI, you can view the connection status of all vCenter and NSX instances that VCF Automation automatically discovers from VCF Operations. See Integrating Data Sources with VCF Operations.

### Supervisor

A VMware vSphere Supervisor is a collection of one or more clusters on a vCenter instance that provide and manage Kubernetes resources. The Supervisor is the component that owns the physical infrastructure resources and runs organization workloads on those resources. Each Supervisor can only be associated with one region.

VCF Automation deploys Supervisors deployed with NSX Virtual Private Cloud (VPC) networking.

### Supervisor VPC Network Stacks

Starting with version 9.1.1, VCF Automation supports two network stack types for Supervisor VPCs:

- Full-stack
  Full-stack is the standard deployment model. VPCs rely on Tunnel Endpoints (TEPs) to create an overlay network. Full-stack VPC networking supports both private and public subnets, NAT, and SNAT.
- VLAN-backed
  Designed for environments where no NSX overlay is configured. All traffic runs directly over VLANs. VLAN-backed VPC networking supports public subnets only.

For more information, see Managing Regions in Your VCF Automation and Deploying Supervisor with VCF Networking with VPC.

### Zone

Zones represent vSphere clusters that provide compute resources for Supervisors. A Supervisor can span across multiple zones. Zones are created and managed in vCenter. See Configuring and Managing vSphere Zones. It is important to use meaningful, user-friendly names because the names assigned to zones in vCenter are visible to end users.

A zone can consist of multiple vSphere clusters, but only one cluster binds to a specific vSphere Namespace in each zone. When enabling a Supervisor, you must choose between two configuration types:

- HA mode: Controller nodes are distributed across three zones, offering high availability. This mode requires exactly three zones.
- Default mode: All controller nodes are placed in the first zone. You can add additional zones to isolate workloads and management functions. This mode provides flexibility for physical separation while maintaining simplicity.

To provide an optimal user experience, configure Supervisors and Namespaces with either one or three zones. You can scale beyond three zones if requirements are driven by workload diversity rather than HA, such as creating dedicated zones for Windows and Linux. Broadcom supports exceeding three zones per Supervisor and Namespace in these cases.

### Region

In VCF Automation, a region combines compute, storage, memory, and networking resources from one or more Supervisors across one or more vCenter instances that are managed by the same NSX Manager instance.

You can create regions for different geographies, business units, or performance needs, but all Supervisors within a region must have homogeneous configurations:

- Identical names and definitions for storage classes
- Matching Kubernetes versions
- A consistent set of IaaS services

VCF Automation does not support heterogeneous configurations within a region. Resources in a region can be dedicated to a single organization or shared across multiple organizations.

If multiple Supervisors are part of the same region and have zones with identical names, such as Zone A, B, C, those zones are combined and presented as a single set of zones to users. For example, if Supervisor 1 and 2 each have Zone A, B, and C, the system presents three zones instead of six. To avoid merging, assign unique zone names across Supervisors.

**Important:**

In VCF Automation 9.1.1, the region homogeneity requirement also pertains to the network stack type of the zones within a region.

### Region Quota of an Organization

A region quota defines the compute and storage resources allocated to an organization for a given region. The key aspects of a region quota are:

- CPU and memory utilization limits which represent the upper bound of quota in a region
- CPU and memory reservations which represent the guaranteed capacity that is allocated in a region
- VM classes which represent the t-shirt-sized classes of compute that an organization has access to
- Storage classes which represent the storage classes and the quota assigned in a region

Key constraints for the region quota are:

- An organization can receive region quotas from multiple regions, but there can be only one quota per region.
- In multi-zone regions, you can assign quotas to all or a subset of the zones

When assigning a region quota:

- You can select to fully allocate the capacity or you can specify the CPU and memory capacity per zone.
- VM classes and storage classes are defined at the region quota level and apply to all zones in that quota.

### VCF Automation Networking

To leverage the networking capabilities of VCF Automation for organizations for all applications, you can add and manage your infrastructure and cloud networking resources through the Provider Management UI.

Before you define networking configurations for regions in the VCF Automation Provider Management UI, you must first set up the connectivity to the physical network:

- Create a Tier-0 gateway or Tier-0 VRF in NSX Manager, see Add an NSX Tier-0 Gateway.
- Create a VLAN-backed segment and configure the uplink teaming policy in NSX Manager, see Create an NSX Segment.
- Create IP allocations corresponding to the physical network.

**Note:**

VCF Automation 9.1.1 supports environments that use VLAN-backed Supervisor VPCs without an NSX overlay. For such deployments, you must configure a distributed transit gateway connected to the target VLAN in your physical infrastructure.

As a provider administrator, you then use the VCF Automation Provider Management UI to perform the following tasks for each region:

1. Determine the set of IPs that organizations have access to and define IP blocks for these IPs. At least one IP block is required per region.
   The IP blocks are used to define the routing for an organization centralized or distributed gateway. You must define in NSX the external IP ranges that are being routed.
2. Define external connections, centralized or distributed, that organizations can use for North-South connectivity.

- You create centralized connections by using your preexisting Tier-0 gateways in an `Active-Active` or `Active-Standby` mode from NSX . For more information about Tier-0 gateways, see Tier-0 Gateways.
- You create distributed connections by using your existing VLAN or EVPN-VXLAN connection.

  **Important:**

  Organizations assigned to a region with VLAN-backed zones must use distributed external connections. Centralized connections, which are Tier-0-based, are not available for such regions.

3. Depending on the type of connection that you want to create, verify that you have edge or VNA clusters available in VCF Automation. Also, ensure that all edge clusters or VNA clusters are synchronized with NSX Manager.
4. Assign networking resources to organizations during organization creation or as a Day-2 operation of setting regional networking for an existing organization.

### Organizations

VCF Automation supports multi-tenancy by using organizations. An organization is a top level entity that represents a line of business or a tenant. Each organization operates in a secure and isolated environment where users can access only allocated services and resources. The default organization name to log into the Provider Management UI is `System`.

You can create two types of organizations. The organizations for all applications are the primary multi-tenancy model within VCF Automation. Organizations for **All Apps** use the underlying vSphere Supervisor infrastructure. From the VCF Automation Provider Management UI, you can create a VM Apps organization using vSphere cluster capabilities which do not require a Supervisor, shared infrastructure, or tenancy isolation. This option is most suitable for VMware Aria Automation customers who want to continue leveraging the capabilities to provision VM-based applications without any disruptions to your business operations.

Regardless of the type, each organization has its own organization portal where users log in to consume resources and services. Every organization has its own identity source that they connect to and can import users from.

### Content Libraries

A VCF Automation provider content library is an abstraction in VCF Automation that provides a single interface for provider administrators to share content with organizations across regions and to manage their content across the regions. Content libraries contain VM images hosted in vCenter that can be used in blueprints or IaaS services. The provider content libraries are read-only to the users of the organizations.

Content libraries host VM images that application teams use when deploying resources. Organizations can either use content libraries that are provided by the provider administrator or create their own organization-specific content libraries.

### VCF Services and Partner Services

VCF services and partner services in VCF Automation are value-added solutions provided by Broadcom and third-party vendors. The VCF Automation services enhance and extend the functionality of the VCF platform. These services integrate natively with the VCF architecture enabling VCF Automation organizations to leverage tailored solutions for automation, scalability, security, and application management. A service can encapsulate UI and API VCF Automation extensions together with their backend services and lifecycle management. For more information, see the Managing Services in Your VCF Automation Provider Management UI topic in the VMware Cloud Foundation Service Administration and Development 9.1 documentation.

### Provider Administrator Workflow Example

As a **Provider Administrator**, you manage the overall infrastructure in VCF Automation. You must create one or more organizations and assign resources to those organizations. You can use this sample workflow of your primary tasks as a **Provider Administrator**.

1. Configure the Provider Management infrastructure
   a. Configure an identity provider. See Managing Identity Providers in VCF Automation.
   b. Configure the VCF Automation access control. See Configuring the VCF Automation Access Control.
   c. Create a Region in Your VCF Automation.
   d. Configure the networking resources. See Managing Networking Resources in VCF Automation.

   e. Upload a Partner Service to Your VCF Automation Provider Management UI.
2. Create and configure an organization
   a. Create an organization for **All Apps** or **VM Apps**. See Create a VCF Automation Organization for All Applications, or Create a VCF Automation Organization for VM Applications.
   b. Configure the Regional Networking of a VCF Automation Organization.
   c. Create a Region Quota for a VCF Automation Organization.

When you set up the necessary infrastructure, **Organization Administrators** can use the first user credentials, that you configure during the organization creation, to log in to their respective organizations by using the Organization UI. See Managing Organizations in VMware Cloud Foundation Automation and Managing Organizations for VM Apps in VMware Cloud Foundation Automation.

# Managing Identity Providers in VCF Automation

You can integrate your VCF Automation with one or more external identity providers (IdPs), and import users and groups to your organizations. You can configure an LDAP server connection at the system or the organization level, a SAML integration at the organization level, an OpenID Connect (OIDC) integration at the organization level, and a VCF Single Sign-On (SSO) integration at the organization level.

Until you configure an identity provider and import users or groups, the only account that can access VCF Automation is the default administrator account created during deployment. To enable access for additional users, you must configure an identity provider and import the users or groups you want to grant access. See the relevant procedures in this section.

Broadcom recommends configuring VCF Single Sign-On (SSO) as the identity provider for VCF Automation. VCF SSO uses the VCF Operations provider identity configuration, making the identity broker the identity provider. VCF SSO is an automatically configured OIDC integration — the configuration is pushed from VCF Operations, so you do not need to manually configure the integration in the VCF Automation Provider Management UI. VCF SSO does not support multi-tenancy. For the steps to enable VCF SSO, see Configure a New VCF Single Sign-On for a VCF Instance. If you are upgrading from VMware Aria Automation 8 to VCF Automation 9.1 and later, both the Provider Administrator and the Organization Administrator must complete the SSO migration from VMware Identity Manager to the identity broker. See Migrate VCF Automation Single Sign-On Configuration from VMware Identity Manager to VCF Identity Broker.

> **Note:**
>
> Configuring VCF Automation for VCF SSO makes most of the steps in the rest of this section unnecessary. After you configure VCF SSO, you can proceed directly to Configuring the VCF Automation Access Control. You must still import the users and groups that you want to grant access to. For more information, see Import Provider Users to Your VCF Automation and Import a Provider Group to Your VCF Automation.

> **Note:**
>
> To determine the correct values and settings and to ensure proper and accurate configuration, see also the product documentation of those identity providers.

You can also use the VCF Operations provider identity configuration, making VIDB the identity provider. See Identity Broker Models and Configuring VCF Single Sign-On. If you already have an OIDC connection for the provider organization, you cannot connect to VCF Single Sign-On (SSO). You must remove the OIDC configuration to configure VCF SSO. If you configure VCF SSO, a **VCF SSO** tab replaces the **OIDC** tab on the **Identity Providers** page.

An organization can define an identity provider that it shares with other applications. Users authenticate to the identity provider to obtain a token that they can then use to log in to the organization. Such a strategy can enable an enterprise to provide access to multiple, unrelated services, including VCF Automation, with a single set of credentials, an arrangement often referred to as single sign-on.

When integrating your VCF Automation organizations with more than one identity provider, you must not have identical user names across IdPs. You can have only one integration per IdP technology. VCF SSO and OIDC use the same technology, so you can configure either VCF SSO or OIDC, but not both. For example, you can have one LDAP, one SAML, and either one OIDC or one VCF SSO integration simultaneously. If you have configured VCF SSO, you can add SAML or LDAP as additional identity providers. The login page displays all configured sign-in options and to make the login more user friendly, you can customize the button labels from the IdP edit pages.

# Managing LDAP Connections in Your VCF Automation

As a **Provider administrator**, you can configure your VCF Automation system organization and any other organizations in the system to use an LDAP server as a source of users and groups. The organizations can use either the system LDAP connection or a private LDAP connection.

> **Note:**
>
> You can use VCF Single Sign-On (SSO) instead of configuring your VCF Automation organizations to use an LDAP server.

VCF Automation has a centralized, tenant-aware storage area for certificate management. This way, VCF Automation centralizes all certificates in one place so that**system administrators** and**organization administrators** can view, audit, and manage all certificates in use by various components in the system. You can use the VCF Automation API to add, update, or remove certificates from the new tenant-aware storage area. See *VCF Automation API Schema Reference*.

When adding or editing a new LDAP server endpoint, you can use the VCF Automation UI to test a remote connection to an endpoint and to establish a trust relationship. See Test the VCF Automation Connection to a Remote Server and Establish a Trust Relationship Using the Provider Management UI. VCF Automation adds any certificate you decide to trust to a centralized certificate storage area.

> **Note:**
>
> To determine the correct values and settings and to ensure proper and accurate configuration, see also the product documentation of those identity providers.

## Configure a System LDAP Connection in Your VCF Automation

To provide VCF Automation and its organizations with shared access to users and groups, you can configure an LDAP connection at a system level.

> **Note:**
>
> You can use VCF Single Sign-On (SSO) instead of configuring your VCF Automation organizations to use an LDAP server.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under Administration, click **Identity Providers**.
3. On the **Identity Providers** page, click the **LDAP** tab.

   The current LDAP settings appear.
4. Click **Configure**.
5. In the **Connection** tab, enter the required information for the LDAP connection.

| Required Information | Description |
|---|---|
| **Server** | The host name or IP address of the LDAP server. |
| **Port** | The port number on which the LDAP server is listening.<br>For LDAP, the default port number is 389. For LDAPS, the default port number is 636. |
| **Base distinguished name** | The base distinguished name (DN) is the location in the LDAP directory where VCF Automation to connect.<br>To connect at root level, enter only the domain components, for example, `DC=example,DC=com`.<br>To connect to a node in the domain tree structure, enter the distinguished name for that node, for example, `OU=ServiceDirector,DC=example,DC=com`.<br>Connecting to a node limits the scope of the directory available to VCF Automation. |
| **Connector type** | The type of your LDAP server. Can be **Active Directory** or **OpenLDAP**. |
| **Use SSL** | If your server is LDAPS, select this check box. |

| Required Information | Description |
|---|---|
| **Authentication method** | Simple authentication consists of sending the user's DN and password to the LDAP server. If you are using LDAP, the LDAP password is sent over the network in plain text.<br>If you want to use Kerberos, you must configure the LDAP connection by using the vCloud API. |
| **User name** | Enter the full LDAP distinguished name (DN) of a service account with domain admin rights. VCF Automation uses this account to query the LDAP directory and retrieve user information.<br>If the anonymous read support is enabled on your LDAP server, you can leave these text boxes blank. |
| **Password** | The password for the service account that connects to the LDAP server.<br>If the anonymous read support is enabled on your LDAP server, you can leave these text boxes blank. |

6. Click the **User Attributes** tab, examine the default values for the user attributes, and, if your LDAP directory uses different schema, modify the values.
7. Click the **Group Attributes** tab, examine the default values for the group attributes, and, if your LDAP directory uses different schema, modify the values.
8. If you want to customize the **Sign in with LDAP** button label that appears on the VCF Automation login page, enter a new custom button text.

   You can enter up to 24 symbols. You can use special characters and accented letters. If you want to revert to the default text, delete the custom label. The default button label is localized, and depending on your browser language settings, the text might appear in a different language. Custom labels always appear as you enter them.
9. Click **Save**.
10. If you selected the **Use SSL** check box, and if the certificate of the LDAPS server is not yet trusted, on the **Trust Certificate** window, confirm if you trust the certificate presented by the server endpoint.
11. To test the LDAP connection settings and the LDAP attribute mappings:
    a) Click **Test**.
    b) Enter the password of the LDAP server user that you configured and click **Test**.

       If connected successfully, a green check mark is displayed.

       The retrieved user and group attribute values are displayed in a table. The values that are successfully mapped to LDAP attributes are marked with green check marks. The values that are not mapped LDAP attributes are blank and marked with red exclamation marks.
    c) To exit, click **Cancel**.
12. To synchronize VCF Automation with the configured LDAP server, click **Sync**.

    VCF Automation synchronizes the user and group information with the LDAP server regularly depending on the synchronization interval that you set in the general system settings.

    Wait a few minutes for the synchronization to finish.

You can import users and groups from the newly configured LDAP server.

## Edit, Test, and Synchronize an LDAP Connection Using Your VCF Automation Provider Management UI

To configure an LDAP connection, you set the details of your LDAP server. You can test the connection to make sure that you entered the correct settings and the user and group attributes are mapped correctly. When you have a successful LDAP connection, you can synchronize the user and group information with the LDAP server at any time.

- If you plan to connect to an LDAP server over SSL (LDAPS), verify that the certificate of your LDAP server is compliant with the Endpoint Identification introduced in Java 8 Update 181. The common name (CN) or the subject alternative name (SAN) of the certificate must match the FQDN of the LDAP server. For more information, see the *Java 8 Release Changes* at https://www.java.com.

- If you want to use SSL, you can test the connection to the LDAP server and establish a trust relationship with it. See Test the VCF Automation Connection to a Remote Server and Establish a Trust Relationship Using the Provider Management UI.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Identity Providers**.
3. Click **LDAP**.

   The current LDAP settings appear.
4. On the **Custom LDAP** tab, click **Edit**.
5. In the **Connection** tab, enter the required information for the LDAP connection.

| Required Information | Description |
|---|---|
| **Server** | The host name or IP address of the LDAP server. |
| **Port** | The port number on which the LDAP server is listening.<br>For LDAP, the default port number is 389. For LDAPS, the default port number is 636. |
| **Base distinguished name** | The base distinguished name (DN) is the location in the LDAP directory where VCF Automation to connect.<br>To connect at root level, enter only the domain components, for example, `DC=example,DC=com`.<br>To connect to a node in the domain tree structure, enter the distinguished name for that node, for example, `OU=ServiceDirector,DC=example,DC=com`.<br>Connecting to a node limits the scope of the directory available to VCF Automation. |
| **Connector type** | The type of your LDAP server. Can be **Active Directory** or **OpenLDAP**. |
| **Use SSL** | If your server is LDAPS, select this check box. |
| **Authentication method** | Simple authentication consists of sending the user's DN and password to the LDAP server. If you are using LDAP, the LDAP password is sent over the network in plain text.<br>If you want to use Kerberos, you must configure the LDAP connection by using the vCloud API. |
| **User name** | Enter the full LDAP distinguished name (DN) of a service account with domain admin rights. VCF Automation uses this account to query the LDAP directory and retrieve user information.<br>If the anonymous read support is enabled on your LDAP server, you can leave these text boxes blank. |
| **Password** | The password for the service account that connects to the LDAP server.<br>If the anonymous read support is enabled on your LDAP server, you can leave these text boxes blank. |

6. Click the **User Attributes** tab, examine the default values for the user attributes, and, if your LDAP directory uses different schema, modify the values.
7. Click the **Group Attributes** tab, examine the default values for the group attributes, and, if your LDAP directory uses different schema, modify the values.
8. If you want to customize the **Sign in with LDAP** button label that appears on the VCF Automation login page, enter a new custom button text.

   You can enter up to 24 symbols. You can use special characters and accented letters. If you want to revert to the default text, delete the custom label. The default button label is localized, and depending on your browser language settings, the text might appear in a different language. Custom labels always appear as you enter them.

9. Click **Save**.
10. If you selected the **Use SSL** check box, and if the certificate of the LDAPS server is not yet trusted, on the **Trust Certificate** window, confirm if you trust the certificate presented by the server endpoint.
11. To test the LDAP connection settings and the LDAP attribute mappings:
    a) Click **Test**
    b) Enter the password of the LDAP server user that you configured and click **Test**.
       If connected successfully, a green check mark is displayed.

       The retrieved user and group attribute values are displayed in a table. The values that are successfully mapped to LDAP attributes are marked with green check marks. The values that are not mapped LDAP attributes are blank and marked with red exclamation marks.
    c) To exit, click **Cancel**.
12. To synchronize VCF Automation with the configured LDAP server, click **Sync**.
    VCF Automation synchronizes the user and group information with the LDAP server regularly depending on the synchronization interval that you set in the general system settings.
    Wait a few minutes for the synchronization to finish.

You can import users and groups from the newly configured LDAP server.

### Configure an Organization LDAP Connection in Your VCF Automation

As a **Provider Administrator**, you can configure a VCF Automation organization to use the system LDAP connection as a shared source of users and groups. You can configure an organization to use a separate LDAP connection as a private source of users and groups.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Organizations**.
3. Click the name of the target organization, and click **Launch Organization Portal**.
4. From the top navigation bar of the VCF Automation organization UI, click **Administer**.
5. In the left navigation pane, click **Identity Providers**, and click **LDAP**.
6. On the **LDAP Options** tab, click **Configure** for initial configuration, or **Edit** for reconfiguring the LDAP connection..
7. Configure the LDAP source of users and groups for this organization, and click **Save**.

| Option | Description |
|---|---|
| **Do not use LDAP** | The organization does not use an LDAP server as a source of organization users and groups. |
| **VCF Automation system LDAP service** | The organization uses the VCF Automation system LDAP connection that you previously configured.<br>See Configure a System LDAP Connection in Your VCF Automation. |
| **Custom LDAP service** | The organization uses a private LDAP server as a source of organization users and groups.<br>Click the **Custom LDAP** tab and Edit, Test, and Synchronize an LDAP Connection Using Your VCF Automation Provider Management UI. |

## Configure Your VCF Automation System to Use a SAML Identity Provider

If you want to import users and groups from a SAML identity provider to your VCF Automation `System` organization, you must configure the `System` organization with this SAML identity provider. Imported users can log in to the `System` organization with the credentials established in the SAML identity provider.

- Verify that you have access to a SAML 2.0 compliant identity provider.

- Obtain an XML file with the following metadata from your SAML identity provider.
  - The location of the single sign-on service
  - The location of the single logout service
  - The location of the service's X.509 certificate

  For information on configuring and acquiring metadata from a SAML provider, consult the documentation for your SAML provider.

To configure VCF Automation with a SAML identity provider, you establish a mutual trust by exchanging SAML service provider and identity provider metadata.

To determine the correct values and settings and to ensure proper and accurate configuration, see also the product documentation of those identity providers.

When an imported user attempts to log in, the system extracts the following attributes from the SAML token, if available, and use them for interpreting the corresponding pieces of information about the user.

- `email address = "EmailAddress"`
- `user name = "UserName"`
- `full name = "FullName"`
- `user's groups = "Groups"`
- `user's roles = "Roles"` (this attribute is configurable)

Group information is used if the user is not directly imported but is expected to log in by virtue of membership in imported groups. A user can belong to multiple groups, so can have multiple roles during a session.

If an imported user or group is assigned the **Defer to Identity Provider** role, the roles are assigned based on the information gathered from the `Roles` attribute in the token. If a different attribute is used, this attribute name can be configured using API and only the `Roles` attribute is configurable. If the **Defer to Identity Provider** role is used, but no role information can be extracted, the user can log in but has no any rights to perform any activities.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Identity Providers**.
3. Click **SAML**, and click **Edit**.

   The current SAML settings appear.
4. From the **Service Provider** tab, download the VCF Automation SAML service provider metadata.
   a) Enter an Entity ID for the `System` organization.

      The Entity ID uniquely identifies your `System` organization to your Identity Provider.
   b) Examine the certificate expiration date and, if expiring soon, regenerate the certificate by clicking **Regenerate**.

      The certificate is included in the SAML metadata, and is used for both encryption and signing. Either or both of these might be required depending on how trust is established between your organization and your SAML IDP.
   c) Click **Retrieve Metadata**.

      Your browser downloads the SAML service provider metadata, an XML file which you must provide to your identity provider.
5. On the **Identity Provider** tab, upload the SAML metadata that you previously received from your identity provider.
   a) Select **Use SAML Identity Provider**.
   b) Either click the **Browse** icon and upload the file, or copy and paste its content in the **Metadata XML** text box.
6. If you want to customize the **Sign in with SAML** button label that appears on the VCF Automation login page, enter a new custom button text.

   You can enter up to 24 symbols. You can use special characters and accented letters. If you want to revert to the default text, delete the custom label. The default button label is localized, and depending on your browser language settings, the text might appear in a different language. Custom labels always appear as you enter them.

7. Click **Save**.

# Configure Your System to Use an OpenID Connect Identity Provider Using Your VCF Automation Provider Management UI

If you want to import users and groups from an OpenID Connect (OIDC) identity provider to your VCF Automation `System` organization, you must configure your `System` organization with this OIDC identity provider. Imported users can log in to the `System` organization with the credentials established in the OIDC identity provider.

> **Note:**
>
> If VCF Single Sign-On (SSO) is configured, a **VCF SSO** tab replaces the **OIDC** tab on the **Identity Providers** page.

OAuth is an open federation standard that delegates user access. OpenID Connect is an authentication layer on top of the OAuth 2.0 protocol. By using OpenID Connect, clients can receive information about authenticated sessions and end-users. The OAuth authentication endpoint must be reachable from the VCF Automation cells which makes it more suitable when you use public identity providers or provider-managed ones.

> **Important:**
>
> Endpoint misconfigurations can lead to user inability to complete the SSO logout from VCF Automation. See Broadcom knowledge base article 394731.

You can allow organizations to generate and issue API access tokens that applications can use on their behalf.

You can configure VCF Automation to automatically refresh your OIDC key configurations from the JWKS endpoint you provide. You can configure the frequency of the key refresh process and the rotation strategy that determines whether VCF Automation adds new keys, replaces the old keys with new, or the old keys expire after a certain period.

> **Note:**
>
> To determine the correct values and settings and to ensure proper and accurate configuration, see also the product documentation of those identity providers.

VCF Automation generates audit events for both successful and failed key refreshes under the event topic `com/vmware/vcloud/event/oidcSettings/keys/modify`. The audit events for failed key refreshes include additional information about the failure.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation panel, click **Administration**.
3. In the left pane, under **Identity Providers**, click **OIDC**.
4. If you are configuring OIDC for the first time, copy the client configuration redirect URI and use it to create a client application registration with an identity provider that complies with the OpenID Connect standard, for example, VMware Cloud Foundation® Identity Broker.

   You need this registration to obtain a client ID and a client secret that you must use during the OIDC identity provider configuration.

5. Click **Configure**.
6. Verify that OpenID Connect is active, and enter the client ID and client secret information from the OIDC server registration.
7. Optional: To use the information from a well-known endpoint to automatically fill in the configuration information, turn on the **Configuration Discovery** toggle and enter a URL at the site of the provider that VCF Automation can use to sent authentication requests to.
8. Click **Next**.
9. If you did not use **Configuration Discovery** in Step 6, enter the information in the **Endpoints** section.
   a) Enter the endpoint and issuer ID information.
   b) If you are using VMware Cloud Foundation® Identity Broker as an identity provider, select **SCIM** as access type.
      For other identity providers, you can leave the default **User Info** selection.
   c) If you want to combine claims from the `UserInfo` endpoint and the ID Token, turn on the **Prefer ID Token** toggle.
      The identity providers do not provide all the required claims set in the `UserInfo` endpoint. By turning on the **Prefer ID Token** toggle, VCF Automation can fetch and consume claims from both sources.
   d) Enter the maximum acceptable clock skew.
      The maximum clock skew is the maximum allowable time difference between the client and server. This time compensates for any small time differences in the time stamps when verifying tokens. The default value is 60 seconds.
   e) Click **Next**.
10. If you did not use **Configuration Discovery** in Step 6, enter the scope information, and click **Next**.
    VCF Automation uses the scopes to authorize access to user details. When a client requests an access token, the scopes define the permissions that this token has to access user information.
11. If you are using **User Info** as an access type, map the claims and click **Next**.
    You can use this section to map the information VCF Automation gets from the user info endpoint to specific claims. The claims are strings for the field names in the VCF Automation response.
12. If you want VCF Automation to automatically refresh the OIDC key configurations, turn on the **Automatic Key Refresh** toggle.
    a) If you did not use **Configuration Discovery** in Step 6, enter the **Key Refresh Endpoint**.
       The **Key Refresh Endpoint** is a JSON Web Key Set (JWKS) endpoint and it is the endpoint from which VCF Automation fetches the keys.
    b) Select how often the key refresh occurs.
       You can set the period in hourly increments from 1 hour up to 30 days.
    c) Select a **Key Refresh Strategy**.

| Option | Description |
|---|---|
| **Add** | Add the incoming set of keys to the existing set of keys. All keys in the merged set are valid and usable.<br>For example, your existing set of keys includes keys A, B, and D. Your incoming set of keys includes keys B, C, and D. When the key refresh occurs, the new set includes keys A, B, C, and D. |
| **Replace** | Replace the existing set of keys with the incoming set of keys.<br>For example, your existing set of keys includes keys A, B, and D. Your incoming set of keys includes keys B, C, and D. When the key refresh occurs, key C replaces key A. The incoming keys B, C, and D become the new set of valid keys without any overlap with the old set. |

| Option | Description |
| --- | --- |
| **Expire After** | You can configure an overlap period between the existing and incoming sets of keys. You can configure the overlapping time using the **Expire Key After Period**, which you can set in hourly increments from 1 hour up to 1 day.<br>The key refresh runs start at the beginning of every hour. When the key refresh occurs, VCF Automation tags as expiring the keys in the existing set of keys that are not included in the incoming set. At the next key refresh run, VCF Automation stops using the expiring keys. Only keys included in the incoming set are valid and usable.<br>For example, your existing set of keys includes keys A, B, and D. The incoming set includes keys B, C, and D. If you configure the existing keys to expire in 1 hour, there is 1 hour of overlap during which both sets of keys are valid. VCF Automation marks key A as expiring and until the next key refresh run, keys A, B, C, and D are usable. At the next run, key A expires and only B, C, and D continue working. |

13. If you did not use **Configuration Discovery** in Step 6, upload the private key that the identity provider uses to sign its tokens.
14. If you want to customize the **Sign in with OIDC** button label that appears on the VCF Automation login page, enter a new custom button text.

    You can enter up to 24 symbols. You can use special characters and accented letters. If you want to revert to the default text, delete the custom label. The default button label is localized, and depending on your browser language settings, the text might appear in a different language. Custom labels always appear as you enter them.
15. Click **Save**.

- Subscribe to the `com/vmware/vcloud/event/oidcSettings/keys/modify` event topic.
- Verify that the **Last Run** and the **Last Successful Run** are identical. The runs start at the beginning of the hour. The **Last Run** is the time stamp of the last key refresh attempt. The **Last Successful Run** is the time stamp of the last successful key refresh. If the time stamps are different, the automatic key refresh is failing and you can diagnose the problem by reviewing the audit events.

## Enable PKCE in VCF Automation

VCF Automation supports Proof Key for Code Exchange (PKCE).

Verify that you configured your system to use an OpenID Connect Identity Provider. See Configure Your System to Use an OpenID Connect Identity Provider Using Your VCF Automation Provider Management UI.

PKCE is an extension to the OAuth 2.0 Authorization Code flow that is used to prevent CSRF and authorization code injection attacks. For more information, see Proof Key for Code Exchange in the OAuth 2.0 documentation.

1. Run the request to retrieve your organization's settings.

   ```
   GET https://vcfa.example.com/api/admin/org/organization_id/settings/oauth
   ```

   The response contains the OAuth settings for your organization.
2. Under `OrgOAuthSettings`, make the following changes.
   a) Modify the `usePkce` element to `true`.
   b) Optional: If your identity provider requires that client credentials be sent as an authorization header when making the API request to retrieve the access token, modify the `sendClientCredentialsAsAuthorizationHeader` element to `true`.

      The default behavior is for the client credentials to be sent in the body of the API request.
3. To update your OAuth settings with your modifications, run a PUT request.

   ```
   PUT https://vcfa.example.com/api/admin/org/organization_id/settings/oauth
   ```

   In the body of the request, include the modified elements of your OAuth settings.

# Remapping VCF Automation Users Between Identity Providers

You can remap VCF Automation users between identity providers.

You can remap individual users from one identity provider (IDP) to another by using the VCF Automation API. You can use the VCF Automation UI for bulk remapping of users between identity providers.

## Remap Users Between Identity Providers Using Your VCF Automation Provider Management UI

You can use the VCF Automation UI for bulk remapping of users between identity providers.

- Verify that the users you want to remap are imported into the organization. See Managing VCF Automation Provider Users and Groups.
- Verify that your role includes the**Group / User: Manage** right.
- Verify that the organization is configured with the identity provider types that you want to remap between.

1. Log in to the VCF Automation Provider Management UI.
2. From the primary left navigation pane, click **Access Control**.
3. Click **Users**.
4. Click **Bulk Update**.
5. Click **Export Users**.

   The users are exported into a CSV file.
6. In the CSV file with the users, change the value in the `providerType` field to identify the new IdP for each user.

   VCF Automation supports the `SAML`, `LDAP`, and `OAUTH` values. To remap a user to an OIDC identity provider, you must change the value in the `providerType` field to `OAUTH`.
7. Optional: To match the user name in the IdP that each user is remapping to, modify the user name.

   For VCF Automation to continue to associate the user's assets with the user when they log in through the new login flow, the ID of the user must remain unchanged. Changes to all other fields are ignored.
8. Save your changes and close the CSV file.
9. In the **Users Bulk Update** wizard, click **Next**.
10. Click **Select CSV File**, browse for the file, and upload it.
11. Click **Next**.
12. Click **Update**.

    When you start the update process, users are remapped one by one. Users for which no changes are detected are skipped. If you close the VCF Automation UI tab while before the process is complete, the update stops.

## Remap a User Between Identity Providers by Using the VCF Automation API

You can remap individual users from one identity provider (IdP) to another by using the VCF Automation API.

- Verify that the users you want to remap are imported into the organization. See Managing VCF Automation Provider Users and Groups.
- Verify that your role includes the**Group / User: Manage** right.
- Verify that the organization is configured with the identity provider types that you want to remap between.

For information about bulk remapping of users between identity providers by using the VCF Automation UI, see Remap Users Between Identity Providers Using Your VCF Automation Provider Management UI.

1. Make a GET request to `/cloudapi/1.0.0/users`.

   VCF Automation returns a list of the users within the organization.

2. Locate the user you want to remap, and retrieve the user information.

   ```
   GET /cloudapi/1.0.0/users/{user_id}
   ```

3. Make a PUT request to `/cloudapi/1.0.0/users/{user_id}`.

   To remap a user, you must change the `providerType` field to identify the new IDP. VCF Automation supports the `SAML`, `LDAP`, and `OAUTH` values. Additionally, to match the user name in the IDP that the user is remapping to, you can modify the user name. For VCF Automation to continue to associate the user's assets with the user when they log in through the new login flow, the ID of the user must remain unchanged.

   **Important:**

   If you are remapping to provider type `LDAP`, VCF Automation validates the user name with the LDAP server before committing the operation. If VCF Automation does not complete this step for any reason, for example, loss of connectivity to the LDAP server, the remapping fails.

   If you are remapping a user to be a local user by specifying provider type `LOCAL`, similar to the process of creating a user, you must provide a password.

4. Verify that VCF Automation returns an `OK` response specifying the newly remapped provider type in the response body.

   To find the user that you want to remap, make the following request.

   Request:

   ```
   GET /cloudapi/1.0.0/users?pageSize=10 HTTP/1.1
   Host: 127.0.0.1:8443
   Accept: application/json;version=37.1
   ```

   Sample response:

   ```
   {
     "resultTotal": 2,
     "pageCount": 1,
     "page": 1,
     "pageSize": 10,
     "associations": null,
     "values": [
       ...,
       {
         "username": "testuser",
         "fullName": "",
         "description": null,
         "id": "urn:vcloud:user:2b038199-0063-4c13-9bba-a3b58d775785",
         "roleEntityRefs": [
           {
             "name": "vApp Author",
             "id": "urn:vcloud:role:85f69506-52a5-3e20-869a-ea18d667e19e"
           }
         ],
         "orgEntityRef": {
           "name": "testorg",
           "id": "urn:vcloud:org:806f0d87-c8b9-47f5-bfbe-3dc73a4c0d14"
         },
         "password": "******",
         "email": "",
         "nameInSource": "testuser",
         "enabled": true,
         "isGroupRole": false,
         "providerType": "LOCAL"
       }
     ]
   ```

```
}
```

To remap `testuser` from `LOCAL` to `LDAP`, make a PUT request.

Request:

```
PUT /cloudapi/1.0.0/users/urn:vcloud:user:2b038199-0063-4c13-9bba-a3b58d775785 HTTP/1.1
Host: 127.0.0.1:8443
Accept: application/json;version=37.1
Content-Type: application/json;version=37.1

Body: {
  "username": "testuser",
  "fullName": "",
  "description": null,
  "id": "urn:vcloud:user:2b038199-0063-4c13-9bba-a3b58d775785",
  "roleEntityRefs": [
    {
      "name": "vApp Author",
      "id": "urn:vcloud:role:85f69506-52a5-3e20-869a-ea18d667e19e"
    }
  ],
  "orgEntityRef": {
    "name": "testorg",
    "id": "urn:vcloud:org:806f0d87-c8b9-47f5-bfbe-3dc73a4c0d14"
  },
  "password": "******",
  "email": "",
  "nameInSource": "testuser",
  "enabled": true,
  "isGroupRole": false,
  "providerType": "LDAP"
}
```

Sample response:

```
{
  "username": "testuser",
  "fullName": "",
  "description": null,
  "id": "urn:vcloud:user:2b038199-0063-4c13-9bba-a3b58d775785",
  "roleEntityRefs": [
    {
      "name": "vApp Author",
      "id": "urn:vcloud:role:85f69506-52a5-3e20-869a-ea18d667e19e"
    }
  ],
  "orgEntityRef": {
    "name": "testorg",
    "id": "urn:vcloud:org:806f0d87-c8b9-47f5-bfbe-3dc73a4c0d14"
  },
  "password": null,
  "email": "",
  "nameInSource": "\\63\\36\\62\\35\\30\\66\\35\\63\\2D\\61\\62\\30\\35\\2D\\34\\37\\64\\33\\2D\
\62\\61\\64\\34\\2D\\39\\32\\64\\35\\32\\37\\30\\36\\62\\39\\39\\33",
  "enabled": true,
  "isGroupRole": false,
  "providerType": "LDAP"
}
```

# Using VCF Automation as an Identity Provider Proxy Server

You can take advantage of the **OIDC Services** to use VCF Automation as a tenant-aware identity provider proxy server.

When VCF Automation is configured as an identity provider proxy server by using the OAuth 2.0 OpenID Connect standard, relying parties can use VCF Automation for tenant-aware authentication of users known to VCF Automation. For detailed information on the OpenID Connect standard, see OpenID Connect Core 1.0.

As an identity provider proxy server, VCF Automation acts as intermediary between the client application (relying party) and the identity provider, and delegates actual authentication to the respective authentication mechanism used by the provider or tenants.

**System administrators** can configure relying parties that enable organizations to use VCF Automation as an identity provider proxy. Starting with version 9.1.1, organization administrators can also configure tenant-specific relying parties exclusively for their organization.

> **Important:**
>
> When making browser-based requests to the `/oidc` endpoints, the default behavior is for browser pages to be loaded from the VCF Automation host. To use OIDC, you can also configure a non-VCF Automation host into the CORS filtering list.

## Authentication Flow

Integration with VCF Automation is implemented through the OAuth 2.0 OIDC authorization code flow standard. For detailed information, see Authentication using the Authorization Code Flow.

When a relying party redirects a user to VCF Automation without an existing client session, VCF Automation prompts the user to log in by identifying the target organization or provider UI. The user authenticates through the configured authentication mechanism, which might involve further redirections to external identity providers. If the browser of the user detects an existing VCF Automation user session, the authentication flow provides an SSO experience and no user interaction is necessary for reauthentication.

After successful authentication, VCF Automation returns an access token and an ID token. The authorization code is valid for 5 minutes. The ID token is valid for one hour. Starting with version 9.1.1, the access token is also valid for one hour.

You cannot use the access token that VCF Automation returns upon successful authentication for accessing the UI portals or for making regular VCF Automation API calls. Starting with version 9.1.1, you can use the token to make regular VCF Automation API calls only if you specified the `vcfa_api scope` when creating the relying party.

**ID Token Details**

The ID token that VCF Automation returns contains the following OpenID standard claims and VCF Automation specific claims.

| Claim | Description |
|---|---|
| `at_hash` | (OpenID standard claim) Access token hash value. |
| `sub` | (OpenID standard claim) The `userId` in VCF Automationin UUID format. |
| `iss` | (OpenID standard claim) Public address of VCF Automation. |
| `preferred_username` | (OpenID standard claim) User name of the user in VCF Automation |
| `nonce` | (OpenID standard claim) String value used to associate a client session with an ID token, and to mitigate replay attacks. Only present if it was initially included in the relying party request. |
| `aud` | (OpenID standard claim) The audience for this token. The value is the client ID of requesting relying party. |

| Claim | Description |
|---|---|
| azp | (OpenID standard claim) Authorized party for the token. The value is the client ID of the relying party. Its value is same as the aud claim. |
| name | (OpenID standard claim) Full name of the user, if known to VCF Automation. |
| phone_number | (OpenID standard claim) Phone number of the user, if known to VCF Automation. |
| exp | (OpenID standard claim) Expiration time. Time after which the ID token is not accepted for processing. |
| iat | (OpenID standard claim) Time at which the ID token was issued. |
| email | (OpenID standard claim) Email address of the user, if known to VCF Automation. |
| roles | (VCF Automation custom claim) An array of the names of the roles the user has in VCF Automation. |
| groups | (VCF Automation custom claim) An array of the names of the groups in which the user belongs in VCF Automation. |
| org_name | (VCF Automation custom claim) Name of the organization in which the user is logged in. |
| org_display_name | (VCF Automation custom claim) Display name of the organization. |
| org_id | (VCF Automation custom claim) The organization ID in UUID format. |

**OpenID Request Scopes**

The scope of the OpenID request is used to specify the privileges requested for an access token.

| Scope Values | Description. |
|---|---|
| openid | Required. OpenID standard scope. |
| profile | OpenID standard scope. Requests access to the end-user default profile claims. |
| email | OpenID standard scope. Requests access to the end-user email address claims. |
| groups | OpenID standard scope. Requests access to the groups that the user is part of in VCF Automation. |
| phone | OpenID standard scope. Requests access to the user phone number claim. |
| vcd_idp | VCF Automation specific scope. Requests access to VCF Automationcustom claims, such as roles, groups, org_name, org_display_name, and org_id. |

| Scope Values | Description. |
| --- | --- |
| `vcfa_api` | Optional scope for versions 9.1.1 and later. When present, VCF Automation API endpoints outside the OIDC authorization flow accept tokens issued for this RP. When absent, tokens are rejected with HTTP 403 on all non-OIDC endpoints.<br>You must specify `vcfa_api` in the scope array only when creating a relying party. You cannot add or remove this scope after creation. Do not include other scope values in the creation request. The six base scopes (`openid`, `profile`, `email`, `phone`, `groups`, and `vcd_idp`) apply automatically. To change whether a relying party grants access to VCF Automation API endpoints, you must delete and recreate the relying party.<br>**Note:**<br>When a client uses the `refresh_token` grant, the resulting access token includes `vcfa_api` only if both conditions are true:<br>• The original authorization for the refresh token included `vcfa_api`.<br>• The refresh request omits the scope parameter or explicitly lists `vcfa_api`.<br>If the refresh request includes a scope parameter that does not list `vcfa_api`, the issued token does not include this scope, even if the original authorization included it. |

**Endpoints**

- You can use the access token returned by VCF Automation to retrieve claims about the authenticated user at the *hostname*`/oidc/UserInfo` endpoint. For details, see UserInfo Endpoint.
- You can retrieve the provider configuration values, including the JWKS endpoint and information about other endpoints and scopes necessary for the OIDC proxy configuration at the well-known configuration URL *hostname*`/oidc/.well-known/openid-configuration`. See View and Edit the OIDC Proxy General Settings in Your VCF Automation.

## Token Exchange Access to VCF Automation Identity Provider Proxy

Programmatic integration with the identity provider proxy functionality of VCF Automation is available through the token exchange flow that is detailed below. This flow does not involve the VCF Automation UI and is suitable for scripted access, such as CLI.

1. Obtain a VCF Automation JWT by either directly logging in or by using an API token.
2. Run a POST request.

   ```
   POST  hostname/oidc/oauth2/token
   ```

   a. Select `x-www-form-urlencoded` for the body of the request.
   b. Include the following parameters in the body of the request.

   ```
   {
        "grant_type": "urn:ietf:params:oauth:grant-type:jwt-bearer",
        "assertion": "VMware_Cloud_Director JWT",
        "client_id": "Relying_party_ID",
        "scope": "openid profile email phone groups vcd_idp",
      }
   ```

3. The response returns both an ID token that includes the OIDC and VCF Automation claims, and an access token that you can use to retrieve claims about the authenticated user at the *hostname*`/oidc/UserInfo` endpoint.

   Encoded ID token example:

```
eyJhbGciOiJSUzI1NiIsInR5NDg4SI6I.................4dHnbU1RQ6Y9Yohgw
```

Decoded ID token example:

```
{
  "at_hash": "1AA1aAA1AAAAAAaAA1A11a",
  "sub": "111111111-1111-1111-1111-11111111",
  "roles": [
    "Organization Administrator"
  ],
  "iss": "https://hostname/oidc",
  "groups": [
    "ALL USERS"
  ],
  "preferred_username": "testuser@vcd-ms1",
  "nonce": "ab123acab",
  "aud": "33333333-3333-3333-3333-33333333333",
  "azp": "22222222-2222-2222-2222-22222222",
  "org_id": "12345678-1234-1234-1234-123456789abc",
  "org_display_name": "oidcorg",
  "name": "test user",
  "phone_number": " ",
  "exp": 1111111111,
  "org_name": "oidcorg",
  "iat": 1111111111,
  "email": "user1@vmware.com"
}
```

User Info response example:

```
{
    "sub": "111111111-1111-1111-1111-11111111",
    "preferred_username": "administrator",
    "name": "administrator user",
    "email": "user1@site.com",
    "phone_number": "0 (111) 222-3333",
    "roles": [ "system administrator" ],
    "groups": [],
    "org_name": "system",
    "org_display_name": "System Organization",
    "org_id": "12345678-1234-1234-1234-123456789abc"
}
```

## Provider and Tenant OIDC Relying Parties

Starting with version 9.1.1, provider and tenant organizations can create and manage OIDC relying parties (RPs) in VCF Automation. Managing RPs at both levels provides granular control over authentication scopes and tenancy access.

The `isGlobal` field on an RP controls the tenancy scope. You must set this field during creation.

The `isGlobal` value determines how VCF Automation resolves the effective operating organization when issuing tokens:

- Global provider-org RP (`isGlobal: true`): Standard tenancy resolution applies. The RP does not alter the operating organization of the system user.
- Restricted provider-org RP (`isGlobal: false`): When a system user requests a token for this RP, VCF Automation sets their effective operating organization to the provider organization. The token does not include cross-tenant scope.
- Tenant-org RP: Always behaves as non-global. VCF Automation scopes the token to the tenant that owns the RP.

**Important:**

You can set `isGlobal` only during RP creation. You cannot change it afterward. If you attempt to modify the field on an existing RP, the API returns an HTTP 400 error. To change the global or non-global designation, you must delete the RP and recreate it with the required value.

**Provider organization RPs**

When you create an RP in the provider organization using API version 9.1.1 or later, set `isGlobal` to control whether the RP is available across all tenant organizations or restricted to the provider organization.

| **`isGlobal` value at creation** | **Behavior** |
|---|---|
| `true` | The RP is global. System users can obtain tokens for this RP regardless of their operating tenant organization. |
| `false` | The RP is restricted to the provider organization. Tokens issued for this RP apply exclusively to the provider organization. |
| Omitted or `null` | Defaults to `true`. |

**Note:**

In API version 9.1.0 and earlier, provider organization RPs are always global. If you set `isGlobal` to false, the API returns an HTTP 400 error. Omitting the field defaults to true. This behavior preserves compatibility for API clients that omit the field.

**Tenant organization RPs**

Starting with version 9.1.1, tenant organizations can create organization-scoped RPs without provider organization involvement. For tenant organization RPs, you must set `isGlobal` to `false` or omit the field.

| **`isGlobal` value at creation** | **Behavior** |
|---|---|
| `false` or omitted | The RP applies only to the tenant organization. |
| `true` | The API returns an HTTP 400 error. Tenant organizations cannot create global RPs. |

## Audience Validation Requirement for Tenanted Relying Parties

Starting with version 9.1.1, the VCF Automation tenanted OIDC service enforces tenancy boundaries through two complementary requirements. VCF Automation never issues a token where the audience or the relying party and the authenticated user belong to different tenant organizations. Relying parties must uphold the complementary requirement: reject any token whose `aud` claim does not include the registered client ID of the relying party.

Every relying party registered in VCF Automation is assigned a UUID that is globally unique across all tenant organizations. This UUID is set as the `aud` (audience) claim in every token VCF Automation issues for that relying party. Because relying party IDs never collide across tenants, a valid `aud` match confirms that the token was issued for your application within your organization.

When a user authenticates through the VCF Automation tenanted OIDC service, VCF Automation verifies that the target relying party exists within a tenant organization accessible to that user before issuing the token. The resulting token carries an audience value that is bound to the tenancy context of the user. A token issued for a relying party in Organization A never carries the audience value of a relying party registered in Organization B.

Relying parties integrating with the VCF Automation tenanted OIDC service must validate the `aud` claim as specified by:

- For ID tokens - OpenID Connect Core 1.0 — ID Token Validation
- For access tokens - RFC 9068 §4 — JWT Profile for OAuth 2.0 Access Tokens

Specifically, a relying party must:

1. Extract the `aud` claim from the received token.
2. Verify that the `aud` claim contains the registered client ID of the relying party.

3. Reject any token where this check fails, regardless of whether the token is otherwise cryptographically valid.

Skipping or weakening this check breaks the tenancy boundary. A compliant OIDC or JWT library performs this validation automatically when configured with the expected audience value — the client ID assigned at relying party registration in VCF Automation.

> **Note:**
>
> VCF Automation provides no additional backstop beyond the `aud` claim. If you omit audience validation, your integration assumes full responsibility for any cross-tenant token acceptance.

## Configure VCF Automation as an OpenID Provider Proxy Server

You can use VCF Automation as a tenant-aware OpenID Connect (OIDC) identity provider proxy server.

Verify that your role includes the **OIDC Server: Manage** right.

After you configure VCF Automation as an OIDC proxy server, when a user attempts to log in to the OIDC relying party (OIDC client), they are redirected to VCF Automation and prompted to enter the name of their organization and their SSO or local credentials. After providing the necessary credentials, the user is directed to the OIDC relying party.

VCF Automation delegates actual authentication to the authentication mechanism used by the provider or organization. This can result in additional redirects to any external Identity Providers that perform authentication for those users.

Starting with version 9.1.1, to use the RP to call VCF Automation API endpoints outside the OIDC authorization flow, such as `/cloudapi`, you must include `vcfa_api` in the scope array during creation through the VCF Automation API. If you omit this scope, VCF Automation rejects the RP tokens on non-OIDC endpoints. You cannot add the scope after creation.

For more information about scopes, see Using VCF Automation as an Identity Provider Proxy Server.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **OIDC Services**.
3. Click **Relying Parties** and click **New**.
4. Enter a relying party name for the client application registration, and make a note of it.
5. Starting with version 9.1.1, if you want this relying party to be usable across all tenant organizations, select the **Is Global** check box.

   When **Is Global** is selected, system users can obtain tokens for this relying party regardless of which tenant organization they operate in. When deselected, tokens are scoped to the provider organization only. You cannot change this setting after the relying party is created. To change the designation, delete the relying party and recreate it.
6. Enter the URI to which to redirect users attempting to log in to the relying party.
7. Click **Save**.
8. Copy the relying party ID and secret, and make note of them.
9. Configure your OIDC relying party to use VCF Automation as an identity provider proxy server with the relying party ID and secret.

   > **Tip:**
   >
   > You can retrieve the provider configuration values, including the JWKS endpoint and information about other endpoints and scopes necessary for the OIDC proxy configuration at the well-known configuration URL *`hostname`*`/oidc/.well-known/openid-configuration`. See View and Edit the OIDC Proxy General Settings in Your VCF Automation.

When a user attempts to log in to the OIDC relying party, the redirection behavior depends on the relying party configuration.

- For a tenant organization relying party, VCF Automation automatically redirects the user to the login page of the organization to provide their credentials.

- For a global relying party, the system automatically redirects the user to VCF Automation and prompts the user to select an organization before providing their credentials.

After a successful authorization, the system redirects the user back to the OIDC relying party.

You can use the vertical ellipsis ( ⋮ ) menu next to the name of the relying party ID to edit and delete the relying party, to regenerate a secret, and to copy the relying party ID.

## View and Edit the OIDC Proxy General Settings in Your VCF Automation

In the VCF Automation **General Settings** tab, you can view the `openid-configuration` endpoint, from which you can retrieve the provider configuration values, including the JWKS endpoint and information about other endpoints and scopes necessary for the OIDC proxy configuration.

Verify that your role includes the **OIDC Server: Manage Settings** right.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **OIDC Services**.
3. Click **General Settings**.

   The general OIDC proxy settings appear, including the well-known configuration URL from which you can retrieve the provider configuration values, the access token lifetime, the ID token lifetime, the authorization code lifetime, and the redirect URL policy.
4. To edit the settings, click **Edit**.

   The refresh tokens are used by the services backing the Organization Portal to perform long-running actions and the refresh token lifetime limits the length of these tasks.

## Managing OIDC Proxy Keys in Your VCF Automation

When you configure VCF Automation to function as an OIDC identity provider proxy, VCF Automation generates a pair of OIDC keys with which it signs the JWT tokens that it issues.

Verify that your role includes the **OIDC Server: Manage** right.

When configured as an identity provider proxy server, VCF Automation automatically generates a single built-in 2048-bit RSA signing key, which the **system administrator** can choose to use or to discard. Any new keys must comply with the minimum key size and the other VCF Automation cryptographic requirements.

When rotating the signing key you must upload the new key, wait at least 24 hours before marking the new key as the active signing key, and keep the previous signing key for a sufficient period before decommissioning it, in order to prevent the invalidation of any tokens issued by the old key.

> **Tip:**
>
> To view the VCF Automation key requirements, in the left navigation bar, click **SSL**.

The relying parties that are using VCF Automation as an OIDC proxy server can retrieve the provider configuration values, including the list of available public keys from the JWKS endpoint listed at `{{hostname}}/oidc/.well-known/openid-configuration`.

### Add an OIDC Proxy Key Set Using Your VCF Automation

You can manually add an OIDC proxy key set to VCF Automation.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **OIDC Services**.
3. Select **Keys**.
4. To manually upload a new OIDC proxy key set, click **New**.
5. Enter a description for the OIDC proxy key set.
   You can edit the key description later, if necessary.
6. Under Public Key, click **Browse Files**, navigate to the public key PEM file, and upload it.
7. Under Private Key, click **Browse Files**, navigate to the private key PEM file, and upload it.
8. Enter the private key passphrase.
9. Click **Save**.

To edit the OIDC proxy key set description, click the vertical ellipsis ( ⋮ ) next to the name of the key set that you want to edit, and select **Edit**.

### Activate a New OIDC Proxy Key Set Using Your VCF Automation

You can use the VCF Automation Provider Management UI to select a new active OIDC proxy key.

- Verify that your role includes the **OIDC Server: Manage** right.
- Verify that you uploaded the key set that you want to make active.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **OIDC Services**.
3. Click **Keys**

   A list of the available key sets appears. To view the key labels, hover over the key names. The currently used key has an **Active** label.

4. Click the vertical ellipsis ( ⋮ ) next to the name of an inactive key set, and click **Make Active**.

### Delete an OIDC Proxy Key Set From Your VCF Automation

If an OIDC key set is no longer in use, you can delete it by using the VCF Automation Provider Management UI.

Verify that your role includes the **OIDC Server: Manage Settings** right.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **OIDC Services**.
3. Click **Keys**

   A list of the available key sets appears. To view the key labels, hover over the key names. The currently used key has an**Active** label.

4. Click the vertical ellipsis ( ⋮ ) next to the name of the key set that you want to remove, and click **Delete**.

## Configure CORS for VCF Automation

VCF Automation populates and updates the CORS filtering list during the VCF Automation configuration. When making browser-based requests to the `/oidc` endpoints, the default behavior is for browser pages to be loaded from the VCF Automation host. To use OIDC, you can also configure a non-VCF Automation host into the CORS filtering list.

- Familiarize yourself with the relevant Provider Management API documentation.
- Verify that you have **system administrator** credentials.

By default, when making browser-based requests to the `/oidc` endpoints, the browser pages load from the VCF Automation host. You must use the Provider Management API to configure Cross-Origin Resource Sharing (CORS) and

include a non-VCF Automation host into the CORS filtering list. Otherwise, requests to the `/oidc` endpoints on the VCF Automation host fail with an `HTTP 403` error.

1. Make a GET request with an `Authorization` header with the JSON Web Token (JWT) and an `Accept` header to the `https://<api_host>/cloudapi/1.0.0/site/settings/cors` API endpoint.

   See the CORS Provider Management OpenAPI category at https://developer.broadcom.com/xapis/.

   Alternatively, you can interact with this API through the API Explorer at `https://<vcfautomation_host>/tm/api-explorer/provider/#cors`.

   If you are using the API Explorer, within the `CORS` section, you must use the following APIs.

   ```
   GET https://<api_host>/cloudapi/1.0.0/site/settings/cors
   PUT https://<api_host>/cloudapi/1.0.0/site/settings/cors
   ```

   The system output is a list containing HTTP and HTTPS entries with IP addresses and DNS names added during the VCF Automation configuration.

2. Add to the list an additional endpoint for `<new_host>` with three entries: FQDN, HTTP, and HTTPS, and make a PUT request to the API endpoint.

   When you perform a REST PUT operation, provide all values of the origins configuration which are currently configured and which you must retain.

   ```
   {
   "values": [
      {
         "origin": "uihost.domain.local"
      },
      {
         "origin": "http://uihost.domain.local"
      },
      {
         "origin": "https://uihost.domain.local"
      }
   ]
   }
   ```

# Configuring the VCF Automation Access Control

By using the VCF Automation Provider Management UI, you can import **Provider Administrators** to VCF Automation individually, or as part of an IdP group. You can also create and modify global roles that determine what rights a user has within their organization.

## Managing VCF Automation Rights and Roles

A right is the fundamental unit of access control in VCF Automation. A role associates a role name with a set of rights. Each organization can have different rights and roles.

VCF Automation uses roles and their associated rights to determine whether a user or group is authorized to perform an operation. Many of the procedures documented in the VCF Automation guides include a prerequisite role. These prerequisites assume that the named role is the unmodified predefined role or a role that includes an equivalent set of rights.

System**Administrators**can use rights bundles and global roles to manage the rights and roles that are available to each organization.

The `System Rights Bundle` is not published to any organization. The system also contains built-in global roles that are published to all organizations managed by the `system` organization. For information about the predefined roles, see Predefined VCF Automation Roles and Their Rights.

### Rights Terminology

**Right**

Each right provides view or manage access to a particular object type in VCF Automation. Rights belong to different categories depending on the objects to which they relate, for example, `Catalog`, `Organization`, and so on. The provider organization contains all rights available in the system. The **System Administrator** defines the rights that are available to each organization. You cannot create or modify the rights included in VCF Automation.

**Rights Bundle**

**Note:**

By default, VCF Automation shows the rights bundles in `Simple Mode` and you can only view the built-in read-only rights bundles and publish the `Orchestrator Rights Bundle` to organizations in which you want to enable VCF Operations orchestrator.

If you enable the **Advanced Rights Bundle Mode** feature flag, you can use rights bundles to manage the rights that are available to each organization. You can manage feature flags by navigating to the **Feature Flags** page from the left navigation pane.

A rights bundle is a set of rights that you as a **System Administrator** can publish to one or more organizations. You can create and publish rights bundles that correspond to tiers of service, separately monetizable functionality, or any other arbitrary rights grouping. Only **System Administrators**can view and manage the rights bundles. You can publish multiple bundles to the same organization.

**Organization Rights**

Organization rights are the full set of rights that are available to an organization. Organization rights can comprise multiple rights bundles, but the **organization administrators** and users see a flat set of rights that they can use to create and modify tenant-specific roles.

### Roles Terminology

**Role**

A role is a set of rights that is assignable to one or more users and groups. When you create or import a user or group, you must assign it a role.

**Provider Roles**

Provider roles are the set of roles that are available only to the provider organization. **System Administrators** can assign provider roles only to provider users. **System Administrators** can create custom provider roles.

**Global Roles**

Global roles are the set of roles available to an organization.

**System Administrators** can create and edit global roles and publish them to one or more organizations. Administrators can assign global roles to users in the organizations to which they are published. **Organization administrators** cannot edit global roles.

**Note:**

Organization users can use only those rights from their roles that are published to their organizations.

**Organization-Specific Roles**

**Organization administrators** can create and edit organization-specific roles, which are local to their organizations. Organization-specific roles can be assigned only to users in the organization to which they belong. Organization-specific roles can contain a subset of the organization rights only.

## Predefined VCF Automation Roles and Their Rights

Each VCF Automation predefined role contains a default set of rights required to perform operations included in common workflows. By default, all predefined global roles are published to every organization in the system.

### Predefined Provider Roles

By default, the provider roles that are local only to the provider organization are the **System Administrator** roles. **System administrators** can create additional custom provider roles.

The **System Administrator** role exists only in the provider organization. The **System Administrator** role includes all rights in the system. For a list of rights available only to the **System administrator** role, see System Administrator Rights in VCF Automation. The **System administrator** credentials are established during installation and configuration. A **System Administrator** can create other system administrator and user accounts in the provider organization.

### Predefined Global Roles

By default, the predefined global roles and the rights they contain, are published to all organizations. **System Administrators** can unpublish rights and global roles from individual organizations. **System Administrators** can edit or delete predefined global roles. **System administrators** can create and publish other global roles.

| | |
|---|---|
| **Defer to Identity Provider** | Rights associated with the predefined **Defer to Identity Provider** role are determined based on information received from the user's OAuth or SAML Identity Provider. To qualify for inclusion when a user or group is assigned the **Defer to Identity Provider** role, a role or group name supplied by the Identity Provider must be an exact, case-sensitive match for a role or group name defined in your organization.<br>• If an OAuth Identity Provider defines the user, the user is assigned the roles named in the `roles` array of the user's OAuth token.<br>• If a SAML Identity Provider defines the user, the user is assigned the roles named in the SAML attribute whose name appears in the `RoleAttributeName` element, which is in the `SamlAttributeMapping` element in the organization's `OrgFederationSettings`.<br>If a user is assigned the **Defer to Identity Provider** role but no matching role or group name is available in your organization, the user can log in to the organization but has no rights. If an Identity Provider associates a user with a system-level role such as **System Administrator**, the user can log in to the organization but has no rights. You must manually assign a role to such users. |
| **Organization Administrator** | After creating an organization, a **System Administrator** can assign the role of **Organization Administrator** to the first user of the organization. A user with the predefined **Organization Administrator** role can manage users and groups in their organization and assign them roles, including the predefined **Organization Administrator** role. Roles created or modified by an **Organization Administrator** are not visible to other organizations. |
| **Organization Auditor** | The **Organization Auditor** is a read-only role that can view anything that an **Organization Administrator** can see, however, they cannot edit or manage users and groups in their organization. |
| **Organization User** | A base-level role that allows users to log in and be added to projects. |

Except the **Defer to Identity Provider** role, each predefined role includes a set of default rights. Only a **System Administrator** can modify the rights in a predefined role. If a **System Administrator** modifies a predefined role, the modifications propagate to all instances of the role in the system.

### Rights in Predefined Global Roles

A **System Administrator** can use the Provider Management UI to view the list of rights included in a role.

1. From the primary left navigation pane, click **Access Control**.
2. Click the **Global Roles** tab.
3. Click the name of the role you want to view.

An **Organization Administrator** can log in to their organization to view the rights in a role or create roles local to the organization.

Various rights are common to multiple predefined global roles. These rights are granted by default to all new organizations, and are available for use in other roles created by the **Organization Administrator**. For a list of the rights in predefined tenant roles, see VCF Automation Rights in Predefined Global Roles.

## System Administrator Rights in VCF Automation

The **System Administrator** role exists only in the provider organization. By default, the **System Administrator** role has all VCF Automation rights.

The **System Administrator** role has all VCF Automation rights. This list consists of the rights available only to **System Administrators**. The **System Administrator** role has also the VCF Automation Rights in Predefined Global Roles.

The rights' names in the table below are the VCF Automation API rights' names. The API and UI rights' names might be different. If you want to see a list of all VCF Automation rights with API rights' names, UI rights' names, right classifications, UI right categories, and so on, see the VCF Automation 9.1.x Rights file in CSV format.

**Table 1804: Rights Available by Default Only to System Administrators**

| New in This Release | API Name of the System Administrator Right |
|---|---|
| | Access Any Namespace with Elevated Privileges |
| | Access Any Namespace: Edit |
| | Access Any Namespace: View |
| | Access Control List: Manage |
| | Access Control List: View |
| | Access Metrics Endpoint |
| | Advisory Definitions: Create and Delete |
| | Advisory Definitions: Read |
| | Allowed Origins: Manage |
| | Allowed Origins: View |
| | Alternate Admin Entity: View |
| | API Tokens: Manage |
| | API Tokens: Manage All |
| | Approvals: Manage |
| | Approvals: View |
| | Audit: Manage |
| X | Avi Load Balancer Controller: Edit |
| X | Avi Load Balancer Controller: View |
| | Billing: View |
| | Blueprint Request: Manage |
| | Blueprint Request: View |
| | Blueprint: Manage |
| | Blueprint: View |
| | Catalog Instance: Manage |
| | Catalog: Manage |
| | Catalog: Request |
| | Catalog: View |
| | Cell Configuration: Edit |
| | Cell Configuration: View |
| | Certificate Library: Manage |
| | Certificate Library: View |
| | Connectivity Profile: Manage |
| | Connectivity Profile: View |
| | Content Library Item: Manage |
| X | Content Library Item: Share content with other organizations |
| | Content Library Item: View |
| | Content Library: Manage |
| | Content Library: View |
| | Content Orchestrator Action: View |

| New in This Release | API Name of the System Administrator Right |
|---|---|
| | Content Orchestrator Workflow: Publish |
| | Content Orchestrator Workflow: View |
| | Custom entity: Create custom entity definitions |
| | Custom entity: Delete custom entity definitions |
| | Custom entity: Edit custom entity definitions |
| | Custom entity: Manage any custom entity definition |
| | Custom entity: View any custom entity definition |
| | Custom entity: View custom entity definitions |
| | Custom Resources: Manage |
| | Custom Resources: View |
| | Datastore: View |
| X | Distributed Firewall: Manage |
| X | Distributed Firewall: View |
| X (Starting from v.9.1.1) | Distributed VXLAN Connection: Manage |
| X (Starting from v.9.1.1) | Distributed VXLAN Connection: View |
| | Edge Cluster: Manage |
| | Edge Cluster: View |
| | Extensibility: Manage |
| | Extensibility: View |
| | Extension Service API Definition: Manage |
| | Extension Service API Definition: View |
| | Extension Services: View |
| | Extensions: View |
| | External Service: Manage |
| | External Service: View |
| X (Starting from v.9.1.1) | File Store: Manage |
| X (Starting from v.9.1.1) | File Store: View |
| | General: Administrator Control |
| | General: Administrator View |
| | General: View Error Details |
| | Global Role: Edit |
| | Global Role: View |
| | Group / User: Manage |
| | Group / User: View |
| | Host: View |
| X | Integrations Active Directory: Manage |
| X | Integrations Active Directory: View |
| | Integrations Orchestrator: Manage |
| | Integrations Orchestrator: View |
| | Integrations: Manage |
| | Integrations: View |
| | Inventory: View |
| | IP Blocks: Manage |
| | IP Blocks: View |

| New in This Release | API Name of the System Administrator Right |
|---|---|
| | LDAP Settings: Manage |
| | LDAP Settings: View |
| | Localization Resources: Manage |
| | Metadata File Entry: Create/Modify |
| | Metrics: Token Exchange |
| | Metrics: View |
| X | Migration Data: Manage |
| | Namespace Class: Manage |
| | Namespace Class: View |
| | Namespace Lifecycle: Manage |
| | Namespace Lifecycle: View |
| X | Namespace Self-Service Settings: Manage |
| X | Namespace Self-Service Settings: View |
| | Namespace Usage: Manage |
| | Namespace Usage: View |
| | Namespace: Manage |
| | Namespace: View |
| X | Network Security Group: Manage |
| X | Network Security Group: View |
| X | Network Service: View |
| | Notifications: Manage |
| | Notifications: View |
| | NSX Manager: Manage |
| | NSX Manager: View |
| X (Starting from v.9.1.1) | Object Store: Manage |
| X (Starting from v.9.1.1) | Object Store: View |
| X (Starting from v.9.1.1) | OIDC Relying Party: Manage |
| X (Starting from v.9.1.1) | OIDC Relying Party: View |
| | OIDC Server: Manage |
| | OIDC Server: View |
| | Orchestrator: Administrator |
| | Orchestrator: Designer |
| | Orchestrator: Viewer |
| | Organization Quotas: Manage |
| | Organization vDC Storage Policy: View Capabilities |
| | Organization: Activate or Deactivate |
| | Organization: Create or Delete |
| | Organization: Edit Federation Settings |
| | Organization: Edit LDAP Settings |
| | Organization: Edit Limits |
| | Organization: Edit Name |
| | Organization: Edit OAuth Settings |
| | Organization: Edit Password Policy |
| | Organization: Edit Properties |

| New in This Release | API Name of the System Administrator Right |
|---|---|
| | Organization: Edit Quotas Policy |
| | Organization: Edit SMTP Settings |
| | Organization: Edit Username Mask |
| | Organization: Perform Administrator Queries |
| | Organization: Traversal |
| | Organization: Use Provider LDAP as Tenant |
| | Organization: View |
| (Renamed in 9.1 from Organization: View Other Orgs) | Organization: View All Orgs |
| | Organization: View Federation Settings |
| | Organization: View LDAP Settings |
| | Organization: View OAuth Settings |
| | Policy: Manage |
| | Policy: View |
| | Project Assignment: Manage |
| | Project Assignment: View |
| | Projects: Manage |
| | Projects: View |
| | Property Group: Manage |
| | Property Group: View |
| | Provider Gateway: Manage |
| | Provider Gateway: View |
| | Provider Network: View |
| | Provider vDC Resource Pool: View |
| | Provider vDC Storage Policy: Limited View |
| | Provider vDC: Limited View |
| | Public Endpoints: Manage |
| X | Region Load Balancer Setting: View |
| | Region Storage Policy: Edit |
| | Region Storage Policy: Limited View |
| | Region Storage Policy: View |
| | Region: Limited View |
| | Region: Manage |
| | Region: Simple View |
| | Region: View |
| | Regional Networking Setting: Manage |
| | Regional Networking Setting: View |
| | Regions: View |
| X | Report: View |
| | Resource Actions: Manage |
| | Resource Actions: View |
| | Resource Class Action: Manage |
| | Resource Class Action: View |
| | Resource Pool: View |
| | Right: Manage |

| New in This Release | API Name of the System Administrator Right |
|---|---|
| | Right: View |
| | Rights Bundle: Edit |
| | Rights Bundle: View |
| | Role: Create, Edit, Delete, or Copy |
| | Secrets: Manage |
| | Secrets: View |
| X | Self-Service Namespace: Manage |
| X | Self-Service Namespace: View |
| | Service Account: Manage |
| X (Starting from v.9.1.1) | Service Account: Role Assign |
| | Service Account: Simple View |
| | Service Account: View |
| | Service Authorization: Manage |
| | Service Configuration: Manage |
| | Service Configuration: View |
| X | Service Engine Group Assignment: Manage |
| X | Service Engine Group Assignment: View |
| X | Service Engine Group Self-Service Settings: Manage |
| X | Service Engine Group Self-Service Settings: View |
| X | Service Engine Group: Manage |
| X | Service Engine Group: View |
| | Service Link: Manage |
| | Service Link: View |
| | Service Resource Type: Manage |
| | Service Resource Type: View |
| | Service Resource: Manage |
| | Service Resource: View |
| | Settings: Manage |
| | Settings: View |
| | Site: View |
| | SSL Settings: Manage |
| | SSL Settings: View |
| | SSL: Test Connection |
| | SSL: Test Connection Manage |
| | Storage Classes: View |
| X | Subnet: Manage |
| X | Subnet: View |
| | Supervisor Inventory: View |
| | System IP Spaces: Manage |
| | System IP Spaces: View |
| | System Organization: Manage |
| | System Organization: View |
| | System Settings: Manage |
| | System Settings: View |

| New in This Release | API Name of the System Administrator Right |
|---|---|
| | System: Manage Proxy Rules |
| | System: View Proxy Rules |
| | Task: Resume, Abort, or Fail |
| | Task: Update |
| | Task: View Tasks |
| | Toggles: Manage |
| | Token: Manage |
| | Token: Manage All |
| X | Transit Gateway Firewall: Manage |
| X | Transit Gateway Firewall: View |
| X | Transit Gateway Services: Manage |
| X | Transit Gateway Services: View |
| X | Transit Gateway: Manage |
| | Transit Gateway: View |
| X | Transparent Proxy: Avi Proxy Usage |
| | Transparent Proxy: NSX Proxy Usage |
| | Transparent Proxy: vCenter Proxy Usage |
| | Transparent Proxy: VCF Operations API Proxy Usage |
| | Truststore: Manage |
| | Truststore: View |
| | UI Plugins: Define, Upload, Modify, Delete, Associate or Disassociate |
| | UI Plugins: View |
| | vApp: Delete |
| | vApp: Edit VM Network |
| | vApp: Edit VM Properties |
| | vApp: Import Options |
| | vApp: Sharing |
| | vApp: Use Console |
| | vApp: View ACL |
| | vApp: VM Check Compliance |
| X | VC: Edit privileged labels<br>(In version 9.1.1 and later, the name is Edit Protected Labels) |
| | vCenter: Attach or Detach |
| | vCenter: Enable or Disable |
| | vCenter: Open in vSphere |
| | vCenter: Refresh |
| | vCenter: View |
| | VCF Infra Endpoint : Manage |
| | VCF Infra Endpoint : View |
| | VCF Service Registry Capability : Manage |
| | VCF Service Registry Capability : View |
| | Virtual Datacenter: Manage |
| | Virtual Datacenter: View |

| New in This Release | API Name of the System Administrator Right |
|---|---|
| | VKSm: Attach/Detach Clusters |
| | VKSm: Manage |
| | VKSm: View |
| | VM Group OS association: Delete |
| | VM Group OS association: Manage |
| | VM Group OS association: View |
| X | VPC Gateway Firewall: Manage |
| X | VPC Gateway Firewall: View |
| X | VPC Network Security Group: Manage |
| X | VPC Network Security Group: View |
| X | VPC Security Profile: Manage |
| X | VPC Security Profile: View |
| X | VPC Self-Service Settings: Manage |
| X | VPC Self-Service Settings: View |
| X | VPC Service Profile: Manage |
| X | VPC Service Profile: View |
| X | VPC Services: Manage |
| X | VPC Services: View |
| | VPC: Manage |
| | VPC: View |
| | vSphere Server: Manage |
| | vSphere Server: Manage Proxy Configuration |
| | vSphere Server: View |

## VCF Automation Rights in Predefined Global Roles

Various VCF Automation rights are common to multiple predefined global roles. These rights are granted by default to all new organizations and are available for use in other roles created by the **organization administrator**.

### Rights Included in the Global Roles in VCF Automation

This list consists of the rights available to user roles in the VCF Automation organizations. For the rights available to **System Administrators**, see System Administrator Rights in VCF Automation.

The rights' names in the table below are the VCF Automation API rights' names. The API and UI rights' names might be different. If you want to see a list of all VCF Automation rights with API rights' names, UI rights' names, right classifications, UI right categories, and so on, see the VCF Automation 9.1.x Rights file in CSV format.

| New in This Release | API Name of the Right | Organization Administrator | Organization Auditor | Organization User |
|---|---|---|---|---|
| | Access Any Namespace: Edit | X | | |
| | Access Any Namespace: View | X | X | |
| | API Tokens: Manage | X | | X |
| | API Tokens: Manage All | X | | |
| | Approvals: Manage | X | | |
| | Approvals: View | X | X | |

| New in This Release | API Name of the Right | Organization Administrator | Organization Auditor | Organization User |
|---|---|---|---|---|
| | Audit: Manage | X | X | |
| | Billing: View | X | X | |
| | Blueprint Request: Manage | X | | |
| | Blueprint Request: View | X | X | |
| | Blueprint: Manage | X | | |
| | Blueprint: View | X | X | |
| | Catalog Instance: Manage | X | | |
| | Catalog: Manage | X | | |
| | Catalog: Request | X | | |
| | Catalog: View | X | X | |
| | Certificate Library: Manage | X | | |
| | Certificate Library: View | X | X | |
| | Connectivity Profile: Manage | X | | |
| | Connectivity Profile: View | X | X | |
| | Content Library Item: Manage | X | | |
| X | Content Library Item: Share content with other organizations | X | | |
| | Content Library Item: View | X | X | |
| | Content Library: Manage | X | | |
| | Content Library: View | X | X | |
| | Content Orchestrator Action: View | X | X | |
| | Content Orchestrator Workflow: Publish | X | | |
| | Content Orchestrator Workflow: View | X | X | |
| | Custom Resources: Manage | X | | |
| | Custom Resources: View | X | X | |
| X | Distributed Firewall: Manage | X | | |
| X | Distributed Firewall: View | X | X | |
| | Extensibility: Manage | X | | |
| | Extensibility: View | X | X | |
| X (Starting from v.9.1.1) | File Store: Manage | X | | |
| X (Starting from v.9.1.1) | File Store: View | X | X | |
| | General: Administrator Control | X | | |
| | General: Administrator View | X | X | |
| | Group / User: Manage | X | | |

| New in This Release | API Name of the Right | Organization Administrator | Organization Auditor | Organization User |
|---|---|---|---|---|
| | Group / User: View | X | X | |
| X | Integrations Active Directory: Manage | X | | |
| X | Integrations Active Directory: View | X | X | |
| | Integrations Orchestrator: View | X | X | |
| | Integrations: Manage | X | | |
| | Integrations: View | X | X | |
| | Inventory: View | X | X | |
| | IP Blocks: Manage | X | | |
| | IP Blocks: View | X | X | |
| | LDAP Settings: Manage | X | | |
| | LDAP Settings: View | X | X | |
| | Metadata File Entry: Create/Modify | X | | |
| | Metrics: View | X | X | X |
| | Namespace Class: Manage | X | | |
| | Namespace Class: View | X | X | |
| | Namespace Lifecycle: View | X | X | |
| X | Namespace Self-Service Settings: Manage | X | | |
| X | Namespace Self-Service Settings: View | X | X | |
| | Namespace Usage: Manage | X | | X |
| | Namespace Usage: View | X | X | X |
| | Namespace: Manage | X | | |
| | Namespace: View | X | X | |
| X | Network Security Group: Manage | X | | |
| X | Network Security Group: View | X | X | |
| X | Network Service: View | X | X | |
| | Notifications: Manage | X | | |
| | Notifications: View | X | X | |
| X (Starting from v.9.1.1) | Object Store: Manage | X | | |
| X (Starting from v.9.1.1) | Object Store: View | X | X | |
| X (Starting from v.9.1.1) | OIDC Relying Party: Manage | X | | |
| X (Starting from v.9.1.1) | OIDC Relying Party: View | X | X | |
| | Orchestrator: Administrator | X | | |
| | Organization: Edit Federation Settings | X | | |

| New in This Release | API Name of the Right | Organization Administrator | Organization Auditor | Organization User |
|---|---|---|---|---|
| | Organization: Edit LDAP Settings | X | | |
| | Organization: Edit Name | X | | |
| | Organization: Edit OAuth Settings | X | | |
| | Organization: Edit Password Policy | X | | |
| | Organization: Edit Properties | X | | |
| | Organization: Edit Quotas Policy | X | | |
| | Organization: Edit SMTP Settings | X | | |
| | Organization: View | X | X | |
| (Renamed in 9.1 from Organization: View Other Orgs) | Organization: View All Orgs | X | X | |
| | Organization: View Federation Settings | X | X | |
| | Organization: View LDAP Settings | X | X | |
| | Organization: View OAuth Settings | X | X | |
| | Policy: Manage | X | | |
| | Policy: View | X | X | |
| | Projects: Manage | X | | |
| | Projects: View | X | X | |
| | Property Group: Manage | X | | |
| | Property Group: View | X | X | |
| X | Region Load Balancer Setting: View | X | X | |
| | Region: Simple View | X | X | |
| | Regions: View | X | X | |
| X | Report: View | X | X | |
| | Resource Actions: Manage | X | | |
| | Resource Actions: View | X | X | |
| | Right: View | X | X | |
| | Role: Create, Edit, Delete, or Copy | X | | |
| | Secrets: Manage | X | | |
| | Secrets: View | X | X | |
| X | Self-Service Namespace: Manage | X | | |
| X | Self-Service Namespace: View | X | X | |
| | Service Account: Manage | X | | |

| New in This Release | API Name of the Right | Organization Administrator | Organization Auditor | Organization User |
|---|---|---|---|---|
| X (Starting from v.9.1.1) | Service Account: Role Assign | X | | |
| | Service Account: Simple View | X | X | |
| | Service Account: View | X | X | |
| X | Service Engine Group Self-Service Settings: Manage | X | | |
| X | Service Engine Group Self-Service Settings: View | X | X | |
| X | Service Engine Group: Manage | X | | |
| X | Service Engine Group: View | X | X | |
| | Settings: Manage | X | | |
| | Settings: View | X | X | |
| | SSL: Test Connection | X | X | |
| | Storage Classes: View | X | X | |
| X | Subnet: Manage | X | | |
| X | Subnet: View | X | X | |
| | Toggles: Manage | X | | |
| X | Transit Gateway Firewall: Manage | X | | |
| X | Transit Gateway Firewall: View | X | X | |
| X | Transit Gateway Services: Manage | X | | |
| X | Transit Gateway Services: View | X | X | |
| X | Transit Gateway: Manage | X | | |
| | Transit Gateway: View | X | X | |
| | Truststore: Manage | X | | |
| | Truststore: View | X | X | |
| | UI Plugins: View | X | X | |
| | vApp: Use Console | X | | X |
| X | VC: Edit privileged labels (In version 9.1.1 and later, the name is Edit Protected Labels) | X | | |
| | VKSm: Manage | X | | |
| | VKSm: View | X | X | |
| X | VPC Gateway Firewall: Manage | X | | |
| X | VPC Gateway Firewall: View | X | X | |
| X | VPC Network Security Group: Manage | X | | |

| New in This Release | API Name of the Right | Organization Administrator | Organization Auditor | Organization User |
|---|---|---|---|---|
| X | VPC Network Security Group: View | X | X | |
| X | VPC Security Profile: Manage | X | | |
| X | VPC Security Profile: View | X | X | |
| X | VPC Self-Service Settings: Manage | X | | |
| X | VPC Self-Service Settings: View | X | X | |
| X | VPC Service Profile: Manage | X | | |
| X | VPC Service Profile: View | X | X | |
| X | VPC Services: Manage | X | | |
| X | VPC Services: View | X | X | |
| | VPC: Manage | X | | |
| | VPC: View | X | X | |

## Managing Rights Bundles in VCF Automation

By default, VCF Automation shows the rights bundles in `Simple Mode` and you can only view the built-in read-only rights bundles and publish the `Orchestrator Rights Bundle` to organizations in which you want to enable VCF Operations orchestrator.

As a **system administrator**, you can activate the feature flag for **Advanced Rights Bundle Mode** and constrain certain organizations from performing certain actions. You can manage feature flags by navigating to the **Feature Flags** page from the left navigation pane.

In `Advanced Mode`, you can create custom rights bundles and publish them to one and more VCF Automation organizations in your cloud. You can edit and delete rights bundles, and unpublish rights bundles from organizations in your cloud.

### Create a Rights Bundle in VCF Automation

You can group a set of rights as a rights bundle which you can publish to one or more VCF Automation organizations in your system.

In the left navigation pane, click **Feature Flags** and enable **Advanced Rights Bundle Mode**.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Rights Bundles** tab, and click **Add**.
4. Enter a name and, optionally, a description for the new rights bundle.
5. Select the rights that you want to associate with this bundle.

   The rights are grouped in categories and subcategories for view or manage access to the object to which they relate.

You can select the rights individually, by view or manage by subcategory, or by view or manage globally.

| Category | Description |
|---|---|
| Access Control | Contains rights for viewing and managing organizations, rights, roles, and users. |
| Administration | Contains rights for viewing and managing general, billing, integration settings, and so on. |
| Compute | Contains rights for viewing and managing organizations, regions, region quotas, vSphere Namespaces, and VM monitoring. |
| Content Hubs | Contains rights for viewing and managing blueprints and workflows. |
| Extensions | Contains rights for viewing and managing VCF Automation extensions. |
| Infrastructure | Contains rights for viewing and managing SDDCs. |
| Libraries | Contains rights for viewing and managing content libraries and content library items. |
| Networking | Contains rights for viewing and managing network resources. |
| Other | Contains rights for viewing and managing services, branding, Kubernetes clusters, Service Manager, and so on. |
| Policies | Contains rights for viewing and managing approval requests. |

6. Click **Save**.

You can publish the newly created rights bundle to one or more organizations in your system. See Publish or Unpublish a Rights Bundle to VCF Automation.

### Clone a Rights Bundle Using VCF Automation

You can use an existing rights bundle as a template for the creation of a new bundle.

- Verify that you have the rights to add new roles to VCF Automation.
- In the left navigation pane, click **Feature Flags** and enable **Advanced Rights Bundle Mode**.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Rights Bundles** tab.
4. Click the vertical ellipsis ( ⋮ ) next to the rights bundle that you want to clone, and click **Clone**.
5. In the **Clone Rights Bundle** window, enter a name and optionally, a description for the cloned bundle.
6. Optional: (Optional) To edit the cloned rights, turn on the **Modify Selected Rights** toggle, and select or deselect the rights you want to change for the cloned role.
7. Click **Save**.

### Publish or Unpublish a Rights Bundle to VCF Automation

You can publish a rights bundle to one or more VCF Automation organizations in your system. After you publish a rights bundle to an organization, the rights in this bundle become part of the organization set of rights.

Organization rights can comprise multiple rights bundles, but the organization administrators and users see a flat set of rights that they can use to create and modify roles.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Rights Bundles** tab.
4. Select the rights bundle, and click **Publish**.
5. To publish the bundle:
   a) Turn on the **Publish to Tenants** toggle.
   b) Select the organizations to which you want to publish the role.
      - If you want to publish the bundle to all existing and newly created organizations in your system, select **Publish to All Tenants**.
      - If you want to publish the bundle to particular organizations in your system, select the organizations individually.
6. To unpublish the bundle:
   - To unpublish the bundle from all organizations in your system, deselect **Publish to Tenants**.
   - To unpublish the bundle from particular organizations in your system, deselect **Publish to All Tenants**, and deselect the organizations individually.
7. Click **Save**.

The rights in the published bundle are available in the selected organizations and can be used in the roles in these organizations.

The rights in the unpublished bundle are removed from the selected organizations and cannot be used in the roles in these organizations.

### View and Edit a Rights Bundle Using VCF Automation

You can view the rights that are included in a rights bundle. You can modify the name, the description, and the rights of a bundle.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Rights Bundles** tab.
4. Click the name of the target bundle.
   You can view the rights that are associated with the bundle by expanding the right categories.
5. Edit the bundle and click **Save**.

If you modified the rights of the bundle, the new set of rights is applied to all organizations to which this rights bundle is published.

### Delete a Rights Bundle From VCF Automation

You can remove a rights bundle that you no longer use in your VCF Automation organizations.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Rights Bundles** tab.
4. Select the rights bundle, and click **Delete**.
5. To confirm, click **Delete**.

## Managing Global VCF Automation Roles

As a **system administrator**, you can create global roles and publish them to one or more VCF Automation organizations that you manage. You can edit and delete existing global roles. You can unpublish global roles from individual organizations that you manage.

After the initial VCF Automation installation and setup, the system contains a set of predefined global roles that are published to all organizations. See Predefined VCF Automation Roles and Their Rights.

### Create a Global Role in Your VCF Automation

You can create a global role that you can publish to one or more VCF Automation organizations in your system.

After the initial VCF Automation installation and setup, the system contains predefined global roles that are published to all organizations. For information about the predefined roles, see Predefined VCF Automation Roles and Their Rights.

You can add custom global roles to your system.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Global Roles** tab.
4. Click **New**.
5. Enter a name and, optionally, a description for the new role.
6. Select the rights that you want to associate with the role.

   The rights are grouped in categories and subcategories for view or manage access to the object to which they relate.

   You can select the rights individually, by view or manage by subcategory, or by view or manage globally.

| Category | Description |
|---|---|
| Access Control | Contains rights for viewing and managing organizations, rights, roles, and users. |
| Administration | Contains rights for viewing and managing general, billing, integration settings, and so on. |
| Compute | Contains rights for viewing and managing organizations, regions, region quotas, vSphere Namespaces, and VM monitoring. |
| Content Hubs | Contains rights for viewing and managing blueprints and workflows. |
| Extensions | Contains rights for viewing and managing VCF Automation extensions. |
| Infrastructure | Contains rights for viewing and managing SDDCs. |
| Libraries | Contains rights for viewing and managing content libraries and content library items. |
| Networking | Contains rights for viewing and managing network resources. |
| Other | Contains rights for viewing and managing services, branding, Kubernetes clusters, Service Manager, and so on. |

| Category | Description |
| --- | --- |
| Policies | Contains rights for viewing and managing approval requests. |

7. Click **Save**.

Upon its creation, the new global role is available only to the VCF Automation `System` organization.

You can publish the newly created role to one or more organizations in your system. See Publish or Unpublish a Global Role to Your VCF Automation.

### Clone a Global Role to Your VCF Automation

You can use an existing global role as a template for the creation of a new role.

Verify that you have the rights to add new roles to VCF Automation.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Global Roles** tab.
4. Click the vertical ellipsis ( ⋮ ) next to the role that you want to clone, and click **Clone**.
5. In the **Clone Global Role** window, enter a name and description for the cloned role.
6. Optional: (Optional) To edit the cloned rights, turn on the **Modify Selected Rights** toggle, and select or deselect the rights you want to change for the cloned role.
7. Click **Save**.

### Publish or Unpublish a Global Role to Your VCF Automation

You can publish a global role to one or more VCF Automation organizations in your system. After you publish a role to an organization, this role becomes a part of the set of roles of the organization.

To unpublish a global role from an organization, verify that no user is assigned with this role in the organization.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Global Roles** tab.
4. If you want to publish a role, click the name of the role, and click **Publish**.
   a) Turn on the **Publish to Tenants** toggle.
   b) Select the organizations to which you want to publish the role.
      - If you want to publish the role to all existing and newly created organizations in your system, select **Publish to All Tenants**.
      - If you want to publish the role to one or more organizations in your system, select the organizations individually.
5. If you want to unpublish a role, click the name of the role, and click **Publish**.
   - To unpublish the role from all organizations in your system, turn off the **Publish to Tenants** toggle.
   - To unpublish the role from specific organizations in your system, turn off the **Publish to All Tenants** toggle, and deselect the organizations individually.
6. Click **Save**.

The published role is available in the selected organizations and can be assigned to users in these organizations. **Organization administrators** cannot edit global roles that are published to their organizations.

The unpublished role is removed from the selected organizations and cannot be assigned to users in these organizations.

### View and Edit a Global Role Using Your VCF Automation

You can view the rights that are included in a global role. You can modify the name, the description, and the rights of a global role.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Global Roles** tab.
4. Click the name of the target role.

   You can view the rights that are associated with the role by expanding the right categories.
5. To modify the name, the description, or the rights of the role, click **Edit**.
6. Edit the role and click **Save**.

If you modified the rights of the role, VCF Automation applies the new set of rights to the users across all organizations that you manage that are assigned with this role.

### Delete a Global Role From Your VCF Automation

You can remove a global role that you no longer use in your VCF Automation organizations.

The global role that you want to delete must not be assigned to any user across all organizations.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Global Roles** tab.
4. Click the name of the global role and click **Delete**.
5. To confirm, click **Delete**.

## Managing VCF Automation Provider Roles

You can create and manage roles in your VCF Automation `System` organization.

### Create a Role in Your VCF Automation Provider Management UI

You can create a role available in your VCF Automation `System` organization.

After the initial VCF Automation installation and setup, the system contains predefined roles that are local to the `System` organization and roles that are global to all organizations. For information about the predefined roles, see Predefined VCF Automation Roles and Their Rights.

You can add custom provider roles to your `System` organization.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Roles** tab, and click **New**.
4. Enter a name, and optionally, a description for the new role.
5. Select the rights that you want to associate with the role.

   The rights are grouped in categories and subcategories for view or manage access to the object to which they relate.

You can select the rights individually, by view or manage by subcategory, or by view or manage globally.

| Category | Description |
|---|---|
| Access Control | Contains rights for viewing and managing organizations, rights, roles, and users. |
| Administration | Contains rights for viewing and managing general, billing, integration settings, and so on. |
| Compute | Contains rights for viewing and managing organizations, regions, region quotas, vSphere Namespaces, and VM monitoring. |
| Content Hubs | Contains rights for viewing and managing blueprints and workflows. |
| Extensions | Contains rights for viewing and managing VCF Automation extensions. |
| Infrastructure | Contains rights for viewing and managing SDDCs. |
| Libraries | Contains rights for viewing and managing content libraries and content library items. |
| Networking | Contains rights for viewing and managing network resources. |
| Other | Contains rights for viewing and managing services, branding, Kubernetes clusters, Service Manager, and so on. |
| Policies | Contains rights for viewing and managing approval requests. |

6. Click **Save**.

The newly created role is available for assigning to users in your `System` organization.

### Clone a Role in Your VCF Automation Provider Management UI

You can use an existing provider role as a template for the creation of a new role.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Roles** tab.
4. Click the vertical ellipsis ( ⋮ ) next to the role that you want to clone, and click **Clone**.
5. In the **Clone Role** window, enter a name and description for the cloned role.
6. Optional: (Optional) To edit the cloned rights, turn on the **Modify Selected Rights** toggle, and select or deselect the rights you want to change for the cloned role.
7. Click **Save**.

### View or Edit a Role in Your VCF Automation Provider Management UI

You can view the rights that are included in a role that is local to your VCF Automation`System` organization. You can modify the name, the description, and the rights of a role.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Roles** tab.
4. Click the name of the target role.

   You can view the rights that are associated with the role by expanding the right categories.

5. To modify the name, the description, or the rights of the role, click **Edit**.
6. Edit the role and click **Save**.

If you modified the rights of the role, the new set of rights is applied to the users that are assigned with this role.

### Delete a Role From Your VCF Automation Provider Management UI

You can remove a role that you no longer use in your VCF Automation `System` organization.

The role that you want to delete must not be assigned to any user.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, select **Access Control**.
3. Click the **Roles** tab.
4. Click the name of the global role and click **Delete**.
5. To confirm, click **Delete**.

## Managing VCF Automation Provider Users and Groups

In VCF Automation, each organization can have only one local user, including the `System` organization. You can import users and groups to your VCF Automation `System` organization.

> **Note:**
> VCF Automation users can share or transfer ownership of entities to other users within the same organization. For this reason, within an organization, any user can see the other users' basic information, such as user name, full name, description, ID, role, and organization.

### Managing Provider Users in Your VCF Automation

You can manage the users in your `System` organization by using the Provider Management UI.

#### Import Provider Users to Your VCF Automation

You can import users to your VCF Automation `System` organization from a previously configured LDAP, SAML, OIDC, or VCF Single Sign-On (SSO) identity provider.

You must configure at least one identity provider before you can import users.

- VCF SSO — Configure a New VCF Single Sign-On for a VCF Instance.
- LDAP — Configure a System LDAP Connection in Your VCF Automation
- SAML — Configure Your VCF Automation System to Use a SAML Identity Provider
- OIDC — Configure Your System to Use an OpenID Connect Identity Provider Using Your VCF Automation Provider Management UI

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Users** tab, and click **Import Users**.
4. From the **Source** drop-down menu, select your identity provider type.
5. Specify the users.

> **Note:**
>
> To log in to VCF Automation, you must assign VCF SSO users the `Defer to Identity Provider` role. You can assign this role from the **Assign Role** drop-down menu during import, or at any time after import by clicking the vertical ellipsis (⋮) next to the user's name on the **Access Control** > **Users** tab and clicking **Edit**.

| Option | Description |
| --- | --- |
| **LDAP** | 1. Enter a full or partial name of a user and click **Search**.<br>2. From the search results, select the users that you want to import.<br>3. From the **Assign Role** drop-down menu, select a role for the imported users. |
| **SAML, OIDC** | 1. Enter the user names of the users that you want to import in the name identifier format supported by the SAML identity provider.<br>Use a new line for each user name.<br>2. From the **Assign Role** drop-down menu, select a role for the imported users. |
| **VCF SSO** | 1. Enter the user names of the users that you want to import in the name identifier format supported by the SAML identity provider.<br>Use a new line for each user name.<br>2. From the **Assign Role** drop-down menu, select a role for the imported users.<br>Starting with version 9.1.1, you can also enter a full or partial name of a user and click **Search**. From the search results, select the users that you want to import. From the **Assign Role** drop-down menu, select a role for the imported users. |

6. Assign one or more roles to the selected users.
7. If you want the roles of the users to become a combination of all group and assigned roles, turn on the **Inherit roles from groups** toggle.

   Stranded users, which do not belong to any groups in VCF Automation and do not have direct role assignment cannot perform any activities. These users experience login failures even with valid credentials.
8. Click **Save**.

You can see the imported users in the list of users.

### Edit a Provider User in Your VCF Automation

You can change the user name, password, and role of a user in your VCF Automation `System` organization.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Users** tab.
4. Click the vertical ellipsis ( ⋮ ) next to the name of the target user, and click **Edit**.
5. Edit the user details and click **Save**.

### Activate or Deactivate a Provider User in Your VCF Automation

After you deactivate a user, the user cannot log in to VCF Automation.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Users** tab.
4. Click the name of the user, and click **Enable** or **Disable**.
5. If you are deactivating a user, click **OK** to confirm.

### Delete a Provider User From Your VCF Automation

You can remove a user from your VCF Automation `System` organization by deleting the user account.

Deactivate the user that you want to delete. See Activate or Deactivate a Provider User in Your VCF Automation.

To delete a stranded user that lost access to the system because their LDAP group was deleted, use the VCF Automation API.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Users** tab.
4. Click the name of the user, and click **Delete**.
5. To confirm, click **Delete**.

### Unlock a Provider User in Your VCF Automation

If you enabled account lockout in your password policy system settings, users might lock their accounts after a certain number of invalid login attempts. Even if the lockout is set with an account lockout interval, you can unlock a user account without waiting for the lock to expire.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Users** tab.
4. Click the name of the user, and click **Unlock**.

## Managing VCF Automation Provider Groups

You can import, edit, and delete groups from your VCF Automation `System` organization by using the Provider Management UI.

### Import a Provider Group to Your VCF Automation

You can import groups to your VCF Automation `System` organization from a previously configured LDAP, SAML, OIDC, or VCF Single Sign-On (SSO) identity provider.

You must configure at least one identity provider before you can import a provider group.

- VCF SSO — Configure a New VCF Single Sign-On for a VCF Instance.
- LDAP — Configure a System LDAP Connection in Your VCF Automation
- SAML — Configure Your VCF Automation System to Use a SAML Identity Provider
- OIDC — Configure Your System to Use an OpenID Connect Identity Provider Using Your VCF Automation Provider Management UI

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Groups** tab, and click **Import Groups**.
4. From the **Source** drop-down menu, select your identity provider type.

   The identity provider types can be **LDAP**, **SAML**, or **OIDC**.

   If you configured only one identity provider, this option is hard-coded.
5. Specify the groups.

   **Note:**

   To log in to VCF Automation, you must assign VCF SSO users the `Defer to Identity Provider` role. You can assign this role from the **Assign Role** drop-down menu during import, or at any time after import by clicking the vertical ellipsis (⋮) next to the user's name on the **Access Control** > **Users** tab and clicking **Edit**.

| Option | Description |
|---|---|
| **LDAP** | 1. Enter a full or partial name of a group, and click **Search**.<br>2. From the search results, select the groups that you want to import. |

| Option | Description |
|---|---|
| | 3. From the **Assign Role** drop-down menu, select a role for the users in the imported groups. |
| **SAML, OIDC** | 1. Enter the names of the groups that you want to import in the name identifier format supported by the identity provider.<br>Use a new line for each group name.<br>2. From the **Assign Role** drop-down menu, select a role for the users in the imported groups. |
| **VCF SSO** | 1. Enter the names of the groups that you want to import in the name identifier format supported by the SAML identity provider.<br>Use a new line for each group name.<br>2. From the **Assign Role** drop-down menu, select a role for the users in the imported groups.<br>Starting with version 9.1.1, you can also enter a full or partial name of a group and click **Search**. From the search results, select the groups that you want to import. From the **Assign Role** drop-down menu, select a role for the users in the imported groups. |

6. Click **Save**.

### Edit a Provider Group in Your VCF Automation

You can edit the description and change the role of the members of a group that you previously imported to your VCF Automation Provider organization.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Groups** tab.
4. Click the name of the group, and click **Edit**.
5. Edit the group details, and click **Save**.

### Delete a Provider Group From Your VCF Automation

You can remove a group from your VCF Automation`System` organization

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Groups** tab.
4. Click the name of the target group and click **Delete**.
5. To confirm, click **Delete**.

## Managing Service Accounts in VCF Automation

You can automate the access of third-party applications to VCF Automation by using service accounts. Service accounts provide secure, API-based, non-interactive access, which is essential for integrating external systems and running automated tasks.

#### Service Account Visibility

If you want to limit the service account information that users can see, you can grant to certain roles only the **Service Account: Simple View** right. When a user with only the **Service Account: Simple View** right makes a GET request on a service account, the `softwareId`, `softwareVersion`, `uri`, and `status` properties return as `null`. Users with the full `Service Account: View` right can see all properties.

### Authorization Process

To provide automated access to VCF Automation, service accounts use the OAuth 2.0 Device Authorization Grant (RFC 8628). After the authorization flow completes, the client application receives an OAuth refresh token and an initial access token (JWT) for its first VCF Automation session. Applications use the refresh token to request new access tokens for subsequent API calls.

The authorization process steps:

1. The client application sends a device authorization request to the VCF Automation device endpoint and receives a `device_code` and a human-readable `user_code`.
2. The client application provides the `user_code` to a system administrator through a separate communication channel, such as a secure message or an IT ticket.
3. A user with the `Service Account: Manage` right logs in to VCF Automation, navigates to the service account, and approves or denies the request by confirming the `user_code`. VCF Automation delivers the resulting refresh token exclusively to the client application that initiated the request. The token is not accessible through any management API, and no other actor, including administrators, can retrieve it.
4. The client application polls the token endpoint with the `device_code` until the request reaches the `GRANTED` state.
5. When the grant is confirmed, the token endpoint issues the OAuth refresh token and an initial access token (JWT) for immediate use.
6. For subsequent API sessions, the client application exchanges its refresh token for a new access token (JWT) at the token endpoint.

### Managed Service Accounts

You can designate a service account as managed by a parent service account by using the `managedBy` property. Managed service accounts support the JWT Bearer grant (`urn:ietf:params:oauth:grant-type:jwt-bearer`), which allows the parent service account to obtain tokens for a child account programmatically without an interactive device flow. Standard service accounts support only the device authorization grant type.

### Token Rotation

The `tokenRotation` property controls token rotation for service account refresh tokens. Unused service account refresh tokens never expire unless you explicitly revoke them.

| **`tokenRotation` Value** | **Behavior** |
|---|---|
| `ALWAYS` | The refresh token rotates automatically on every use. |
| `NEVER` | The refresh token never rotates. |
| `ONCE_ON_NEXT_LOGIN` | The token rotates on the next authentication, and then the setting automatically resets to `NEVER`. |
| `NONE` | The rotation is not configured. |

### Roles

Service accounts support multiple roles through the roles array property. When registering with the VCF Automation OAuth endpoint, specify roles in the scope property as space-separated URL-encoded URNs in the format `urn:vcloud:role:`*`[roleName]`*.

**Service Account Statuses**

The status property reflects the state of the OAuth refresh token. This property is visible only to users with the `Service Account: View` right.

| Status | Description |
|---|---|
| `CREATED` | The account is in its initial state after creation. No refresh token exists and no device authorization requests exist. |
| `REQUESTED` | One or more device authorization requests are outstanding. No refresh token exists. |
| `GRANTED` | An administrator grants an outstanding request. The client application must poll the token endpoint to retrieve the refresh token. |
| `ACTIVE` | The client application retrieves its refresh token and can authenticate with VCF Automation. |

You can use the `Grant` and `Deny` actions only when the account is in the `REQUESTED` state. You can revoke access when the account is in the `GRANTED` or `ACTIVE` state.

**Restrictions**

Because service accounts provide third-party application access, they have restrictions when authenticated:

- Cannot create API tokens. The API token management right is not available to service account sessions.
- Cannot manage arbitrary service accounts. A service account can manage only itself and the service accounts it directly owns through the `managedBy` property.

## Create a Service Account Using Your VCF Automation Provider Management UI

You can create an account for automated access to VCF Automation by using the Provider Management UI.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Access Control**.
3. Click the **Service Accounts** tab, and click **New**.
4. Enter a name for the service account.
5. From the **Assign Role** drop-down menu, select a role for the service account.

   The list of available roles comprises the local system organization roles or if you are in another organization, the global roles published to the organization in addition to any local roles in the organization.
6. Enter a software ID for the service account or generate and enter one using the **Generate Software ID** button.

   Service accounts must have software IDs which are unique identifiers, in UUID format, representing the software that is connecting to VCF Automation. This ID would be the same for all versions and instances of a piece of software.

   For larger solutions, to retain control over the identity of your service accounts, do not use the **Generate Software ID** option, and generate your own software ID.
7. Optional: (Optional) Enter the software version of the system using the service account.

   The software version is an optional vendor-specified informational piece of metadata that is associated with the service account. To track when a piece of software changes, VCF Automation uses the software version. The software version might be useful for identifying a service account.
8. Optional: (Optional) Enter a client URI.

   The client Uniform Resource Identifier (URI) is a URL to the webpage of the vendor and provides information about the client.

9. Click **Next**.
10. Review the service account information, and click **Finish**.

The service account appears on the **Service Accounts** page with status `Created`.

You can create a service account also by using the VCF Automation API. The API request uses the same API endpoint as creating a user API token, but the presence of the `software_id` field indicates the intent to create a service account.

Sample request:

```
POST /oauth/provider/register

Accept:application/json

Content-Type:application/json

Authorization:Bearer eyJhbGciOiJSUzI...7g7rA

Body: {

    "client_name": "exampleServiceAccount",

    "software_id": "bc2528fd-35c4-44e5-a55d-62e5c4bd9c99",

    "scope": "urn:vcloud:role:System%20Administrator",

    "client_uri": "https://www.companyname.com",

    "software_version": "1.0"

}
```

Sample response:

```
{

"client_name": "exampleServiceAccount",

"client_id": "734e3845-1573-4f07-9b6c-b493c9042187",

"grant_types": [

"urn:ietf:params:oauth:grant-type:device_code"

],

"token_endpoint_auth_method": "none",

"client_uri": "https://www.company_name.com",

"software_id": "bc2528fd-35c4-44e5-a55d-62e5c4bd9c99",

"software_version": "1.0",

"scope": "urn:vcloud:role:System%20Administrator"

}
```

Copy the client ID that appears in the service account details. To grant access to the service account, you must use the client ID.

### Grant Access to a Service Account Using Your VCF Automation Provider Management UI

After you create a service account and the application requests authorization to receive an access token, you can grant the token by using the Provider Management UI.

1. Copy the client ID from the service account details in the Provider Management UI..
2. Verify that the application requesting the account makes an OAuth 2.0 Device Authorization Grant RFC-compliant request to the `https://site.cloud.example.com/oauth/provider/device_authorization` API endpoint. For more information on device authorization requests, see RFC 8628 section 3.1.

| Key | Value |
|---|---|
| `client_ID` | *`Generated_Client_ID`* |

Once the application requests access, the service account status in the Provider Management UI changes to `Requested`. The application receives the device code, user code, and some additional information.

Run the following `curl` command to initiate the authorization request. Replace the `client_id` with your service account client ID.

Sample request:

```
curl --location 'https://<provider-host>/oauth/provider/device_authorization' \
--header 'Accept: application/json;version=40.0' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--data-urlencode 'client_id=<client-id>'
```

The system returns a `user_code` and a `device_code`. You use the `user_code` in the portal to grant access.

Sample response:

```
{
   "device_code": "<device-code>",
   "user_code": "<user-code>",
   "verification_uri": "https://<provider-host>/provider/access-control/service-accounts",
   "expires_in": 3600,
   "interval": 60
}
```

The device must poll at the frequency that is specified in the above response (in seconds) `/oauth/provider/token` as per the RFC. The device must use the device code until it receives the tokens from VCF Automation, or the request times out.

Sample request:

```
curl --location 'https://<provider-host>/oauth/provider/token' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--data-urlencode 'grant_type=urn:ietf:params:oauth:grant-type:device_code' \
--data-urlencode 'client_id=<client-id>' \
--data-urlencode 'device_code=<device-code>'
```

The response includes the `access_token`. The application must use this token for all subsequent API requests.

Sample response before granting:

```
{
    "error": "authorization_pending",
    "error_description": "Device authorization request pending",
    "error_uri": null,
    "minorErrorCode": "authorization_pending",
    "message": "Device authorization request pending",
    "stackTrace": null
}
```

Sample response after granting:

```
{
   "access_token": "<access-token>",
```

```
    "token_type": "Bearer",
    "expires_in": 3600,
    "refresh_token": "<refresh-token>"
}
```

If you do not confirm or deny an access request, the user code times out. The timeout period appears in the response of the device authorization request.

VCF Automation grants a primary API token to the application only if the application and the administrator use the device code and user code corresponding to each other.

3. Get the user code from the application. You must enter the code in step 4.

**Note:**

If the timeout period expires during this procedure, the service account status in the Provider Management UI changes back to `Created`, and you must start the procedure again.

1. Log in to the VCF Automation Provider Management UI.
2. From the left navigation panel, select **Access Control**.
3. Select the **Service Accounts** tab.
4. Click **Review Access Requests**.
5. Enter the user code for the application that you obtained in prerequisite 3, click **Lookup**, and verify the requested access details.
6. Grant access to the application.

   If you deny access to the application, the service account status in the Provider Management UI changes back to `Created`.

The service request status changes to `Granted`. VCF Automation grants the application linked to the service account its primary API token in the form of an API token. Included in the response, as required by the RFC, is an OAuth access token representing a user session for immediate use by the service account. If the application does not use the OAuth access token immediately, the session times out as per the configured idle session timeout. The service account might also explicitly log out, which is recommended not only for security reasons, but also provides a good test run for the service account to make an API call to VCF Automation. Once the application fetches the API token, the status changes to `Active`.

- To change the assigned service account role, software ID, software version, client URI, or quota restrictions, select a service account and click **Edit a Service Account**. The changes take effect at the next token refresh.
- To revoke service account access so that the granted API token granted becomes invalid, click **Revoke**. VCF Automation terminates all active sessions. Revoking an API token does not delete the service account, however, the status of the account changes to `Created`. If the application has already requested access again, the status of the service account changes to `Requested`. You must once again follow the procedure to grant access to the service account for the account to become `Active`.

## Generate an API Access Token Using Your VCF Automation Provider Management UI

Generate an API access token to authenticate automation against VCF Automation. You can use API access tokens with your existing authentication methods, including two-factor authentication.

Authenticating with an API token uses the Refreshing an Access Token standard specified in OAuth 2.0 RFC 6749 Section 6 to grant access to VCF Automation as an OAuth application. The API token you generate in this procedure serves as the refresh token in the OAuth 2.0 exchange. The API returns a new, standard VCF Automation access token. Client applications use this token to authenticate subsequent API calls to VCF Automation.

To make an OAuth 2.0 RFC-compliant request, familiarize yourself with Request for Comments (RFC) 6749 Section 6 information about refreshing an access token.

Access tokens are artifacts that client applications use to make API requests on behalf of a user. Applications need access tokens for authentication. When an access token expires, to obtain access tokens, applications can use API tokens. By default, API tokens expire after 90 days.

When using access tokens, applications cannot perform certain tasks.

- Change the user password
- Perform user management tasks
- Create more tokens
- View or revoke other tokens

When accessing VCF Automation by using an API access token, applications have only view rights for the following resources.

- User
- Group
- Roles
- Global roles
- Rights bundles

Applications accessing VCF Automation by using an API access token do not have the following rights.

- Token: Manage
- Token: Manage All

Similar to generating a user API token, you can create a service account by using the VCF Automation API. The API request for creating a service account uses the same API endpoint as creating a user API token, but the presence of the `software_id` field indicates the intent to create a service account.

1. Log in to the VCF Automation Provider Management UI.
2. In the upper right corner of the navigation bar, click your user name, and click **My Account**.
3. Under the **API Tokens** section, click **New**.
4. Enter a name for the token, and click **Create**.

   The generated API token appears. You must copy the token because it appears only once. After you click **OK**, you cannot retrieve this token again, you can only revoke it.
5. Make an OAuth 2.0 RFC-compliant request to the `https://`*`site.cloud.example.com`*`/oauth/provider/token` API endpoint.

| Key | Value | Type | Required |
|---|---|---|---|
| grant_type | `refresh_token` | String | Yes |
| refresh_token | `Generated_API_Token` | String | Yes |

   The request returns an access token that applications can use to perform tasks in VCF Automation. The token is valid even after the user logs out. When an access token expires, the application can obtain more access tokens by using the API token.

   Request:

```
POST https://host_name/oauth/provider/token
Accept: application/json
Content-Type: application/x-www-form-urlencoded

grant_type=refresh_token&refresh_token=Generated_API_Token
```

   Response:

```
HTTP/1.1 200 OK
Content-Type: application/json

{
   "access_token":"Generated_Access_Token",
   "token_type":"Bearer",
```

```
   "expires_in":2592000,
   "refresh_token":null
}
```

**Note:**

The `expires_in` value in the response is expressed in seconds. A value of 2592000 equals 30 days. The access token expires after this period, at which point the application must request a new access token using the API token. This 30-day access token lifetime is distinct from the 90-day API token lifetime described in the context section.

```
curl -X POST \
  https://host_name/oauth/provider/token \
  -H "Accept: application/json" \
  -H "Content-Type: application/x-www-form-urlencoded" \
  -d "grant_type=refresh_token&refresh_token=Generated_API_Token"
```

Request using the generated access token:

```
GET https://host_name/cloudapi/1.0.0/orgs
Accept: application/json;version=40.0
Authorization: Bearer Generated_Access_Token
```

Response:

```
{"resultTotal":1,"pageCount":1,"page":1,"pageSize":25,"associations":
[{"entityId":"urn:vcloud:org:a11a11a1-1111-1111-1a11","associationId":"a11a11a1-1111-1111-1a11@a11a1a11-1111-1111-1a
[{"id":"urn:vcloud:org:a11a1a11-1111-1111-1a11-
a1a11aa11a1","name":"System","displayName":"System","description":"System
 Bucket","isEnabled":true,"managedBy":null,"canManageOrgs":true,"maskedEventTaskUsername":"system","directlyManaged
```

If the token request fails, the API returns an HTTP error status code. For the list of possible error codes and the error response schema, see the VCF Automation API reference.

- If you want tenants to be able to generate tokens, you must grant tenant roles the **Manage user's own API token** right.
- By default, tenant users see only the tokens they create. To allow organization administrators to see and revoke the tokens of the other tenant users in the organization, you must grant them the **Manage all users' API tokens** right. Administrators with the **Manage all users' API tokens** right can see only the names of other users' the tokens, not the tokens themselves.
- To revoke any of your tokens, navigate to the **User preferences** page, and click the vertical ellipsis next to the token.
- To revoke the tokens of other users, in the top navigation bar, under **Administration**, navigate to the access control settings for users. When selecting a specific user, you can also see their access tokens and revoke them.
- If you must identify events triggered by the use of an API access token, in the event log the following line appears in the `additionalProperties` section of an event.

  `"currentContext.refreshTokenId": "<UUID_of_the_token_that_performed_the_action>",`

# Enable Cross-Organization Content Sharing in the VCF Automation Provider Management UI

You can enable specific organizations to share catalog items, such as blueprints, VCF Operations orchestrator workflows, and VM images, as well as content library VM images, with other organizations in your VCF Automation environment.

Verify that you have at least one provider content library in your environment. See Create a Content Library.

1. Log in to the VCF Automation Provider Management UI as a **system administrator**.
2. In the left navigation pane, click **Access Control**.
3. Click the **Rights Bundles** tab.
4. Click the **Privileged Organization Access Bundle** name, and click **Publish**.

   The **Privileged Organization Access Bundle** includes the **Organization: View All Organizations** and **Share content with other organizations** rights to enable content sharing. Granting the **Organization: View All Organizations** right enables an organization to view and list the available set of organizations in VCF Automation. Granting the **Share content with other organizations** right, enables the organization to share content. By default, VCF Automation publishes the **Privileged Organization Access Bundle** only to the provider consumption organization, if one exists.
5. Turn on the **Publish to Tenants** toggle.
6. Select the organizations to which you want to publish the role.
   - If you want to publish the bundle to all existing and newly created organizations in your system, select **Publish to All Tenants**.
   - If you want to publish the bundle to specific organizations in your system, select the organizations individually.
7. Click **Save**.

To stop an organization from sharing its content with other organizations, deselect the organization from the list of organizations to which the **Privileged Organization Access Bundle** is published.

> **Important:**
>
> When you unpublish the rights bundle from an organization with shared items or delete such an organization, the catalog items are no longer visible to other organizations. The copied VM images remain available to all organizations because they are copied to the provider content library.

When you log in to an organization by using the **Launch Organization Portal** button in the Provider Management UI, even if the organization does not have the necessary rights published, you can see the options to share catalog images and VM images to other organizations. The options appear because users with **system administrator** credentials have all VCF Automation rights, including sharing content across organizations. Because VCF Automation performs rights bundle validations, you can share only VM images as a **system administrator** in the organization context. If the rights bundle is not published to the organization, **organization administrators** cannot see the options to share catalog images and VM images to other organizations.

# Getting Started with the VCF Automation Provider Management UI

The VCF Automation Provider Management UI is a dedicated interface for Provider Administrators.

For more information about the terms and components of VCF Automation Provider Management, see Overview of VCF Automation Provider Management.

## Log in to VCF Automation Provider Management UI

As a **provider administrator**, you can access the VCF Automation Provider Management UI by using a web browser.

You must have the **system administrator** rights to access the Provider Management UI.

1. In a browser, enter the Provider Management UI URL of your VCF Automation site.

   The Provider Management UI URL `https://vcfa.example.com/provider` opens the login screen for the `System` organization.
2. Depending on your environment settings, choose one of the following.
   - If there is no interaction, after 5 seconds the Provider Management UI redirects you to the VCF SSO, if configured. If you click **Stop** or select another option from the drop-down menu, you can sign in with a local or SAML user.
   - To sign as a local user, enter the system administrator user name and password, and click **Sign In**.

  - To log in for the first time, you must use the default administrator account with a local user. Enter username `admin` and the password which you set up during the installation.
- After you log in, if there are more than 25 minutes of inactivity, you are automatically logged out of the VCF Automation Provider Management UI. The logout does not affect long-running operations, such as upload, for example.
- Integrate your VCF Automation with one or more external identity providers (IdPs), and import users and groups to your organizations. See Managing Identity Providers in VCF Automation.
- Configure access for other users and groups. See Managing Provider Users in Your VCF Automation and Managing VCF Automation Provider Groups.

## Quick Start Your VCF Automation Setup

You can go through a simplified wizard for the creation of an organization for all applications with certain predefined settings. **Quick Start** is an alternative to manually setting up all aspects of your first organization in VCF Automation.

- Verify that you complete the VMware Cloud Foundation wizard for setting up your vCenter and NSX infrastructure. See Getting Started with VMware Cloud Foundation.
- Verify that the matching NSX Manager for the Supervisor is registered in your environment. See View and Manage Your NSX Manager Instances.
- Verify that you have a Tier-0 gateway or Tier-0 VRF gateway created in NSX Manager. For instructions on creating a Tier-0 gateway, see Add an NSX Tier-0 Gateway. For instructions on creating a Tier-0 VRF gateway, see Step 4 in Deploy an NSX VRF-Lite with BGP.

  **Note:**

  If you want to add a Tier-0 gateway that is backed by a VRF gateway in NSX, you must also create a VRF gateway that is linked to the Tier-0 gateway.
- Verify that your vCenter instance has a Supervisor with VPC networking enabled. See Deploying Supervisor with VCF Networking with VPC.
- Verify that you have edge clusters configured and available in your environment.

You have two primary options when setting up your tenant organization:

- The **Quick Start** is designed for efficiency by automatically creating a fully functional tenant organization along with all the necessary resources. The **Quick Start** is the recommended choice for single-tenant setups or exploring and testing VCF Automation. You can modify the resulting organization for production use.
- The **Manual Setup** appears as a check list of tasks in the UI and is intended to remind you of the tasks that you must carry out to realize the full capabilities of the Provider Management UI. When you complete a task, it automatically turns into a check mark.

  **Important:**

  The **Quick Start** is not applicable to VLAN-backed environments, which are supported in VCF Automation 9.1.1 and later. To use the **Quick Start** option, you must have a full-stack Supervisor. For VLAN-backed environments, use the **Manual Setup** option, see Manually Set Up Your VCF Automation.

Creating an organization by using the **Quick Start** option does not have any limitations compared to the **Manual Setup** and you can change the configuration of the organization after its creation.

You can use the **Quick Start** option only once. The **Quick Start** option disappears after you set up the first organization and region either through the **Quick Start** or the **Manual Setup** options. The **Quick Start** uses the first available tier-0 gateway in NSX, and creates a provider gateway for that tier-0. To create an IP space, VCF Automation uses an IPv4 block with external visibility in NSX. If you want to ensure that VCF Automation uses the tier-0 gateway and IP blocks generated during the workload domain configuration, do not add any additional tier-0 gateways or IP blocks. When creating the organization, VCF Automation automatically sets up the organization's quota and networking, including the transit gateway and VPC.

1. Log in to the VCF Automation Provider Management UI.
2. Under **Quick Start**, click **Get Started**.
3. Enter a name for your first organization, and click **Next**.
4. Enter a name for the region, and select a Supervisor.

   The list shows all available Supervisors for this NSX Manager instance.
5. Click **Next**.
6. Select one or more storage classes.

   The list shows all storage classes across the selected Supervisors.
7. Review the configuration of the organization, and click **Create and Provision Organization**.

   The **Organization Provisioning in Progress** dialog opens where you can view and follow the creation process.
8. When the task finishes, click **OK**.

Use the VCF Automation Provider Management UI to make further changes to the configuration of your organization.

- To set up the initial administrator for the organization, see Edit the First User Details for a VCF Automation Organization.
- To finish setting up the organization, you can use the first user details to log in to the Organization UI by navigating to `https://FQDN_VCF_Automation`. See Organization Management.
- Configure one or more identity providers. See Managing Identity Providers in VCF Automation Organizations.
- The **Quick Start** does not create users for the organization it creates. You must configure the organization users and organization groups manually. See Managing Predefined User Roles in VCF Automation.
- Create content libraries at the provider level or directly within the organization. See Create a Content Library in the Provider Management guide or Managing Content Libraries in VCF Automation in the Organization Management guide.
- Review the default region quota. See Region Quota of a VCF Automation Organization.
- Review the default networking settings, including the provider gateway, IP space, Edge cluster. See Managing Networking Resources in VCF Automation.

## Manually Set Up Your VCF Automation

To realize the full capabilities of the Provider Management UI you must set up VCF Automation.

- Verify that you complete the VMware Cloud Foundation wizard for setting up your vCenter and NSX infrastructure. See Getting Started with VMware Cloud Foundation.
- Verify that the matching NSX Manager for the Supervisor is registered in your environment.
- To use and configure centralized connections, verify that you have a Tier-0 gateway or Tier-0 VRF gateway created in NSX Manager. For instructions on creating a Tier-0 gateway, see Add an NSX Tier-0 Gateway. For instructions on creating a Tier-0 VRF gateway, see Step 4 in Deploy an NSX VRF-Lite with BGP.

  **Note:**

  If you want to add a Tier-0 gateway that is backed by a VRF gateway in NSX, you must also create a VRF gateway that is linked to the Tier-0 gateway.
- To use and configure distributed connections, verify that you have a VLAN configured in your infrastructure.
- Verify that your vCenter instance has a Supervisor with VPC networking enabled. See Deploying Supervisor with VCF Networking with VPC.

When you set up the necessary infrastructure, **Organization Administrators** can use the first user credentials, that you configure during the organization creation, to log in to their respective organizations by using the Organization UI. See Managing Organizations in VMware Cloud Foundation Automation and Managing Organizations for VM Apps in VMware Cloud Foundation Automation.

1. Configure an identity provider. See Managing Identity Providers in VCF Automation.
2. Configure the VCF Automation access control. See Configuring the VCF Automation Access Control.
3. Create a Region in Your VCF Automation.
4. Create an organization for **All Apps** or **VM Apps**. See Create a VCF Automation Organization for All Applications, or Create a VCF Automation Organization for VM Applications.
5. Configure the Regional Networking of a VCF Automation Organization.
6. Create a Region Quota for a VCF Automation Organization.
7. Configure the networking resources. See Managing Networking Resources in VCF Automation.
8. Upload a Partner Service to Your VCF Automation Provider Management UI.

## Troubleshoot Failed Access to the VCF Automation User Interface

If you are using an external load balancer, to view and update the valid IP addresses and DNS entries for the VCF Automation cells in your VCF Automation environment, you can use the VCF Automation API.

You cannot access the VCF Automation Provider Management UI after a successful login or after changing the DNS entries.

After you enter your credentials in the login screen, the following error message appears: `Failed to Start. An error was encountered during initialization. This can be caused by issues such as accessing the application via an unsupported public URL or poor connectivity.`

The access to the VCF Automation UI might be limited even if the `Public Addresses` fields are properly configured. See Adding Cloud Accounts.

1. Make a `GET` request with an authorization header with the JSON Web Token (JWT) and an accept header to the `https://{api_host}/cloudapi/1.0.0/site/settings/cors` API endpoint.

   The system output is a list that should contain HTTP and HTTPS entries with IP addresses and DNS names for all cells in the server group. It should also contain the public host name and IP address that the load balancer uses.

   Each endpoint in the list must have three entries:

   - FQDN
   - HTTP
   - HTTPS

   Example:

   ```
   {
    "values": [
       {
        "origin": "vcd.domain.local"
       },
       {
        "origin": "http://vcd.domain.local"
       },
       {
        "origin": "https://vcd.domain.local"
       }
     ]
   }
   ```

2. Verify that for every endpoint in the list, there are three entries and make a PUT request to the API endpoint.

   Verify that when you perform a REST PUT operation, you provide all values of the origins configuration which are currently configured and which you need to retain.

# Viewing and Managing Your Infrastructure Resources in VCF Automation

VCF Automation derives its resources from an underlying virtual infrastructure. VCF Automation displays the vCenter and NSX Manager instances discovered from VCF Operations that you can view and manage.

VCF Automation uses one or more vCenter environments to back its regions. Each vCenter instance must have at least one Supervisor enabled.

> **Note:**
>
> VCF Automation must manage a vCenter instance, Supervisors, vSphere clusters, and so on, exclusively. When you use a vCenter instance for VCF Automation and other purposes, a vCenter administrator might accidentally move VCF Automation resources. This can cause VCF Automation to operate incorrectly, including losing virtual machines that VCF Automation manages.

**Managing Networking Resources**

After creating a Tier-0 gateway in NSX Manager, you can make your networking configurations in VCF Automation. As a Provider Administrator, you can define IP spaces and provider gateways and assign resources to organizations.

As a provider administrator, you can make networking configurations for your organizations. For example, you can define IP blocks and external connections, which you can later use to assign networking resources to organizations.

## View the vCenter Instances in Your VCF Automation

You can see a list of all your vCenter instances, discovered from VCF Operations, and their current connection status.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Administration**, click **Connections**.
3. Click the **vCenter** tab.

A list of all vCenter instances appears. The list contains the following information for each vCenter instance.

| | Description |
|---|---|
| **Name** | The name of the vCenter instance in VCF Automation. |
| **Description** | Description of the vCenter instance. |
| **State** | **Enabled** or **Disabled**. See Deactivate and Unregister a vCenter Instance in VCF Automation. |
| **Connected** | Connected or not to VCF Automation. See Reconnect a vCenter Instance to VCF Automation. |
| **Username** | Username for login to this vCenter instance. |
| **License Status** | The licensing status of the vCenter instance. `Licensed` indicates that the vCenter instance is properly licensed. `Not Licensed` indicates that the vCenter license is either invalid or the license is expired. |
| **VCF Instance** | The VCF Instance managing this vCenter instance. |

## Deactivate and Unregister a vCenter Instance in VCF Automation

After you delete a workload domain using VCF Operations, you must deactivate and unregister the vCenter instance from VCF Automation.

1. Decommission a vCenter instance in VCF Operations. See Delete a Workload Domain.
2. Delete the regions created on the vCenter instance. See Create a Region in Your VCF Automation.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Administration**, click **Connections**.
3. Click the **vCenter** tab.
4. Click the vertical ellipsis ( ⋮ ) next to the name of the target vCenter instance, and select **Disable**.

   When you deactivate a vCenter instance, the status of the instance changes to `Disabled`, and you cannot create any regions on it. Existing regions are not affected.
5. When the vCenter instance is deactivated, click the vertical ellipsis ( ⋮ ) next to the name of the target vCenter instance, and select **Unregister**.

   Unregistering a vCenter instance, deletes the instance from VCF Automation.

## Reconnect a vCenter Instance to VCF Automation

If a vCenter instance appears as disconnected, or if you modified the connection settings, you can try to reset the connection to VCF Automation.

**Note:**

During establishing the new connection, the vCenter instance is unavailable for operations.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Administration**, click **Connections**.
3. Click the **vCenter** tab.
4. Click the vertical ellipsis ( ⋮ ) next to the name of the target vCenter instance, and select **Reconnect**.
5. To confirm, click **Reconnect**.

## Refresh a vCenter Instance in VCF Automation

If you make a change to the underlying vCenter infrastructure and you need instant confirmation of the change in VCF Automation, you can refresh the vCenter instances. By default, VCF Automation performs an automatic refresh of the vCenter instances every 10 minutes.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Administration**, click **Connections**.
3. Click the **vCenter** tab.
4. Click the vertical ellipsis ( ⋮ ) next to the name of the target vCenter instance, and select **Refresh**.
5. To confirm, click **Refresh**.
6. If you want to refresh the vCenter storage classes, click the vertical ellipsis, and select **Refresh Policies**.
7. To confirm, click **Refresh**.

## View Your VMware vSphere Supervisors

You can use the VCF Automation Provider Management UI to view your Supervisors.

A Supervisor is a control plane that spans across one or more zones, each zone has one cluster.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Supervisors**.
3. To expand the view and show the full Supervisor details, click the button to the left of the Supervisor name.

   Information about the vCenter, Supervisor zones, and Reserved VM classes appears. If the Supervisor is part of a region in VCF Automation, the region also appears in the details.

Starting with version 9.1.1, information about the VPC network stack of each Supervisor zone is also displayed. For more information about the full-stack and VLAN-backed network stacks, see Supervisor Architecture with VCF Networking with VPC.

### View and Manage Your NSX Manager Instances

You can see a list of all your NSX Manager instances, discovered from VCF Operations, and their current connection status.

If an NSX Manager instance is not properly registered, it might not appear in the list. You can use the VCF Automation API to register the NSX Manager instance.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Administration**, click **Connections**.
3. Click the **NSX Manager** tab.

   A list of all **NSX Manager** instances appears. You can see the name, description, and status of the **NSX Manager** instance.
4. If you want to see the URL of the **NSX Manager** instance, click its name.
5. If you want to edit the **NSX Manager** name, description, URL, username, or password, on the **General** page, click **Edit**.

# Managing Regions in Your VCF Automation

In VCF Automation, a region combines the compute, memory, and networking resources from the underlying infrastructure.

#### Region Configuration Requirements

VCF Automation creates a region by grouping one or more Supervisors from one or more vCenter instances, if they are all configured with the same NSX instance. If the vCenter instances are associated with different NSX instances, you cannot add their Supervisors to the same region. In a typical VCF Automation deployment, you can use multiple regions to meet diverse service level requirements, or connect to infrastructure hosted in different VCF Instances within a single VCF Operations environment.

When you define a region, you must also select the storage classes that you want to make available across all Supervisors in the region.

> **Important:**
>
> All Supervisors within a region must have a homogeneous configuration including:
>
> - Identical names and configurations for storage classes
> - Identical names and configurations for VM classes
> - Matching Kubernetes versions
> - A consistent set of IaaS services
> - Starting with version 9.1.1: Identical network stack type - full-stack or VLAN-backed.
>
> VCF Automation does not support non-homogeneous region configurations for these elements. All specified classes must be accessible to all zones within the region. Reserved VM classes used for GPUs are an exception to this requirement and do not need to be homogeneous across the region.

The uniformity of all Supervisors ensures consistent and reliable user experience across the region. When you add a storage class or VM to a region, you must not rename it in any of the Supervisors.

Regions can contain Supervisors associated with different zones. For example, a region can include one Supervisor in Zone A and another Supervisor in Zone B.

To deploy Supervisor Services, VCF CLI plug-ins, and VKS standard packages on VKS clusters in a region, you must install Regional Harbor to serve as the central container registry for VCF.

### Mixed Tenancy Mode and Resource Sharing

Starting with version 9.1, `VM Apps` organizations and `All Apps` organizations can share a vCenter instance that has a Supervisor with VPC networking enabled. This resource sharing enables customers leveraging VM Apps organizations to explore modern `All Apps` functionalities without provisioning additional hardware. To activate resource sharing, navigate to the **Feature Flags** page from the left navigation pane and activate the **Mixed Tenancy Mode** feature flag.

While the feature flag enables vCenter sharing across organization types, VCF Automation does not enforce strict resource isolation in this configuration. Specifically, `VM Apps` organizations might gain read-only visibility into `All Apps` resources. Conversely, `All Apps` organizations remain isolated from `VM Apps` resources.

Beyond visibility concerns, the **Mixed Tenancy Mode** forces both organizations to share a single NSX or vCenter instance, creating a risk of API bottlenecks and management-plane contention. Use **Mixed Tenancy Mode** only for migration phases or trials where cross-visibility is acceptable. Do not use this configuration for production environments requiring strict multi-tenancy.

### Supervisor VPC Network Stack Types

Starting with version 9.1.1, each zone in a vSphere Supervisor has a `supported_networking_stack` property that indicates the VPC network stack type of the zone. There are two types of VPC network stacks.

- Full-stack VPC
  The underlying NSX networking relies on Tunnel Endpoints (TEPs) to create an overlay network.
- VLAN-backed VPC
  No overlay is required. All traffic runs directly over VLANs. This configuration is designed for environments running entirely VLAN-based infrastructure where NSX overlay networking is not available.

For each zone, the VPC network stack type is visible in the VCF Automation Provider Management UI.

1. Navigate to **Infrastructure Overview** > **Regions** and select a region.
2. Click the **Zones** tab.
   The information about the zone includes a network stack column indicating whether the VPC network stack is full-stack or VLAN-backed.

   **Note:**

   You can also view zone network stack details by navigating to **Infrastructure** > **Supervisors**and expanding a Supervisor to review the zone list.

As a provider administrator, you must know whether a zone is VLAN-backed or full-stack before you assign the zone to an organization, because all zones per region must have the same network stack type. Mixing VLAN-backed and full-stack zones in one organization is not supported. Also, you cannot add a zone without a network stack to a region. To prevent incompatible configurations, zone assignments are automatically validated when you create or modify region quotas.

You can view the supported networking options for each of the network stack types in the following table.

**Table 1805: Networking Capabilities per VPC Network Stack**

| Characteristic | Full-stack VPC | VLAN-Backed VPC |
|---|---|---|
| Public subnets | Supported | Supported |
| Private subnets | Supported | Not supported |
| VLAN extension subnets | Supported | Supported |
| NAT/SNAT | Supported | Not supported |
| Private transit gateway IP blocks | Supported | Not supported |
| Centralized connections (Tier-0) | Supported | Not supported |

| Characteristic | Full-stack VPC | VLAN-Backed VPC |
|---|---|---|
| Distributed VLAN connections | Supported | Supported |
| IPSEC VPN | Supported | Not supported |
| DHCP Relay mode Subnets | Supported | Not supported |

## Create a Region in Your VCF Automation

To make vSphere compute, memory, and storage resources available to VCF Automation, you must create a region.

- Verify that your vCenter instance has a Supervisor enabled.
- Verify that the following conditions are met across the participating vCenter instances and Supervisors:
  - All instances are configured to use the same NSX instance
  - All storage classes must:
    - Exist on all instances
    - Have identical names
    - Have the same configuration, for example, the same policies and backing storage.
  - All VM classes must:
    - Exist on all instances
    - Have identical names
    - Have consistent configuration, for example, CPU, memory and reservations
    - Have consistent zone naming (optional for capacity pooling)
  - If you want to group zones across Supervisors into a shared capacity pool:
    - The zone names must be the same across all vCenter instances.
  - All Supervisors must run the same Kubernetes version to ensure workload compatibility
  - The same set of IaaS services, for example, VM Service, Storage Service, Load Balancer Service, must be enabled and configured consistently across Supervisors.
- If you want to install Supervisor Services, VKS standard packages, and VCF CLI plug-ins, install Harbor. See Install Regional Harbor Through VCF Automation. Regional Harbor provides a central container registry for VCF and contains the necessary images to deploy Supervisor Services, VCF CLI plug-ins, and VKS standard packages on VKS clusters.

  **Important:**

  VCF Automation does not verify that a storage class or VM class is present in all vCenter instances.

Before creating organizations for your tenants, you must create a region and regional quotas that define the consumption of resources. Because a region constrains the vSphere capacity and services available to tenants, **Provider Administrators** commonly create regions that furnish different classes of service, as measured by performance, capacity, and features. Tenants can then be provisioned with regional quotas that deliver specific classes of service defined by the configuration of the backing region. Before you create a region, consider the set of vSphere capabilities that you plan to offer your tenants.

**Important:**

If the region uses a vCenter instance version 9.0 or earlier, organization administrators cannot create project content libraries.

Starting with version 9.1.1, VCF Automation supports VLAN-backed infrastructures. Mixing full-stack and VLAN-backed zones within a single region is not allowed.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Regions**.
3. Click **New**.
4. Enter a name and, optionally, a description for the region.
5. Select the NSX Manager instance that integrates with the vCenter instances you want to use for the region.

   A single NSX Manager instance must be integrated with all vCenter instances within a region.

6. Select one or more Supervisors.

   The list shows all available Supervisors for this NSX Manager instance.

7. Select one or more storage classes.

   The list shows all storage classes across the selected Supervisors.

   **Important:**

   The storage classes within a region must be present and have the same names across all vCenter instances within a region.

   VCF Automation does not verify that a storage class or VM class is present in all vCenter instances.

8. Click **Submit**.

- To view the region details, such as, description, status, Supervisors, NSX Manager instance, VPC networking stack, and CPU and memory utilization, click the button to the left of the region name to expand the region details pane.
- To view the **Supervisors**, **VM Classes**, **Storage Classes**, and **Reserved VM Classes** applicable to this region, click the region name and click the relevant tab in the secondary left navigation pane.

The **Zones** tab displays the zones from all Supervisors added to the region. If you added two Supervisors with three zones each, the page displays a total of six zones. If a region includes more than one Supervisor and the zones in the Supervisors have identical names, VCF Automation displays the combined memory and CPU capacity of the zones with the same name. To see the zone details and capacity coming from each vCenter, expand the zone view by clicking the button to the left of the zone name.

The **Reserved VM Classes** are a special VM class that defines one or more vGPU PCI devices in addition to the CPU and memory. These VM classes are reserved for each zone in which the Provider Administrator has actual matching GPU capacity available. You can reserve a VM class in vCenter. VCF Automation detects the reserved counts of these VM classes per zone and shows them in the VCF Automation Provider Management UI under the **Reserved VM Classes** section of a region. The counts of these **Reserved VM Classes** are further given to organizations through region quotas.

- To edit a region, click the name of the region and click **Edit**. See Edit a Region.
- Delete a region
  a. Remove all region quotas created from the region. See Delete a VCF Automation Region Quota.
  b. Click the name of the region and click **Delete**.

## Edit a Region

You can use the Provider Management UI to edit the region settings. You cannot edit the region name and NSX Manager instance.

If the entire capacity of a region is shared with an organization through a region quota, any changes you make to the region automatically apply also to the region quota. You can edit a region to remove a storage class, even when a provider content library uses that storage class to host the published vCenter content library. If namespaces on a Supervisor originate from a full allocation region quota, you cannot remove that Supervisor by editing the region. If one or more namespaces use these resources from a full allocation region quota, you cannot remove a storage class or VM class. You cannot remove the resources from the region until you either remove the namespaces that use the resources or edit the namespaces so they no longer use these resources.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Regions**.
3. Click the name of the region that you want to edit.
4. Click **Edit**.
5. (Optional) Change the description of the region.
6. (Optional) Increase or decrease the scope of the region by selecting or deselecting Supervisors.

   You cannot remove a Supervisor from a region if resources from this Supervisor are already allocated to the organizations as region quotas. You must delete the region quotas before removing the Supervisor from the region.

7. (Optional) Select or deselect the storage classes applicable to the region.

   The list shows all storage classes across the selected Supervisors.

   **Important:**

   The storage classes within a region must be present and have the same names across all vCenter instances within a region.

   VCF Automation does not verify that a storage class or VM class is present in all vCenter instances.

   You cannot remove a storage class from a region if the resources from this storage class are already allocated to the organizations as region quotas. You must delete the region quotas before removing the storage class from the region or you must update the region quotas to remove the storage class from it.

8. Click **Submit**.

## Modify the Metadata for a VCF Automation Region

You can use the Provider Management UI to add, edit, and delete metadata for a region in VCF Automation.

To edit or delete metadata, verify that your user account has the **Create, update, and delete Regions** right. For read access, your account must have the **View Regions** right.

You can use metadata to tag, categorize, and automate the management of tenant and provider resources, enabling policy-driven operations such as billing, compliance, workload placement, and service customization across multi-tenant environments.

By using metadata, you can associate user-defined `key=value` pairs with a region. You can use metadata in VCF Automation API query filter expressions.

You can edit only the value of an existing metadata item.

**Table 1806: Provider Metadata**

| Parameter | VCF Automation Metadata Limits |
|---|---|
| Key length | Up to 256 UTF-8 characters |
| Metadata entries per resource | Up to 200 entries, with a maximum limit of 300. |
| Value length | Up to 1,000 characters. The system discards entries that exceed this limit. |
| Supported types | `String`, `Bool`, and `Integer` |
| Search patterns | • `?metadata=org:vmware\|test:key:8=='string'` – searches for a string entry. When searching for a string, always enclose the value in single quotes (').<br>• `?metadata=test.key.3==42` – searches for a number entry.<br>• `?metadata=test.key.3==*` - searches for a key.<br>• `?metadata=test.key.*==*` - searches for a key starting with `test.key`.<br>• `?metadata=test.key.3=lt=43`<br>• `?metadata=test.key.4=='str*'` – searches for a string entry with a value starting with `str`. There are no limitations on the size of the FIQL expression or the number of different key-values in the expression. |
| Bulk creation | Supports bulk metadata creation. |

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Regions**.
3. Click the name of the region that you want to edit.
4. In the secondary left navigation pane, click **Metadata** and click **Edit**.
5. If you want to add a key-value pair, click **Add**, enter a key and a value, and select a type for the new key-value pair.

   The available metadata types are **Text**, **Yes or No**, and **Number**.
6. If you want to remove a key-value pair, in the right end of the row, click the **Delete** icon.
7. Click **Save**.

# Create a Content Library

You can use the Provider Management UI to create a provider content library shared with all organizations in this VCF Automation instance. The provider content libraries are read-only to the users of the organizations.

Create a Region in Your VCF Automation

A VCF Automation provider content library is an abstraction in VCF Automation that provides a single interface for cloud providers to manage their content across regions. You create a provider content library by selecting one or more regions and a storage class from each region. To store the actual content in each vCenter taking part in the region, when you create a provider content library, a content library is also automatically created in vCenter on appropriate storage, compatible with the selected storage class. The content is automatically synced among all of the backing vCenter content libraries. For this purpose, one of the vCenter content libraries is randomly picked as a publisher and is set to the `local` type. The rest of the vCenter libraries subscribe to that publisher are set to the `subscribed` type. Because of this underlying synchronization mechanism, you must ensure that the vCenter instances can communicate with each other and that there is network connectivity between the vCenter instances. The connectivity between the vCenter instances is critical to the distribution to all vCenter instances of the regions that are part of a VCF Automation provider content library.

From your provider consumption organization, you can promote project-scoped content library images to a provider content library so that the images become available to all your organizations. See Add a VM Image to a Content Library in VCF Automation.

### Tenant Content Libraries

By default, each tenant project is limited to 200 content libraries. As a service provider, you can use the VCF Automation API to adjust this threshold for all tenants using the `contentLibrary.maxLibrariesPerProject` setting. This configuration is a global system setting that does not affect provider content libraries.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Content Libraries**.
3. On the **Content Library** tab, click **New**.
4. Enter a name, and optionally, a description of the content library.
5. To subscribe to an external library, turn on the toggle.

   You can subscribe to any content library compliant with the VMware Content Subscription Protocol (VCSP). VCSP is the underlying communication standard used by VMware Content Libraries to synchronize and share VM templates, OVA and OVF packages, and ISO images between different vCenter instances.
6. Click **Next**.
7. If you want to subscribe to an external library, enter the library URL and password, and click **Next**.
8. Select one or more regions and a corresponding storage class for each region.

   **Important:**

   The storage classes within a region must be present and have the same names across all vCenter instances within a region.

VCF Automation does not verify that a storage class or VM class is present in all vCenter instances.

9. Click **Next**.
10. Review the content library settings, and click **Confirm**.

VCF Automation creates the provider content library in each selected region and shares it with all organizations.

To edit, refresh, or delete a provider content library, click the vertical ellipsis next to the content library name.

- Edit a provider content library by clicking the vertical ellipsis next to the content library name and clicking **Edit**.
  You can edit a VCF Automation provider content library to increase or decrease its scope. For example, you can add a new region to increase the distribution radius of the content library, alternatively, you can remove a region. You can also change the storage class of a content library to trigger a migration of content from an old to a new storage.
- Refresh a provider content library by clicking the vertical ellipsis next to the content library name and clicking **Refresh**.
  You can use the refresh option to reconcile changes from underlying vCenter instances. If a new content library item is added directly in the vCenter content library backing the provider content library, the refresh action propagates this new item to the provider content library. If an item is directly deleted from a vCenter content library, the refresh option removes the item from the provider content library.
- Delete a provider content library by clicking the vertical ellipsis next to the content library name and clicking **Delete**.

## Upload a Content Library Item

You can use the Provider Management UI to upload an item to a provider content library that is not subscribed. All items within a provider content library and the content library itself are automatically shared with all organizations.

Create a Content Library

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Content Libraries**.
3. On the **Content Library Items** tab, click **Upload**.
4. Upload a source file.

   You can upload an OVA, ISO, or an OVF file and all files that the OVF references.
5. From the drop-down menu, select a content library to which to add the item.
6. Enter a name, and optionally, a description for the content library item.
7. Click **Submit**.

If you want to delete an item, click the vertical ellipsis next to the content library name, and click **Delete**.

# Create an Infrastructure Policy in VCF Automation

You can create an infrastructure policy that defines the VM placement on specific hosts. By adding policy enforcement criteria, you can target a subset of VMs based on their attributes, ensuring a correct workload placement.

1. vSphere supports the creation of multiple zones. To use them in combination with infrastructure policies, these criteria must be met:
   - Verify that one or more vSphere clusters back each zone.
   - Verify that one or more hosts in each vSphere cluster within a single zone have the tags defined in the vCenter compute policies to ensure the same vCenter compute policy compatibility.
2. In the vSphere Client, create a tag and tag category. See What Are vSphere Tags and How to Use Them.
3. In the vSphere Client, create a **VM-Host affinity** compute policy.
   See Create a VM-Host Affinity Policy.

**Note:**

After the creation of a compute policy, the synchronization between the vSphere Client and VCF Automation might take a few minutes.

Infrastructure policies build upon vCenter compute policies to enforce placement and resource rules in the vSphere Supervisor context. You can use the Provider Management UI to view the available vCenter compute policies. See View the vCenter Compute Policies.

After you assign infrastructure policies to a tenant, the tenant can create a namespace, selecting one or more zones. Namespace creation fails if the selected zones are not compatible with the policies. The system automatically applies all assigned mandatory policies. The tenant can also select from the available optional policies to meet the requirements of the namespace.

- VCF Automation does not require a namespace to comply with the policies within a single zone. To meet all policy requirements, the namespace might span across multiple zones. If a zone has multiple clusters, the infrastructure policies must be available in all the clusters of that zone. For the infrastructure policy to be available, the vCenter compute policy defined in the infrastructure policy must be compatible with at least one host in this cluster. Compatibility requires that the host tag defined in the policy is assigned to that host.
- A VM with multiple policy requirements must have all policies fulfilled within a single zone and a single cluster in that zone.
- If a single zone cannot fulfill the VM policy requirements, even if the namespace is placed across multiple zones, the VM placement fails.
- Zones can host a unique combination of available policies. VCF Automation automatically places workloads, such as VMs, only when you do not provide an explicit zone. For namespaces, you must always provide one or more zones, and you must ensure the selected zones are compatible with all assigned mandatory and optional infrastructure policies.

You can dynamically manage the compute capacity within the defined zones.

- You can add additional vSphere clusters to an existing zone.
- To ensure policy consistency for future namespace placement, you must configure new clusters with the same infrastructure policies as the existing clusters in that zone.
- You must tag at least one host in every vSphere cluster within a zone to match the policies defined for that zone. vCenter enforces the infrastructure policies using your host tagging.
- When you remove a cluster from a zone and migrate the part of the namespace containing the workloads to another cluster within the same zone, the destination cluster must be able to satisfy the full set of infrastructure policies that govern those migrated workloads.
- You can modify the infrastructure policies available in a specific zone by adding new policies for placement. After adding new policies to a zone, you must update the underlying vCenter configuration by ensuring the corresponding policy tags are applied to at least one host per cluster in all clusters in that zone.

**Note:**

You cannot edit the infrastructure policies after their creation.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Infrastructure Policies**.
3. Click **New**.

   If there are no infrastructure policies in VCF Automation, click **Create Infrastructure Policy**.
4. Enter a name for the policy, and optionally, a description.
5. Select a vCenter compute policy to serve as the foundation for the infrastructure policy creation.
6. If you want the policy to be mandatory for the VMs in the namespaces in which the policy is applicable, select the **Make this a mandatory policy when assigned to a region in the organization** check box.

   If a policy is not mandatory, during the namespace creation process, organization administrators can choose to select or deselect that policy as applicable to the namespace.

**Note:**

If existing namespaces are present in a region, you cannot assign mandatory infrastructure policies to the region quota. You can assign only optional policies for that region. To add mandatory policies, you must do so before creating any namespaces for the specific region.

7. (Optional) Apply the policy enforcement criteria to the infrastructure policy.

   You can add rules to determine which VMs the policy manages. You can define conditions by selecting an attribute, such as **Guest OS**, **Guest OS Family**, or **Custom Label**, and an operator. You can use the expression preview to verify the intended rules.

8. Click **Submit**.

- To view the details of a policy, on the **Infrastructure Policies** page, click the policy name.
  On the details page, the **Sync Status** column provides information on whether the policy is properly enforced to all the vCenter instances where VCF Automation detects the vCenter compute policy. You can sync from the vertical ellipsis in the grid to manually trigger the synchronization in certain cases.
  - Deleted vCenter compute policy - If you delete the compute policy in vCenter, the synchronization status for that vCenter becomes `PENDING_DELETION`. Manually syncing the infrastructure policy causes the system to clean up the vCenter policy object. The synchronization fails if VCF Automation detects that namespaces placed on that particular vCenter use the infrastructure policy.
  - vCenter compute policy added to a new vCenter - If you add the vCenter compute policy to a new vCenter, the infrastructure policy details show the new vCenter with a `PENDING_CREATION` synchronization status. When you trigger the synchronization operation, the system creates the infrastructure policy object on the new vCenter to sync the rules and other details from VCF Automation to the new vCenter instance.
- To delete an infrastructure policy, click the vertical ellipsis ( ⋮ ) next to the policy name and click **Delete**. When you delete a policy, VCF Automation removes it from all namespaces to which it applies. If the vCenter instance is unavailable, you can force the policy deletion to remove the policy from VCF Automation.
- When you navigate to a namespace, you can see all infrastructure policies that apply to the namespace, including the mandatory and opt-in policies. You can view the VM details to see its enforced infrastructure policies.

## View the vCenter Compute Policies

After you create one or more vCenter compute policies using the vSphere Client, you can view the policies in the VCF Automation Provider Management UI

You can use vCenter compute policies to define and guide VM placement and resource usage across hosts within a cluster.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Infrastructure Policies**.
3. Click the **vCenter Compute Policies** tab.

# Managing Networking Resources in VCF Automation

To leverage the networking capabilities of VCF Automation for organizations for all applications, you add to and manage your infrastructure and cloud networking resources through the VCF Automation Provider Management UI, and then assign them to the organizations in your environment.

## Managing IP Blocks in VCF Automation

You can use external IP blocks to meet your IP address management needs. Additionally, you can include one or multiple CIDRs in the IP block, while at the same time exclude one or more IPs from the pool of available IPs. IP blocks provide a structured approach to allocating public IP addresses to different organizations, enabling connectivity to external networks.

IP blocks represent IPs used in this local data center, south of the provider gateway. The IP blocks that you define are used by Organization Administrators as external IP addresses for configuring services and public networks. For example, Organization Administrators can allocate CIDRs and individual IP addresses for NAT and load balancing. This allocation is

subject to the provider-defined quota for the respective organization. Organizations can also create subnets from these IP blocks to ensure direct workload connectivity without NAT.

As a provider administrator, you create IP blocks and use them for the external connections that you create and configure. You can assign an IP block to multiple external connections as long as these external connections are not assigned to the same organization. An IP block consists of a set of CIDR blocks and IP ranges that are reserved and dedicated to VCF. The IP addresses in an IP block can only be IPv4.

As a provider administrator, you can benefit from an extended set of functionalities that are related to the definition and usage of external IP blocks.

- You can have up to 10 CIDRs and up to 10 IP ranges inside a single IP block.
- You can specify reserved IP ranges to be excluded from the pool of IP addresses and CIDRs available for allocation.
- You can use existing IP blocks from an external IPAM provided by Infoblox. For more information, see Register Infoblox NIOS DDI with NSX.
- You can hide the content of the IP block from your organizations and only expose the definition of the objects and the quotas applying to it.
  For example, you can share with an organization an IP block with a quota of 5 IPs without exposing the IP addresses or CIDRs inside the IP block.

  **Note:**

  Quota is defined for all IP blocks, hidden or not.

Starting with version 9.1.1, external IP blocks also work with the distributed VXLAN connection type to provide IPs for advanced networking services, such as Load Balancer and NAT. When an organization administrator assigns a Load Balancer VIP or configures NAT, the VIP or NAT IP is allocated as a `/32` address from the external IP block that you as a provider administrator define and assign to the organization. The Route Controller automatically advertises these `/32` service IPs to the physical Border Gateway as BGP EVPN Type-5 routes. No additional routing configuration is required.

## Create an External IP Address Block

You can create an IP block, formerly known as an IP Space, to be used by multiple VCF Automation organizations and you can control organization usage through a quota system. An IP block can only be IPv4.

- Verify that your role includes the **System IP Spaces: Manage** and **System IP Spaces: View** rights.

  **Note:**

  The **System Administrator** role has these rights by default.
- Verify that you have a region with more than one Supervisor, see Create a Region in Your VCF Automation.
- To use an external IPAM, verify that connection to Infoblox is configured in NSX Manager, see Register Infoblox NIOS DDI with NSX.

As a provider administrator, you can add an IP block that can be used by several organizations. The consumption of IP addresses can be controlled by a quota on the IP block, and optionally, overridden at organization level.

**Important:**

An IP block can overlap with other IP blocks as long as the overlapping blocks are not assigned to the same organization.

Starting with version 9.1.1, if you want to create an IP block to support Load Balancer VIPs or NAT IPs for advanced networking services over a VXLAN external connection, you must observe the following best practices:

- Use a dedicated IP range for service IPs. Avoid mixing Load Balancer VIP and NAT IP ranges with VM subnet ranges, because the Route Controller advertises Load Balancer VIPs and NAT IPs as `/32` host routes.
- Size the IP block to accommodate the expected peak number of concurrent Load Balancer VIPs and NAT IPs across all organizations that share this IP block.
- After assigning the IP block to an organization, set per-organization quotas to limit VIP and NAT IP consumption.

**Important:**

The IP ranges used for Load Balancer VIPs and NAT IPs must be defined as external IP blocks in NSX Manager before you create the IP block in VCF Automation. Verify that these ranges are reachable from the Border Gateway before assigning them to an organization.

1. Log in to the VCF Automation Provider Management UI.
2. In the **Infrastructure** section of the left-hand navigation menu, select **Networking** > **External IP Blocks**.
3. Click the **New** button.

   The **Create External IP Blocks** wizard opens.
4. In the **Configure General Settings** tile, provide the general configuration for the IP block.
   a) Provide a name and, optionally, a description for the IP block.
   b) From the **Region** drop-down menu, select a region for the IP block.
   c) (Optional) Activate the **Hide the External IP Block details from organizations** toggle.
   d) Click **Next**.
5. In the **IP Addresses** tile, define the IP block scope and click **Next**.

   The internal scope of an IP block is defined by all CIDRs and IP ranges that represent the exact span of IP addresses that contains all ranges and blocks.

| Option | Description |
|---|---|
| **Option** | Steps |
| **Use an external IPAM** | 1. Activate the **Use External IPAM** toggle.<br>2. Select a CIDR from the drop-down menu.<br>**Note:**<br>The selected Infoblox network container maps to the external IP block that you are creating.<br>3. Click **Next**. |
| **Do not use an external IPAM** | 1. Enter at least one CIDR or IP address range and click **Add**.<br>**Note:**<br>In one IP block, you can use both CIDRs and IP adress ranges.<br>You can enter up to 10 CIDRs and 10 IP ranges.<br>2. (Optional) Enter IP ranges to exclude from the block.<br>**Note:**<br>The **Reserved IP Ranges** field specifes the IP ranges that are excluded from the pool of IP addresses that is available to the organizations.<br>3. Click **Next**. |

6. (Optional) In the **Default Quota Settings** tile, configure the quota defaults for the IP block.
   a) Enter a default maximum number of single IPs.

      Instead of providing a specific number, you can select the **Unlimited** check-box.
   b) Enter a default maximum number of CIDRs for all organizations within the IP bock.

      Instead of providing a specific number, you can select the **Unlimited** check-box.
   c) Specify the default largest subnet size of the CIDR that you want to allow within the IP block.

      This option ensures that you do not allocate subnets that are too large, which can result in unused IP addresses. However, you can also select the **No Restriction** check-box, if you do not want to impose restrictions on the subnet size.

d) Click **Next**.

7. In the **Review and Complete** tile, review the IP block settings, and click **Create**.

The new IP block appears in the list of external IP blocks.

You can configure an IP quota that applies equally to all organizations. You can also specify a quota applicable to a specific organization, see Edit the IP Quota Settings for an IP Block.

## Edit the General Settings of an IP Block

You can use the VCF Automation Provider Management UI to modify the name, description, scope, IP ranges, and IP prefixes of an existing IP block.

Verify that your role includes the **System IP Spaces: Manage** and **System IP Spaces: View** rights.

1. Log in to the VCF Automation Provider Management UI.
2. In the **Infrastructure** section of the left-hand navigation menu, select **Networking** > **External IP Blocks**.
3. Click the name of the IP block that you want to edit.
4. In the secondary left-hand menu for the IP block, click **General** and edit any of the general IP block settings.
   - In the **Summary** section, click **Edit** to edit the name and, optionally, the description of the IP block.
   - In the **External IP Block Addresses** section, click **Edit** to edit the CIDRs, IP ranges, and the reserved IP ranges within the IP block.
5. (Optional) You can view the external connections that are associated with the selected IP block.
   - In the secondary left-hand menu for the IP block, click **Centralized Connections** to view the centralized connections that use the selected IP block.
   - In the secondary left-hand menu for the IP block, click **Distributed Connections** to view the distributed VLAN connections that use the selected IP block.
6. To edit the default quota of the IP space, from the secondary left pane, select **Quotas**, and click **Edit**.
   a) Enter the maximum number of IP addresses for each organization.
   b) Enter the maximum number of CIDRs for each organization.
   c) Specify the largest subnet size that you want to allow within the IP Space.

      This option ensures that you do not allocate subnets that are too large, which can result in unused IP addresses.

## Edit the IP Quota Settings for an IP Block

For any IP block, you can update the default IP quota that applies equally to all organizations that use the respective IP block through the assignment of an external connection. Additionally, you can create a custom IP quota, which is applicable only to a specific organization.

- Verify that your role includes the **System IP Spaces: Manage** and **System IP Spaces: View** rights.
- Verify that you have IP blocks created, see Create an External IP Address Block.
- Configure the Regional Networking of a VCF Automation Organization

1. Log in to the VCF Automation Provider Management UI.
2. In the **Infrastructure** section of the left-hand navigation menu, select **Networking** > **External IP Blocks**.
3. Click the name of the IP block to which you want to add an IP quota.
4. In the secondary left-hand menu for the IP block, click **Quotas** and select your task.

| Option | Description |
|---|---|
| **Edit the default IP quota** | 1. In the **Default IP Quota Settings** section, click **Edit**.<br>2. In the **Edit Default IP Quota** dialog box, update the default IP Quota values.<br>a. Enter the maximum number of singe IPs or select the **Unlimited** check box. |

| Option | Description |
| --- | --- |
| | b. Enter the maximum number of CIDRs or select the **Unlimited** check box.<br>c. From the **Maximum Subnet Size** drop-down menu, select a new size of the CIDR that you want to allow within the IP block or select the **No Restriction** check box.<br>**Note:**<br>This option ensures that you do not allocate subnets that are too large, which can result in unused IP addresses.<br>d. Click **Save**. |
| **Create a custom IP quota** | 1. In the **Custom IP Quota Settings** section, click **New**.<br>2. In the **Create Custom Quota** dialog box, configure the custom IP quota settings.<br>a. From the **Organization** drop-down menu, select the organization to which you want to apply the IP quota.<br>b. Enter the maximum number of IP addresses for the organization or select the **Unlimited** check box.<br>c. Enter the maximum number of CIDRs for the organization or select the **Unlimited** check box..<br>d. From the **Maximum Subnet Size** drop-down menu, select a new size of the CIDR that you want to allow within the IP block or select the **No Restriction** check box.<br>**Note:**<br>This option ensures that you do not allocate subnets that are too large, which can result in unused IP addresses.<br>e. Click **Save**. |

# Managing External Connections in VCF Automation

As a VCF Automation provider administrator, you configure external connections to establish external network connectivity for organizations, for example to the Internet or the data center. After you create an external connection, you assign it to one or more organizations.

External connections can be of the following types:

- Centralized connections, which leverage VCF networking Tier-0 or Tier-0 VRF gateways, and associate them with IP addresses from IP blocks that can be advertised from those gateways.
- Distributed VLAN connections, which leverage VLANs.
- Distributed VXLAN connections, which integrate with the physical Ethernet VPN (EVPN)-VXLAN fabric through a special cluster called a Route Controller, see Create a Route Controller.

  **Important:**

  Distributed VXLAN connections are available for version 9.1.1 and later.

## Centralized Connections

A centralized external connection in VCF Automation is a Tier-0 gateway integrated with an IP block to manage external connectivity for your organizations. Centralized connections are backed by existing Tier-0 or Tier-0 VRF gateways in either in `Active-Active` or in `Active-Standby` mode. Associating IP blocks with a centralized connection provides control over default service configurations, routing rules, and the dynamic routing configuration. Alternatively, you can allow organizations to advertise their own private transit gateway IP blocks, in which case the centralized connection does not require an IP block.

Centralized connections use edge clusters to get the full set of stateful networking services, including NAT, load balancing, IPSec VPN, and gateway firewall.

> **Important:**
>
> Starting with version 9.1.1, VCF Automation supports VLAN-backed regions. You cannot use centralized connections for organizations in VLAN-backed regions.

#### Distributed VLAN Connections

Using distributed external connections provides organizations with VLAN-based external connectivity and ensures high-performance distributed routing. Distributed external connections link the organization's transit gateway directly to a VLAN in the physical infrastructure.

You can add multiple distributed VLAN connections to a single organization configured with a distributed transit gateway. Also, you can assign a single distributed VLAN connection to multiple organizations. You can also create a dedicated VLAN connection, which an organization administrator can use to create a VLAN extension subnet directly in their organization.

#### Distributed VXLAN Connections

Starting with VCF Automation 9.1.1, you can create a distributed VXLAN connection in the VCF Automation Provider Management UI. By configuring and using a distributed VXLAN connection, workload VM traffic is routed directly to gateways in the physical EVPN fabric, eliminating the need for a VLAN spanning the entire physical network.

Distributed VXLAN connections support advanced networking services, such as NAT, Load Balancer VIPs, and Auto SNAT. These services are delivered through a VNA cluster that you select during organization creation or while setting up the regional networking of an organization. External IP blocks serve as the source pools for Load Balancer VIPs and SNAT/DNAT IPs.

Use this type of external connection in environments with an existing EVPN-VXLAN fabric that requires direct VM-to-physical-network routing without edge nodes.

For more information about distributed VXLAN connections and the Route Controller cluster, see Distributed VXLAN Connectivity.

#### Assigning External Connections to Organizations

After you create an external connection, you assign it to one or multiple existing organizations through the regional networking settings. For more information, see Setting Up Regional Networking for an Organization in VCF Automation.

An organization can use only one connection type per region — centralized, distributed VLAN, or - for version 9.1.1 only - distributed VXLAN. Mixing connection types within a single region is not supported.

## Create a New Centralized Connection

Centralized connections define external network connectivity through NSX Tier-0 or Tier-0 VRF gateway. They provide connectivity to designated remote networks or a default gateway and facilitate the advertisement of subnets and IPs from IP blocks associated with this connection. Centralized connections enable a full set of stateful networking services within organizations, such as NAT, load balancing, IPSEC VPN, and Gateway Firewall.

- Create a Region in Your VCF Automation
- Verify that you have a Tier-0 gateway or Tier-0 VRF gateway created in NSX Manager. For instructions on creating a Tier-0 gateway, see Add an NSX Tier-0 Gateway. For instructions on creating a Tier-0 VRF gateway, see Step 4 in Deploy an NSX VRF-Lite with BGP.

**Note:**

If you want to add a tier-0 gateway that is backed by a VRF gateway in NSX, you must also create a VRF gateway that is linked to the tier-0 gateway.

- Verify that you have external IP blocks in your environment, see Managing IP Blocks in VCF Automation.

You can create a centralized connection to serve as an organization's default connection for all external traffic and connection to the Internet. You can also create and assign additional centralized connections to the organization, for specific pruposes, such as providing access to a CIDR hosting shared services.

When you create a centralized external connection, you can configure a Source NAT rule directly for the connection. You can also define inbound remote networks to establish external reachability for the connection. You can create a dedicated centralized connection, which allows the organization that uses the connection to advertise its own transit gateway IP blocks up to the connection.

1. Log in to the VCF Automation Provider Management UI.
2. In the **Infrastructure** section of the left-hand navigation menu, select **Networking** > **External Connections**.
3. Click the **Centralized Connections** tab and click **New**.
4. In the **Configure General Settings** tile, provide the general configuration for centralized connection.
   a) Enter a name and, optionally, a description for the new connection.
   b) Select a region from the drop-down menu.
   c) Click **Next**.
5. In the **Select T0 Gateway** tile, select the Tier-0 gateway for the centralized connection and click **Next**.

   You can only see a list of the available gateways to choose from. The Tier-0 gateways are pre-created and configured in NSX.
6. In the **Associated External IP Blocks** tile, select one or multiple IP blocks to become available to the organization after the connection is associated with an organization, and click **Next**.
   - Select the **Allow Organization to advertise their Private TGW networks through the Connection** check box to create a private centralized connection that you do not want to share to multiple organizations.

     **Important:**

     Selecting this check box means that the organization that you assign that centralized connection to can advertise private transit gateway IP blocks through that connection. If you decide to allow the organization to advertise its private TGW networks, you can skip selecting an IP block from the list.
   - Select the **Insert Provider Outbound NAT on Centralized Connection** check box to use a provider NAT directly on the connection.

     **Note:**

     When you select that option, the transit gateway associated with that connection uses a separate NAT gateway.
7. (Optional) In the **Provider Outbound NAT** tile, select an IP block for the Provider Outbound NAT and click **Next**.
   - Select the **Enable Logging** check box to enable logging for the Provider Outbound NAT.

   This tile appears only if you select the **Insert Provider Outbound NAT on Centralized Connection** check box at Step 6.

   The IP block that you select for the Provider outbound NAT can be different from the IP block that you select for the connection at Step 6.
8. (Optional) In the **Remote Networks** tile, specify the CIDRs for the remote networks for the centralized connection.

   You can have multiple centralized connections associated with a single transit gateway. In such cases, you must specify the CIDRs to be used for external networks for that particular connection.

If you only assign one connection to a transit gateway, you do not have to specify CIDRs for remote networks for that connection. As you start adding more connections to the same transit gateway, you must ensure that the CIDRs for the remote networks for the different connections do not overlap.

9. In the **Review and Complete** tile, review the settings for the new distributed connection and click **Create**.

- To edit the name and description of the centralized connection or to view the connected transit gateways, see Edit a Centralized External Connection.
- To delete a centralized connection, follow the steps.
  a. Verify that no transit gateways are connected to the provider gateway.
  b. On the left of the name of the provider gateway you want to delete, click the vertical ellipsis ( ⋮ ), and click **Delete**.

## Create a New Distributed VLAN Connection

By creating a distributed VLAN connection and then assigning it to your organizations, you can use your existing VLAN infrastructure to provide external connectivity to your organizations. In the distributed connectivity model, to provide stateful network services, such as Load Balancer, you must have a VNA cluster set up.

- Create a Region in Your VCF Automation.
- Verify that you have a VLAN configured in your infrastructure.
- Verify that you have external IP blocks created, see Managing IP Blocks in VCF Automation.

Creating distributed connections allows you to connect workloads to networks already existing in the datacenter, for instance on a VLAN. A distributed VLAN connection can be assigned to multiple organizations. You can also create a dedicated distributed VLAN connection, which you assign to a single organization. When you dedicate a distributed VLAN connection to a single organization, the organization administrator can use the dedicated connection and create VLAN extension subnets directly within the organization. This option gives you more flexibility in the way you design and configure your network infrastructure.

So, the distributed VLAN connection operates in three modes:

- You use overlay connections and VLAN to advertise the external traffic.
- You create a VLAN subnet and then share it with your organizations.
- You create a dedicated VLAN connection to assign to a single organization, whose organization administrator can subsequently create a VLAN subnet in the organization VPC.

1. Log in to the VCF Automation Provider Management UI.
2. In the **Infrastructure** section of the left-hand navigation menu, select **Networking** > **External Connections**.
3. Click the **Distributed Connections** tab and click **New**.
4. In the **Configure General Settings** tile, provide the general configuration for distributed connection.
   a) Enter a name and, optionally, a description for the new connection.
   b) Select a region from the drop-down menu.
   c) Click **Next**.
5. In the **Connection details** tile, enter the VLAN ID and the gateway address in a CIDR format, and click **Next**.
6. (Optional) Activate the **Create a dedicated VLAN** toggle to create a VLAN connection that can connect to a single VPC subnet for bridging.
7. In the **Associate External IP Block** tile, select the IP block to use with that distributed connection and click **Next**.

   The IP blocks are filtered based on the gateway CIDR that you sepcify at Step 5. The gateway CIDR 5 must be part of the selected IP block.

   The selected IP block is used exclusively for the newly created distributed connection. You cannot use the same IP block to create another distributed connection.
8. In the **Review and Complete** tile, review the settings for the new distributed connection and click **Create**.

The new distributed connection appears in the list of connections.

Assign the connection to an organization, see Configure the Regional Networking of a VCF Automation Organization.

## Create a New Distributed VXLAN Connection for Version 9.1.1

Starting with version 9.1.1, you can create a distributed VXLAN connection and then assign it to your organizations. Distributed VXLAN connections enable you to use your existing physical EVPN-VXLAN fabric. By using a distributed VXLAN connection, workload VM traffic routes directly to gateways in the physical EVPN fabric, eliminating the need for a VLAN spanning the entire physical network.

- Enable support for advanced networking services for distributed VXLAN connections in NSX Manager, see Enable Advanced Networking Services for Distributed VXLAN Connections.
- Verify that a Route Controller is created in NSX Manager, see Create a Route Controller.
- Verify that the Route Controller cluster VIP is reachable from your BGP EVPN border gateway routers.
- If you require NAT or Load Balancer services over the VXLAN connection, verify that a VNA cluster is available in the target region.

  **Important:**

  vSphere Kubernetes Service (VKS) requires a Load Balancer.

Unlike a distributed VLAN connection, which maps to a specific physical VLAN, a VXLAN connection uses a Layer 3 VNI to extend connectivity to an EVPN Border Gateway (BGW). The connection integrates with the physical EVPN-VXLAN fabric through a Route Controller. When you create a distributed VXLAN connection, you specify the Layer 3 VNI, Route Target (RT), and Route Distinguisher (RD) that identify the tenant VRF on the Route Controller for BGP EVPN advertisements. For more information about VXLAN connections and the Route Controller cluster, see Distributed VXLAN Connectivity.

Distributed VXLAN connections support the following advanced networking services:

- NSX Load Balancer VIPs
- Avi Load Balancer in `Active-Standby` mode
- Centralized NAT

If you want to enable an organization to use advanced networking services over a VXLAN external connection, you must assign a VNA cluster to the organization at the time of setting up the regional networking for the organization. Also, you must configure an external IP block for the load balancer VIPs and NAT IPs. For more information about VNA clusters, see Managing Edge and VNA Clusters in VCF Automation. For more information about external IP blocks, see Managing IP Blocks in VCF Automation.

You can add multiple distributed VXLAN connections to a single organization configured with a distributed transit gateway. Each organization maps to a separate tenant VRF on the Route Controller, identified by the Layer 3 VNI.

1. Log in to the VCF Automation Provider Management UI.
2. In the **Infrastructure** section of the left-hand navigation menu, select **Networking** > **External Connections**.
3. Click the **Distributed VXLAN Connections** tab and click **New**.

   **Important:**

   If support for distributed VXLAN connection is not enabled in NSX Manager, you get an error at Step 3.

4. In the **Configure General Settings** tile, provide the general configuration for distributed connection.
   a) Enter a name and, optionally, a description for the new connection.
   b) Select a region from the drop-down menu.
   c) Click **Next**.
5. In the **Select Route Controller** tile, select a Route Controller from the list and click **Next**.
6. In the **Connection Details** tile, enter the VXLAN details and click **Next**.

| Parameter | Description | Example |
|---|---|---|
| EVPN L3 VNI | The Layer 3 VNI used to identify the organization VRF in the EVPN fabric. Must match the Layer 3 VNI on the physical EVPN gateway for this organization. | 65000 |
| Route Distinguisher | Uniquely identifies the Route Controller cluster within the EVPN domain. Each distributed VXLAN connection must use a unique route distinguisher value. | 65000:20 |
| Import Route Target | BGP extended community that determines which EVPN routes the RC accepts from the physical gateway. Must match the export route target configured on the gateway side. | 65000:30 |
| Export Route Target | BGP extended community appended to routes the Route Controller advertises to physical gateways. Must match the import route target on the gateway side. | 30:20 |

7. In the **Associate External IP Blocks** tile, select an IP block for the connection.

   The organizations using that connection use IP addresses from the pool of IPs defined by the selected IP block.

   The selected IP block is used exclusively for the newly created distributed connection. You cannot use the same IP block to create another distributed connection.
8. In the **Review and Complete** tile, review the settings for the new distributed connection and click **Create**.

The new distributed VXLAN connection appears on the **Distributed VXLAN Connections** tab.

Assign the connection to an organization, see Configure the Regional Networking of a VCF Automation Organization.

## Edit a Centralized External Connection

For every centralized external connection, you can edit settings, such as its name and description. You can also associate the connection with an external IP block, modify the provider outbound NAT configuration, and specify the remote networks for the connection. Also, you can view all transit gateways that are associated with the distributed connection.

- Verify that you have a centralized external connection created, see Create a New Centralized Connection.
- Verify that you have IP blocks created, see Managing IP Blocks in VCF Automation.

1. Log in to the VCF Automation Provider Management UI.
2. In the **Infrastructure** section of the left-hand navigation menu, select **Networking** > **External Connections**.
3. On the **Centralized Connections** tab, select a connection from the list.
4. Select your task.

| Task | Steps |
|---|---|
| **Edit the general settings of the selected external connection** | 1. From the left-hand navigation menu for the selected external connection, click **General.**<br>2. Under **Summary**, click **Edit**.<br>3. In the **Edit Centralized Connection** dialog box, edit the name or the description of the external connection. |

| | |
|---|---|
| | 4. Click **Save**. |
| **Associate the external connection with an external IP block** | 1. From the left-hand navigation menu for the selected external connection, click **IP Blocks.**<br>2. Under **IP Blocks**, click **New**.<br>3. In the **Associate External IP Blocks** dialog box, select an IP block from the list.<br>4. Click **Save**. |
| **Edit the Provider Outbound NAT configuration** | 1. From the left-hand navigation menu for the selected external connection, click **IP Blocks**.<br>2. Under **Provider Outbound NAT**, click **Edit**.<br>3. In the **Edit NAT Configuration** dialog box, modify the existing NAT configuration for the connection.<br>a. If NAT is not enabled for the external connection, activate the **Enabled** toggle.<br>b. Select an external IP block from the list.<br>c. (Optional) Enable logging by activating the **Logging** toggle.<br>d. Click **Save**. |
| **Modify the remote networks that the external connection can reach** | 1. From the left-hand navigation menu for the selected external connection, click **Remote Networks**.<br>2. Click **Edit**.<br>3. In the **Edit Remote Networks** dialog box, enter a CIDR and click **Add**.<br>**Note:**<br>Repeat the step to add multiple CIDRs.<br>4. Click **Save**. |
| **View the transit gateways that are connected to the external connection** | 1. From the left-hand navigation menu for the selected external connection, click **Connected TGWs**.<br>**Note:**<br>You see a list of all transit gateways the selected external connection is associated with. For each trasit gateway, you also see the organization to which it belongs. |

## Edit a Distributed VLAN Connection

You can see and modify the general settings, such as name, description, and VLAN details, for every distributed external connection in your environment. Also, you can view all transit gateways that are associated with a distributed connection.

- Verify that you have a distributed VLAN external connection created, see Create a New Distributed VLAN Connection.
- Verify that you have IP blocks created, see Managing IP Blocks in VCF Automation.

1. Log in to the VCF Automation Provider Management UI.
2. In the **Infrastructure** section of the left-hand navigation menu, select **Networking** > **External Connections**.
3. On the **Distributed Connections** tab, click a connection from the list.
4. Select your task.

| Task | Steps |
|---|---|
| **Edit the general settings of the selected external connection** | 1. From the left-hand navigation menu for the selected external connection, click **General.**<br>2. Under **Summary**, click **Edit**.<br>3. In the **Edit Distributed Connection** dialog box, edit the name or the description of the external connection.<br>4. Click **Save**. |
| **Edit the VLAN details for the selected external connection** | 1. From the left-hand navigation menu for the selected external connection, click **General.**<br>2. Under **VLAN Details**, click **Edit**.<br>3. In the **Edit Distributed Connection** dialog box, enter a VLAN ID and a getaway CIDR IP address<br>4. Click **Save**. |
| **View the transit gateways associated with the selected external connection** | From the left-hand navigation menu for the selected external connection, click **Connected TGWs**.<br>**Note:**<br>You see a list of all transit gateways the selected external connection is associated with. For each transit gateway, you also see the organization to which it belongs. |

## Edit a Distributed VXLAN Connection for Version 9.1.1

For each VXLAN connection in your environment, you can view general settings, such as name, description, region, and you can also modify the VXLAN details or change the Route Controller for the connection.

- Verify that you have a distributed VXLAN external connection created, see Create a New Distributed VXLAN Connection for Version 9.1.1.
- Verify that you have IP blocks created, see Managing IP Blocks in VCF Automation.

1. Log in to the VCF Automation Provider Management UI.
2. In the **Infrastructure** section of the left-hand navigation menu, select **Networking** > **External Connections**.
3. On the **Distributed VXLAN Connections** tab, click a connection from the list.
4. Select your task.

| Task | Steps |
|---|---|
| **View the general settings of the selected external connection** | 1. From the left-hand navigation menu for the selected external connection, click **General.**<br>2. Under **Summary**, you can view the following information about the selected external connection: name, description, status, region, and type. |
| **Edit the VXLAN details for the selected external connection** | 1. From the left-hand navigation menu for the selected external connection, click **General.**<br>2. Under **VXLAN Details**, click **Edit**.<br>3. In the **Edit Distributed VXLAN Connection** dialog box, modify any of the parameters of the distributed VXLAN connection.<br>4. Click **Save**. |
| **Select a new Route Controller for the selected external connection** | 1. From the left-hand navigation menu for the selected external connection, click **General.**<br>2. Under **Route Controller Details**, click **Edit**.<br>3. In the **Edit Distributed VXLAN Connection** dialog box, select a Route Controller from the list.<br>4. Click **Save**. |
| **View or modify the IP blocks associated with the selected external connection** | 1. From the left-hand navigation menu for the selected external connection, click **IP Blocks.**<br>2. To associate a new IP block with the connection, click **New**.<br>3. To remove an existing association between an IP block and the connection, click the vertical ellipsis on the left of the IP block name and click **Remove Association** |
| **View the transit gateways associated with the selected external connection** | From the left-hand navigation menu for the selected external connection, click **Connected TGWs**.<br>**Note:**<br>You see a list of all transit gateways that the selected external connection is associated with. For each transit gateway, you also see the organization to which it belongs. |

# Managing Edge and VNA Clusters in VCF Automation

Depending on the type of external connection that you want to add to an organization, you specify and use an edge or VNA cluster to provide stateful networking services to your organization.

Both edge clusters and VNA clusters are NSX constructs. You do not configure or provision edge or VNA clusters within VCF Automation. VCF Automation simply discovers the available edge and VNA clusters for your environment and lists them in the VCF Automation Provider Management UI.

An edge cluster is a grouping of multiple NSX Edge transport nodes that host Tier-0 and Tier-1 gateways. A VNA cluster is a collection of edge-based virtual network appliances that deliver edge services to VPCs connected with a distributed transit gateway by enabling service-based traffic redirection.

When you assign a centralized connection to an organization, you must specify an edge cluster to provide stateful networking services.

When you assign a distributed VLAN or VXLAN connection to an organization, you specify a VNA cluster to provide stateful networking services, such as NAT and load balancer.

**Important:**

NAT is available only in full-stack regions. In VLAN-backed environments, VNA clusters provide load balancer services only. For more information about full-stack and VLAN-backed regions, see Managing Regions in Your VCF Automation.

Starting with version 9.1.1, the VNA cluster also hosts the Service VMs that process north-south traffic for advanced networking services for organizations that use EVPN-VXLAN distributed external connections. When a workload VM in the VPC must be reached through a load balancer VIP or NAT IP from outside the data center, traffic is routed by the BGW directly to the ESX host in the VNA cluster where the active Service VM runs.

For more information about edge and VNA clusters, see Create an NSX Edge Cluster and Add Edge Nodes and Create a Virtual Network Appliance Cluster.

## View Details About Edge and VNA Clusters in the VCF Automation Provider Management UI

You can view a list of the available edge and VNA clusters, and you can verify that they are synchronized with NSX.

The list of edge and VNA clusters appears automatically in the VCF Automation Provider Management UI when you register an NSX Manager. To modify an edge or VNA cluster, you must use NSX and then synchronize VCF Automation with NSX.

For each edge cluster, you can view the following information:

- Cluster health
- Number of nodes
- Average CPU and memory usage
- Number of organizations that use the respective cluster
- Number of VPCs
- Cluster type
- The region where the cluster can be used

For each VNA cluster, you can view the following information:

- Cluster health
- Number of nodes
- Number of organizations that use the respective VNA cluster
- The region where the cluster can be used

1. Log in to the VCF Automation Provider Management UI.
2. In the **Infrastructure** section of the left-hand navigation menu, select **Networking** > **Edge/VNA Clusters**.

   In the **Edge and VNA Clusters** pane, you see two tabs, **Edge Clusters** and **VNA Clusters**.

   On the **Edge Clusters** tab, you see a list of all edge clusters in your environment together with essential information about each edge cluster.

   On the **VNA Clusters** tab, you see a list of all VNA clusters in your environment together with essential information about each VNA cluster.
3. Select your task.

| Option | Description |
|---|---|
| **View details about an edge cluster** | 1. Click the **Edge Clusters** tab.<br>2. Click a cluster from the list.<br>3. In the left-hand navigation for the selected cluster, click **General** to view general information about the cluster and information about its default QoS configuration.<br>**Note:**<br>You can edit the default QoS configuration for the cluster, see Edit the Default QoS Configuration of an Edge Cluster. |

| Option | Description |
| --- | --- |
| | 4. Click **Node Health** to view all transport nodes associated with the selected cluster, together with information about node health and resource consumption. |
| **View details about a VNA cluster** | 1. Click the **VNA Clusters** tab.<br>2. Click a cluster from the list.<br>3. In the left-hand navigation for the selected cluster, click **General** to view general information about the cluster.<br>4. Click **Node Health** to view all transport nodes associated with the selected cluster, together with information about node health and resource consumption. |

4. (Optional) To start synchronization with NSX, click the **Sync** button.

### Edit the Default QoS Configuration of an Edge Cluster

Using the VCF Automation Provider Management UI , you can set limits on the traffic rates for the organizations that use a selected edge cluster. The Quality of Service (QoS) profile applies to the Virtual Private Cloud (VPC) traffic in the VPCs deployed on the selected edge cluster.

Because the default QoS profile is used by all organizations that are associated with the selected edge cluster, any change to the values of the default profile might affect multiple organizations.

To set limitations, you can specify the permitted information rate and the burst size for both the ingress and egress traffic. Any traffic that does not conform to the QoS policy, is dropped. QoS profiles can be set for all traffic types, for example, unicast, BUM, IPv4.

1. Log in to the VCF Automation Provider Management UI.
2. In the **Infrastructure** section of the left-hand navigation menu, select **Networking** > **Edge/VNA Clusters**.
3. Click the **Edge Clusters** tab.
4. Select a cluster from the list and click its name.
5. In the left-hand menu for the selected cluster, click **General**.
6. In the **Default QoS Configuration** section, click the **Edit** button and make the desired modifications in the dialog box that opens.
   a) Deselect the **Unlimited** check box for any of the fields whose value you want to modify.

      The **Unlimited** check box is selected by default.
   b) Enter the desired committed bandwidth limits for the ingress and egress traffic.
   c) Enter the desired burst size for the ingress and egress traffic.
   d) Click **Save**.

## Managing VLAN Extension Subnets in VCF Automation

As a provider administrator, you can use the VCF Automation Provider Management UI to create a VLAN extension subnet and share that subnet with different organizations in your environment.

When you create a subnet for the first time, the provider consumption organization, if not already enabled, is automatically enabled and configured with networking. The provider consumption organization is a provider-managed organization that is used for storing and managing provider resources that can be shared with all the organizations in your VCF Automation instance. You can enable the `Provider Consumption Org` feature flag to see the provider consumption organization in the list of your organizations and use it for sharing content. If you do not enable the `Provider Consumption Org` feature flag, the provider consumption organization is enabled, but you cannot access it and cannot use it. For more information about the provider consumption organization, see Create a Provider Consumption Organization in VCF Automation.

Starting with version 9.1.1, VCF Automation supports VLAN-backed environments. In VLAN-backed environments, VLAN extension subnets have an expanded role. Because private subnets are not available in VLAN-backed deployments, VLAN extension subnets are the mechanism that gives Supervisor control-plane VMs and workloads direct Layer 2

connectivity to the physical network. Organization administrators can use VLAN extension subnets as the default network for PodVMs or VM Service VMs in a namespace, replacing the auto-created private subnet that is used in a full-stack environment.

### Create a VLAN Extension Subnet

You can create a VLAN extension subnet that you can share with the organizations in your environment.

When you create a subnet, a dedicated VLAN connection and a VPC are automatically created for the provider consumption organization. The created subnet consumes the VLAN connection in that VPC.

1. Log in to the VCF Automation Provider Management UI.
2. In the **Infrastructure** section of the left-hand navigation menu, select **Networking** > **Subnets**.
3. Click the **New** button.

   The **New Subnet** wizard opens.
4. In the **Configure General Settings** tile, provide the general configuration for the new subnet.
   a) Provide a name and, optionally, a description for the subnet
   b) From the **Region** drop-down menu, select a region for the subnet.
   c) Click **Next**.
5. In the **VLAN Details** tile, define the VLAN specifics.
   a) Enter a VLAN ID.
   b) Enter a Gateway CIDR for the VLAN.
   c) Click **Next**.
6. In the **Review and Complete** tile, review the subnet settings and click **Create**.

The new subnet appears in the list of available subnets in the VCF Automation Provider Management UI. Creating a subnet automatically creates several other resources that you can see in the VCF Automation Provider Management UI:

- A distributed external connection with the name `shared-subnet-<subnet_name>`
  The distributed connection is dedicated to the subnet. You cannot directly delete the distributed connection.
- An external IP block with the name `shared-subnet-<subnet_name>`
  The external IP block is also fully dedicated to the newly created subnet. You cannot directly delete that external IP block.

Share the VLAN subnet with an organization, see Share a Subnet With a VCF Automation Organization.

## Modify the Default Private IP Blocks for the Default VPCs and Transit Gateways of VCF Automation Organizations

You can modify the default system-generated values for the IP blocks for the default VPC and the default transit gateway, which get created when you complete configuring the regional networking for an organization.

> **Important:**
>
> Starting with version 9.1.1, VCF Automation supports VLAN-backed deployments. The following task is not relevant to organizations in VLAN-backed regions. For organizations assigned to VLAN-backed zones, the private IP block configuration is not available in the UI. So, if you do not see the private IP block options for an organization, check the network stack of the zones assigned to that organization.

1. Log in to the VCF Automation Provider Management UI.
2. In the **Infrastructure** section of the left-hand navigation menu, select **Administration** > **General Settings**.
3. In the **Networking** section, click **Edit**.

   The **Edit Networking** dialog box opens.

4. Enter the new default values for the private IP blocks for VPCs and Transit Gateways.
5. Click **Save**.

The new IP blocks that you enter become the default ones for VPCs and Transit Gateways. You can override those default values when you configure the regional networking for an organization, see Configure the Regional Networking of a VCF Automation Organization.

# Managing VCF Automation Organizations

You can use the VCF Automation Provider Management UI to create, configure, and manage VCF Automation organizations.

An organization is a top-level entity used to group and manage resources, users, policies, IaaS services, and catalog entities, while maintaining a secure boundary from other organizations. It offers a centralized structure for billing, access control, and resource sharing, facilitating collaboration and governance across teams and departments.

Use VCF Automation Provider Management UI to manage organizations, set quotas and VM and storage classes to determine how users consume resources that are allocated to an organization.

## Create a VCF Automation Organization for All Applications

From the VCF Automation Provider Management UI, you can create an organization using the underlying vSphere Supervisor infrastructure. The organizations for all applications are the primary multi-tenancy model within VCF Automation.

- Verify that your vCenter instance has a Supervisor enabled.
- If you want to assign a custom role to the first user of the organization, create a custom role. See Create a Global Role in Your VCF Automation.
- If you want to add infrastructure policies to the region quota when creating an organization, verify that you have at least one infrastructure policy in your environment. See Create an Infrastructure Policy in VCF Automation.
- When organization types share a vCenter and NSX Local Manager pair between the two organization types, you must create the organization for VM Apps before the organization for All Apps. You must also configure the organization for VM Apps, including mapping the vCenter and NSX Manager cloud accounts, before you configure a VCF Automation region that uses the same vCenter and NSX Manager.
  During the upgrade from VMware Aria Automation to VCF Automation, the legacy environment becomes an organization for VM Apps. This converted organization for VM Apps deploys as the first organization in the new VCF Automation instance.

Organizations for all applications use vSphere Namespaces within vSphere Supervisor to abstract the users and groups of the organizations from the underlying vSphere and NSX infrastructure. Within an **Organization for All Apps**, there is further isolation at the project and namespace levels. You can use this organization to run a wide variety of applications, including virtual machines (VMs), Kubernetes, data services, and so on. You can also create multiple tenants with secure infrastructure isolation, quotas, and chargeback.

> **Note:**
>
> Using the VCF Automation Provider Management UI, you can create only one organization for VM applications in your VCF Automation. After one organization for VM applications is created, the **Create Organization** wizard skips the step for selecting an organization type and continues with the wizard for the creation of an organization for all applications.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click **Create Organization**.
4. If applicable, click **Organization for All Apps**, and click **Next**.
5. In the **General Details** tile, enter a name, and optionally, a description of the organization.

   The name is the unique identifier that forms the URL for accessing the organization. You can use only alphanumeric characters, underscores, periods, commas, brackets, and the dollar sign - `_ . ( ) , $`.

6. Click **Create and Continue**.

   After you click **Create and Continue**, you cannot cancel the creation of the organization. If you want to delete the organization, you must disable it before you can delete it. If you want to continue setting up the organization later, you can close the wizard.

7. In the **Assign a Region Quota** tile, select a region from the drop-down menu.

   After you assign the region and region quota and continue, you cannot change the region. To change the region, you must delete the organization and create a new one.

8. Select how much of the region capacity you want to share with the organization.

| Option | Description |
|---|---|
| **Share the entire capacity with this organization** | When you create a full allocation region quota, it includes all available zones, memory limits, CPU limits, storage classes, and VM classes of the region. Any further changes to the region automatically apply also to the region quota. By default, full allocation region quotas include zero memory and CPU reservations for each zone. You can assign CPU and memory reservations as a Day 2 operation on a full allocation region quota. You cannot overprovision CPU and memory reservations from a region.<br>You can share the full capacity of the region with more than one organization. |
| **Share a portion of the capacity with this organization** | Select this option to share defined resources with the organization. |

9. If you selected **Share the entire capacity with this organization**, select one or more infrastructure policies for the region quota.
   a) Click **Assign and Continue**.
   b) If your environment has any infrastructure policies, select one or more policies for the region quota, and click **Assign and Continue**.

      The list shows only infrastructure policies that are compatible with the region. You can assign both mandatory and optional infrastructure policies. Mandatory infrastructure policies are not automatically assigned to all region quotas.

10. If you selected **Share a portion of the capacity with this organization**, give the organization access to infrastructure resources.
    a) If you want to assign specific Supervisors, to the organization, turn off the **Add all available supervisors** toggle and select one or more Supervisors.
    b) Add one or more zones and assign CPU and memory limits and reservations.

       If you want to apply the same limits and reservations across all zones, select the **Add all zones and apply the same limits and reservations across all zones.** check box.
    c) Select one or more VM classes and storage classes applicable to the organization.

       The list shows all VM classes and storage classes compatible with the region. You cannot assign a guaranteed VM class to a region quota that has zero reservation. When adding storage classes, you can also edit the storage limit for the storage class.
    d) If your environment has any infrastructure policies, select one or more policies for the region quota, and click **Assign and Continue**.

       The list shows only infrastructure policies that are compatible with the region. You can assign both mandatory and optional infrastructure policies. Mandatory infrastructure policies are not automatically assigned to all region quotas.

e) Click **Assign and Continue**.

11. If applicable, add the reserved VM classes that limit vGPU use for the organization and click **Continue**.
12. In the **Configure Networking** tile, set the networking configuration for the selected region for the organization.
    - To create an organization that can consume centralized connections for its VPCs, select **Centralized Connection**.
    - To create an organization that can consume distributed VLAN connections for its VPCs, select **Distributed VLAN Connection**.
    - To create an organization that can consume distributed VXLAN connections integrated with the physical EVPN-VXLAN fabric, select **Distributed VXLAN Connection**.

    The available connection options depend on the types of clusters - edge or VNA - that are configured for your region.

    If you have only edge clusters in your region, the only type of external connection that you can configure for the organization is centralized.

    If you have only VNA clusters in your region, the only type of connection that you can configure for the organization is distributed.

    If you have both edge and VNA clusters in your region, you can choose what type of connection to configure for the organization.

    **Note:**

    In VCF Automation 9.1.1, for regions that contain VLAN-backed zones, select the **Distributed VLAN Connection** option.
13. Select a connection from the list of available external connections.
14. Configure the advanced networking settings for the selected connection type.

| Option | Description |
|---|---|
| **Connection Type** | Steps |
| **Centralized** | 1. Specify the edge cluster to use for the cluster.<br>**Note:**<br>You can use the edge cluster of the Tier-0 centralized connection. Alternatively, you can select a specific edge cluster from your environment.<br>2. Review the private IP blocks used for the default VPC and the default transit gateway that are created with the organization.<br>3. (Optional) Click **Override settings** to edit the default private IP blocks used for the default VPC and transit gateway. |
| **Distributed** | 1. Select a VNA cluster from the list.<br>2. Review the private IP blocks used for the default VPC and the default transit gateway that are created with the organization.<br>3. (Optional) Click **Override settings** to edit the default private IP blocks used for the default VPC and transit gateway.<br>**Important:**<br>For organizations assigned to VLAN-backed zones in version 9.1.1 and later, the private IP blocks are not visible and available in the UI. So, skip Step 2 and Step 3 for VLAN-backed organizations. |

15. Click **Assign and continue** to set up the regional network configuration for the organization.

    You can click **Skip** to create the organization without setting up its regional networking configuration at that stage. You can configure regional networking for the organization later, see Configure the Regional Networking of a VCF Automation Organization.

16. In the **Add First User** tile, add the first user of the organization.

    You can add only one first user to the organization. Click the **Organization Administrator** role, or a role that can manage the organization and has user and IdP configuration management rights. After you create the organization, the first user logs in to it and connects it to an identity provider.

    a) Enter a user name.
    b) Enter and confirm the user password.

       The password must be at least 15 characters long. It must contain at least one character of each of the following categories:

       - lowercase letter
       - uppercase letter
       - number
       - special character from the following list: `!"#$%&'()*+,-./:;<=>?@[]^_`{|}~")`

    c) From the drop-down menu, select the user role.

17. Click **Add User and Finish**.

Configure the Regional Networking of a VCF Automation Organization

## Create a VCF Automation Organization for VM Applications

From the VCF Automation Provider Management UI, you can create one organization using vSphere cluster capabilities which does not require a Supervisor, shared infrastructure, or tenancy isolation. This option is most suitable for VMware Aria Automation customers who want to continue leveraging the capabilities to provision VM-based applications without any disruptions to your business operations.

- Verify that the **VM Apps Organization Creation** feature flag is activated. You can manage feature flags by navigating to the **Feature Flags** page from the left navigation pane.
- Verify that your VCF Automation deployment does not have any existing organizations for VM applications.

> **Note:**
>
> Using the Provider Management UI, you can create only one organization for VM applications in your VCF Automation. When an organization for VM applications is created, the **Create Organization** wizard skips the step for selecting an organization type and continues with the wizard for the creation of an organization for all applications. You can use the VCF Automation API to create multiple organizations for VM applications. See Create an Organization.

If you are upgrading from VMware Aria Automation 8.18 and if you have multiple tenants, these tenants appear as multiple `VM App` organizations but you cannot use the VCF Automation Provider Management UI UI to create more `VM App` organizations.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click **Create Organization** and click **Organization for VM Apps**.
4. Click **Next**.
5. Enter a name, and optionally, a description of the organization.

   The name is the unique identifier that forms the URL for accessing the organization. You can use only alphanumeric characters, underscores, periods, commas, brackets, and the dollar sign - `_ . ( ) , $`.

6. Click **Create and Continue**.

   After you click **Create and Continue**, you cannot cancel the creation of the organization. If you want to delete the organization, you must disable it before you can delete it.

7. Add the first user of the organization.

   You can add only one first user to an organization. At a minimum, select the **Organization Owner** role, which grants the account the rights to manage the organization and configure users and IdP settings. After you create the organization, the first user logs in to the organization and connects it to an identity provider.

   **Important:**

   Assign the first user the **Assembler Administrator**, **Orchestrator Administrator**, and **Service Broker Administrator** roles as well. Assigning these roles ensures the first user has full access to the organization even if IdP connectivity problems occur.

   a) Enter a user name.
   b) Enter and confirm the user password.

      The password must be at least 15 characters long. It must contain at least one character of each of the following categories:

      - lowercase letter
      - uppercase letter
      - number
      - special character from the following list: `!"#$%&'()*+,-./:;<=>?@[]^_`{|}~")`
   c) From the drop-down menu, select the user role.

8. Click **Add User and Finish**.

## Create a Provider Consumption Organization in VCF Automation

As a provider administrator, you can create a provider consumption organization that you can use for easy access to the Organization UI for consuming your own infrastructure resources.

From the provider consumption organization, you can publish content to other organizations for all appliacations. For example, you can use this organization to publish blueprints and catalogs to the other organizations. As a provider administrator you can use the provider consumption organization to provision your own workloads.

The provider consumption organization does not have access to the VCF Automation Provider Management UI and cannot manage regions, global roles, rights bundles, and so on. When creating the provider consumption organization, VCF Automation configures it to use the same IdPs as the `System` organization.

1. If the `System` organization is configured to use VIDB, VCF Automation performs all the steps of updating the VIDB's client to also allow redirects to the provider consumption organization.
2. If the `System` organization is configured to use any other IdP, the configuration is incomplete and you can update the clients in their IdPs to also contain the redirect URI for the provider consumption organization.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Administration**, click **Feature Flags**.
3. Click the vertical ellipsis ( ⋮ ) next to the **Provider Consumption Org** feature flag, and select **Enable**.

In the list of organizations, a new organization with the `Provider Consumption Org` name appears.

- If you want to remove the provider consumption organization, you can turn off the feature flag to deactivate it. Deactivating the organization does not delete the provider consumption organization, but it cannot be accessed anymore through the UI or the Provider Management API. Turning the flag on makes the organization available again.
- You can change the branding of a provider consumption organization. See Changing the VCF Automation Portal Branding.
- Create a Region Quota for a VCF Automation Organization
- Configure the Regional Networking of a VCF Automation Organization
- Getting Started with Organizations in VCF Automation

# Setting Up Regional Networking for an Organization in VCF Automation

As a provider administrator, you define the external connections for all organizations within your VCF Automation instance. External connections link the organization's network to the external network. The transit gateways that are associated with provider-created external connections can be centralized or distributed.

In a single region, an organization can have either centralized transit gateways or distributed transit gateways, but not both types at the same time. An organization can use only one connection type per region — centralized, distributed VLAN, or distributed VXLAN. Mixing connection types within a single region is not supported. For information about each connection type, see Managing External Connections in VCF Automation.

### Assigning Connections to One or Multiple Organizations

You can assign a single external connection to multiple organizations, or you can assign a connection to a single organization only. The behavior of a connection that is assigned exclusively to a single organization depends on the connection type.

- Centralized connections
  You can create a centralized connection that permits an organization to advertise its private transit gateway IP blocks externally through the connection. Such a centralized connection becomes dedicated to the organization that advertises its private IP blocks and cannot be assigned to any other organizations. For more information, see Create a New Centralized Connection.
- Distributed VLAN connections
  You can create a distributed VLAN connection that is dedicated to a single organization. When you create a dedicated VLAN connection and assign it to an organization, the organization administrator can create a VLAN extension subnet within their respective organization. Afer you assign a dedicated VLAN connection to an organization, you cannot afterwards reassign the connection to any other organization. For more information, see Create a New Distributed VLAN Connection.

### Connection to the Default Transit Gateway of an Organization

When you complete setting up the regional networking configuration for an organization, a default transit gateway is automatically created in NSX. You can set up the networking configuration for the organization at the time of organization creation, see Create a VCF Automation Organization for All Applications. Alternatively, you can add a default centralized or distributed transit gateway to an organization as a Day-2 operation. At this stage, you also configure the remaining network settings. For more information, see Configure the Regional Networking of a VCF Automation Organization.

In either case, the connection that you use to set up the regional networking configuration gets assigned to the organization's default transit gateway. You can add an unlimited number of external connections to the organization later. You can also remove external connections from the organization, except for the default connection, that is, the one associated with the organization's default transit gateway. You cannot delete the default connection for an organization.

> **Important:**
>
> The default centralized connection for an organization must be associated with at least one IP block. You cannot select a centralized connection that is enabled to advertise its private IPs as the default connection for an organization.

## Configure the Regional Networking of a VCF Automation Organization

You can use the Provider Management UI to set up the regional networking configurations of an organization.

- Verify that you have an organization for all applications, see Create a VCF Automation Organization for All Applications.
- Verify that you have external connections created in your environment, see Managing External Connections in VCF Automation.
- If you intend to use advanced networking services over a distributed VXLAN external connection, verify that a distributed VXLAN external connection exists in your environment and that you have an external IP block covering the ranges to be used for Load Balancer VIPs and NAT IPs.

**Important:**

Advanced networking services over a distributed VXLAN connections are available for VCF Automation 9.1.1 and later.

To create VPCs and connect to the Internet, organizations need networking resources per region. You can use the organization regional network configuration to assign these resources to an organization.

Each organization in VCF Automation ties to an *NSX Project* that represents the organization in NSX Manager. You can configure regional networking for each organization per region.

Organizations can use the default VPC and default VPC connectivity profile to connect networking workloads using the VCF Automation Organization UI. For more information about VPCs, see Virtual Private Cloud in NSX and Managing Networking in VCF Automation Organizations.

If you have set up the regional networking for an organization at the time of organization creation, you can make additions and modifications to the initial configuration, such as the following:

- Add a connection to the organization.
- View the external IP blocks that are associated with the organization.
- Change the edge cluster for your connection.
- Share one or multiple subnets with the organization.
- Override the default QoS settings in the VPC Connectivity Profile.

1. Log in to the VCF Automation Provider Management UI.
2. In the **Infrastructure** section of the left-hand navigation menu, click **Organizations**.
3. Click the name of an organization for all applications.

   The **Organizations for All Apps** do not have the `VM Apps` label. For more information about the different types of organizations, see Organization Management.
4. From the left-hand navigation menu for the organization, select **Networking**.
5. From the drop-down menu at the top of the **Networking** panel, select a region.
6. Click the **Set Up Networking** button and follow the prompts to configure the region networking for the selected organization.

   The **Set Up Networking** button and the workflow that follows appear only if you have not configured any networking settings for the organization at the time of its creation.

   a) In the **Connection Type** tile, select the type of connection for the organization and click **Next**.

      The available connection options depend on the types of clusters, edge or VNA, that are configured for your region.

      If you have only edge clusters in your region, the only type of external connection that you can configure for the organization is centralized.

      If you have only VNA clusters in your region, the only type of connection that you can configure for the organization is distributed.

      If you have both edge and VNA clusters in your region, you can choose what type of connection to configure for the organization depending on your needs.

   b) In the **Connection** tile, select an external connection from the list and click **Next**.

   c) In the **VPC Service Configuration** tile, select the edge cluster or VNA cluster in which you want the VPC services of this organization to run and click **Next**.

      If the connection you create is centralized, you can use the default edge cluster of the Tier-0 centralized connection. Alternatively, you can select a specific edge cluster from your environment that is suitable for running VPC services.

      If the connection you create is a distributed VLAN or a distributed VXLAN connection, you must select a VNA cluster. For distributed VXLAN connections, the VNA cluster provides advanced networking services, such as NSX Load Balancer VIPs, Avi Load Balancer, and NAT.

**Important:**

You cannot change the VNA cluster assigned to an organization after regional networking is configured. Select the VNA cluster carefully, considering the expected volume of north-south Load Balancer and NAT traffic.

d) In the **Default Private IP Blocks** tile, review and, optionally, update the default private IP blocks for the transit gateway and the VPC, and click **Next**.

Click the **Override settings** button to define new values for the private IP blocks for the transit Gateway and the VPC for that specific organization.

To change the values used for all organizations by default, see Modify the Default Private IP Blocks for the Default VPCs and Transit Gateways of VCF Automation Organizations.

**Important:**

For VLAN-backed regions, the **Default Private IP Blocks** tile is not visible in the UI.

e) In the **Review and Complete** section, review the network settings and click **Create**.

Setting up the regional networking configuration for an organization creates an external connection for the organization for the specified region. Completing this configuration workflow creates an *NSX Project*, a default NSX transit gateway, a default VPC, a default VPC connectivity profile, and a default outbound SNAT rule that provides connectivity up to the Tier-0 gateway.

## Add an External Connection to a VCF Automation Organization

In VCF Automation 9.1, an organization can have multiple external connections. As a provider administrator, you add an external connection to an organization to make the connection available for the organization. The Organization Administrator then associates the external connection with a transit gateway of their choice.

Verify that the organization is already configured with at least one external connection, see Configure the Regional Networking of a VCF Automation Organization.

If an organization's regional networking configuration uses a centralized transit gateway, the additional connections that you can add to the organization can only be centralized. If the regional networking configuration uses a distributed gateway, the additional connections that you can add to the organization can only be distributed VLAN connections. You cannot add both centralized and distributed connections in a single region in an organization.

**Note:**

You can add dedicated VLAN connections to organizations that are configured with either a distributed or a centralized transit gateway.

1. Log in to the VCF Automation Provider Management UI.
2. In the **Infrastructure** section of the left-hand navigation menu, click **Organizations**.
3. Click the name of an organization for all applications.

   The **Organizations for All Apps** do not have the `VM Apps` label. For more information about the different types of organizations, see Organization Management.
4. From the left-hand navigation menu for the organization, select **Networking**.
5. From the drop-down menu at the top of the **Networking** panel, select a region.
6. On the **Connections** tab, click **New connection**.
7. In the **Add New Connection** dialog box, select a connection from the list and click **Add**.
8. Click **Sync** to refresh the view.

The new connection is added to the region network configuration for the organization and it appears in the list of connections on the **Connections** tab.

To delete the connection, click the vertical ellipsis ( ⋮ ) next to the name of the connection and select **Remove**.

## Add a Dedicated VLAN Connection to a VCF Automation Organization

You can add a new dedicated VLAN connection to an organization, so that the organization administrator can afterwards create a VLAN extension subnet directly within their organization. The organization administrator can share this subnet with one or multiple existing or newly created namespaces.

- Verify that the organization is already configured with at least one external connection, see Configure the Regional Networking of a VCF Automation Organization.
- Verify that you have a dedicated VLAN connection created, see Create a New Distributed VLAN Connection.

For information about how the organization administrator can create a subnet for their organization, see Create a Subnet in a VCF Automation Organization.

1. Log in to the VCF Automation Provider Management UI.
2. In the **Infrastructure** section of the left-hand navigation menu, click **Organizations**.
3. Click the name of an organization for all applications.

   The **Organizations for All Apps** do not have the `VM Apps` label. For more information about the different types of organizations, see Organization Management.
4. From the left-hand navigation menu for the organization, select **Networking**.
5. From the drop-down menu at the top of the **Networking** panel, select a region.
6. On the **Connections** tab, click **New Dedicated VLAN Connection**.

   The **New Dedicated VLAN Connection** button appears only if you have dedicated VLAN connections in your environment.
7. In the **New Dedicated VLAN Connection** dialog box, select a VLAN connection and click **Add**.

A new **Dedicated VLAN Connections** pane appears on the **Connections** tab. The VLAN connection that you selected is added to the list of dedicated VLAN connections for that organization.

## Change the Edge Cluster Associated With a VCF Automation Organization

You can change the edge cluster for the external connection that you assign to an organization.

Verify that the organization is already configured with at least one external connection, see Configure the Regional Networking of a VCF Automation Organization.

You can only change the edge cluster for a centralized external connection. If your organization is configured to use a distributed transit gateway, the VNA cluster that you selected at the time of setting up the regional networking cannot be changed.

1. Log in to the VCF Automation Provider Management UI.
2. In the **Infrastructure** section of the left-hand navigation menu, click **Organizations**.
3. Click the name of an organization for all applications.

   The **Organizations for All Apps** do not have the `VM Apps` label. For more information about the different types of organizations, see Organization Management.
4. From the left-hand navigation menu for the organization, select **Networking**.
5. From the drop-down menu at the top of the **Networking** panel, select a region.
6. On the **Edge Cluster** tab, click **Edit**.
7. In the **Edit Edge Cluster of Networking Setup for Region** dialog box, make your selection.
   - Use the default edge cluster associated with the Tier-0 centralized connection.
   - Select the **Select a specific edge cluster** radio button and select a cluster from the list.

8. Click **Save**.

## Share a Subnet With a VCF Automation Organization

In the VCF Automation Provider Management UI, you can create VLAN extension subnets, which you can later share with multiple organizations in your environment. When you share a subnet with an organization, the organization does not own and manage the subnet, but only uses it for their networking needs.

- Verify that the organization is already configured with at least one external connection, see Configure the Regional Networking of a VCF Automation Organization.
- Verify that you have a VLAN extension subnet created, see Create a VLAN Extension Subnet.

1. Log in to the VCF Automation Provider Management UI.
2. In the **Infrastructure** section of the left-hand navigation menu, click **Organizations**.
3. Click the name of an organization for all applications.

   The **Organizations for All Apps** do not have the `VM Apps` label. For more information about the different types of organizations, see Organization Management.
4. From the left-hand navigation menu for the organization, select **Networking**.
5. From the drop-down menu at the top of the **Networking** panel, select a region.
6. On the **Subnets** tab, click **Share**.
7. In the **Share Subnet** dialog box, select a subnet from the list and click **Share**.

## Override The Default QoS Configuration in the VPC Connectivity Profile for a VCF Automation Organization

You can override the default QoS configuration for the default VPC connectivity profile for an organization. By default, the committed bandwidth and the burst size for both the ingress and egress traffic are set to `unlimited`.

- Verify that you have an organization for all applications, see Create a VCF Automation Organization for All Applications.
- Verify that you have external connections created in your environment, see Managing External Connections in VCF Automation.

1. Log in to the VCF Automation Provider Management UI.
2. In the **Infrastructure** section of the left-hand navigation menu, click **Organizations**.
3. Click the name of an organization for all applications.

   The **Organizations for All Apps** do not have the `VM Apps` label. For more information about the different types of organizations, see Organization Management.
4. From the left-hand navigation menu for the organization, select **Networking**.
5. From the drop-down menu at the top of the **Networking** panel, select a region.
6. Click the **Settings** tab.
7. Under **Default VPC Connectivity Profile**, click **Edit**.

   The **Edit VPC Connectivity Profie** dialog box opens.

8. To override the QoS default settings, activate the **Override Edge Cluster Defaults** toggle for the ingress or egress traffic.
9. Deselect the **Unlimited** check box for the values that you want to modify.
10. Enter the committed bandwidth limit that you want to set for the ingress or egress traffic.
11. Enter the burst size for the ingress or egress traffic.
12. Click **Save**.

## Setting Up the Regional Avi Load Balancer for an Organization in VCF Automation

VCF Automation discovers Avi controllers and their associations with vCenter and NSX Manager. As a provider administrator, you can use the VCF Automation Provider Management UI to govern Avi resources by specifying quotas for the organizations. You can also enable organization administrators to manage Service Engine Groups (SEGs) based on specific application requirements within their respective organization.

To configure an organization to use Avi Load Balancer, all the underlying region Supervisors must be configured to use Avi Load Balancer instead of the default NSX load balancer.

You manage the Avi resource quota for organizations per region. The quota configurations map directly to the corresponding quota limits defined in the settings of the Avi controller.

While setting up the regional Avi Load Balancer for an organization, you specify if Avi runs in provider-managed mode or organization-managed mode. When Avi Load Balancer is configured in provider-managed mode, you create and manage SEGs and assign them to the organization. You configure SEGs in Avi by using the specific VCF Automation zones. If Avi Load Balancer is configured in organization-managed mode, SEGs are created and managed by the organization administrator of the respective organization.

You can use VCF Automation Provider Management UI to perform the following Avi-related actions.

- View and manage your Avi Load Balancer instance.
- Configure the regional Avi Load Balancer for an organization.
- Edit the Avi resources allocated to an organization.
- Add a SEG assignment to an organization, if Avi Load Balancer runs in provider-managed mode.

For detailed information about Avi Load Balancer and instructions on how to perform these actions, see Managing Load Balancing for All Apps Organizations in VCF Automation.

## Edit an Organization for All Applications in VCF Automation

From the VCF Automation Provider Management UI, you can edit the settings of an organization using the VMware vSphere Supervisor capabilities.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of an organization for all applications.

   The **Organizations for All Apps** do not have the `VM Apps` label. For more information about the different types of organizations, see Organization Management.
4. (Optional) Edit the general organization settings.
   a) (Optional) Edit the organization name that appears in VCF Automation.

      You cannot edit the unique identifier that forms the URL for accessing the organization.

b) (Optional) Edit the description of the organization.
c) Click **Save**.

5. (Optional) Allow or disallow the organization to subscribe to external content libraries or to Canonical content libraries.
   a) In the secondary left pane, click **Settings**, and click **Edit**.
   b) (Optional) Activate or deactivate the **Allow this organization to subscribe to external content libraries** toggle.
   c) (Optional) Activate or deactivate the **Allow this organization to subscribe to Canonical content libraries** toggle.

      VCF Automation delivers canonical Ubuntu LTS images in the Solutions Catalog. You, as the Provider Administrator can share these images by allowing organizations to subscribe to the Canonical content libraries.
   d) Click **Save**.
6. (Optional) Activate the validation of uploaded content library items.

   When this option is active, the upload files of the organization remain in a temporary storage area awaiting your validation. You must write an extension against the blocked tasks, and scan the directory where the files are located.

   a) From the secondary left pane, click **Settings**, and click **Edit**.
   b) Turn on the **Enable blocking of content library items upload** toggle, and click **Save**.

## Edit an Organization for VM Applications in VCF Automation

From the VCF Automation Provider Management UI, you can edit the settings of an organization using vSphere cluster capabilities.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of an organization with `VM Apps` label.
4. Click **Edit**.
5. (Optional) Edit the organization name that appears in VCF Automation.

   You cannot edit the unique identifier that forms the URL for accessing the organization.
6. (Optional) Edit the description of the organization.
7. Click **Save**.

## Activate or Deactivate a VCF Automation Organization

Deactivating a VCF Automation organization prevents users from logging in to the organization and terminates the sessions of users that are currently logged in.

As a **Provider Administrator**, you can allocate resources, add networks, and so on, even after you deactivate an organization.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the vertical ellipsis ( ⋮ ) next to the name of the organization, and click **Enable** or **Disable**.
4. To confirm, click **Disable**.

## Delete a VCF Automation Organization

You can use the VCF Automation Provider Management UI to delete an organization and permanently remove it from VCF Automation.

1. Deactivate the organization, see Activate or Deactivate a VCF Automation Organization.
2. IF you want to delete an **Organization for All Apps,** delete all region quotas of the organization, see Delete a VCF Automation Region Quota.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the vertical ellipsis ( ⋮ ) next to the name of the organization, and click **Delete**.
4. To confirm, click **Delete**.

## Region Quota of a VCF Automation Organization

A region quota defines the compute and storage quota that an organization gets against a given region. Region quotas are applicable only to organizations for all applications.

One organization can get a quota from more than one region. For a multi-zone region, you can give quotas to organizations for all of the zones or a subset of the zones. When creating a region quota for an organization, you must give a quota for at least one zone from the region. You can assign compute capacity and vGPU Reserved VM class capacity only on a per zone basis. vGPU Reserved VM class capacity is optional and applicable only for environments where vGPUs are present. Storage classes and VM classes are not given on a per-zone basis and apply to all the zones that are part of the region quota.

If you want to add infrastructure policies to the region quota, verify that you have at least one infrastructure policy in your environment. See Create an Infrastructure Policy in VCF Automation.

> **Note:**
>
> You cannot set a region quota for `VM Apps` organizations, and you cannot set a region quota with unlimited capacity.

Region quotas constrain the ability of organization users to consume storage and processing resources. One organization cannot get more than one region quota from a given region. An organization can get many region quotas if each region quota is coming from a different region.

#### Storage Use for Project Content Libraries

Because VCF Automation copies project content libraries across all Supervisors within the scope of a region quota, the number of Supervisors directly impacts storage use. To determine the total storage consumption for a single content library item, multiply the image size by the total number of Supervisors or data centers in the regions that include that content library. For example, consider a project content library is available in Region A with two Supervisors and in Region B with one Supervisor. If you upload a 10-GB image to this library, the image consumes a total of 30 GB of the region quotas. Because consumes 10 GB per Supervisor, the upload uses 20 GB of the region quota in Region A and 10 GB of the region quota in Region B.

> **Important:**
>
> You must ensure that the region quota has sufficient capacity to accommodate the multiplied storage requirements of library items across all associated Supervisors.

## Create a Region Quota for a VCF Automation Organization

You must create one or more region quotas to a VCF Automation organization for it to create namespaces and content libraries.

If you want to add infrastructure policies to the region quota, verify that you have at least one compatible infrastructure policy in your environment. See Create an Infrastructure Policy in VCF Automation.

A region quota defines the compute and storage quota that an organization gets against a given region. Region quotas are applicable only to organizations for all applications.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of an organization for all applications.

   The **Organizations for All Apps** do not have the `VM Apps` label. For more information about the different types of organizations, see Organization Management.
4. In the secondary left pane, click **Region Quota**.
5. Click **New**.
6. Select a region for the quota.
7. Select how much of the region capacity you want to share with the organization.

| Option | Description |
|---|---|
| **Share the entire capacity with this organization** | When you create a full allocation region quota, it includes all available zones, memory limits, CPU limits, storage classes, and VM classes of the region. Any further changes to the region, such as capacity or zone modifications, automatically apply to the region quota. By default, full allocation region quotas include zero memory and CPU reservations for each zone. You can assign CPU and memory reservations as a Day 2 operation on a full allocation region quota. You cannot overprovision CPU and memory reservations from a region.<br>You can share the full capacity of the region with more than one organization. |
| **Share a portion of the capacity with this organization** | Select this option to share defined resources with the organization. |

8. If you selected **Share the entire capacity with this organization**, select one or more infrastructure policies for the region quota.

   The list shows only infrastructure policies that are compatible with the region. You can assign both mandatory and optional infrastructure policies. Mandatory infrastructure policies are not automatically assigned to all region quotas.
9. If you selected **Share a portion of the capacity with this organization**, give the organization access to certain infrastructure resources.
   a) If you want to assign specific Supervisors, to the organization, turn off the **Add all available supervisors** toggle and select one or more Supervisors.
   b) Add one or more zones and assign CPU and memory limits and reservations.
   c) Click **Assign and Continue**.
   d) Select one or more VM classes, storage classes, and infrastructure policies applicable to the organization.

      The list shows all VM classes, storage classes, and infrastructure policies compatible with the region. You cannot assign a guaranteed VM class to a region quota that has zero reservation. When adding storage classes, you can also edit the storage limit for the storage class. You can assign both mandatory and optional infrastructure policies. Mandatory infrastructure policies are not automatically assigned to all region quotas.
10. Click **Assign and Continue**.
11. If applicable, add the reserved VM classes that limit vGPU use for the organization and click **Continue**.

- Edit a Region Quota
- Edit the Zones of a Region Quota
- Edit the VM Classes, Storage Classes, Reserved VM Classes, and Infrastructure Policies of a Region Quota

## Edit a Region Quota

After you create a region quota, you can use the VCF Automation Provider Management UI to change the settings and the resource allocation through the region quota.

If you want to add infrastructure policies to the region quota, verify that you have at least one compatible infrastructure policy in your environment. See Create an Infrastructure Policy in VCF Automation.

When you edit the region quota, you can select between two allocation options:

- Full allocation - you share the entire capacity of the region with the organization. If a region quota has a partial allocation and you change the allocation to share the entire capacity of the region, the CPU and memory limits can increase but the pre-set reservations remain unchanged.
- Partial allocation - you control the resource distribution by sharing a portion of the capacity with the organization. You can edit an existing full allocation region quota and convert it to a partial region quota. Such a quota retains the existing quota values, but the quota no longer reflects the direct changes to the region.

**Note:**

Tenant content libraries are pinned to certain vCenter instances in certain regions. For this reason, if you add or remove Supervisors from a region quota, the scope of the affected content libraries expands or shrinks accordingly. VCF Automation copies project content libraries across all vCenter instances in the scope of the region quota. You can remove any Supervisor, but the region quota defining the scope of the project content library must have at least one Supervisor. Organization content libraries have one published or local copy and multiple subscribed copies across vCenter instances.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of an organization for all applications.

   The **Organizations for All Apps** do not have the `VM Apps` label. For more information about the different types of organizations, see Organization Management.
4. In the secondary left pane, click **Region Quota**.
5. Expand the region quota view by clicking the button to the left of the region name.
6. Under the **General** section, click **Edit**.
7. Select how much of the region capacity you want to share with the organization.

| Option | Description |
|---|---|
| **Share the entire capacity with this organization** | When you create a full allocation region quota, it includes all available zones, memory limits, CPU limits, storage classes, and VM classes of the region. Any further changes to the region automatically apply also to the region quota. By default, full allocation region quotas include zero memory and CPU reservations for each zone. You can assign CPU and memory reservations as a Day 2 operation on a full allocation region quota. You cannot overprovision CPU and memory reservations from a region.<br>You can share the full capacity of the region with more than one organization. |
| **Share a portion of the capacity with this organization** | Select this option to share defined resources with the organization. |

8. If you selected **Share the entire capacity with this organization**, select one or more infrastructure policies for the region quota.

   The list shows only infrastructure policies that are compatible with the region. You can assign both mandatory and optional infrastructure policies. Mandatory infrastructure policies are not automatically assigned to all region quotas.

9. If you selected **Share a portion of the capacity with this organization**, give the organization access to certain infrastructure resources.
   a) If you want to assign specific Supervisors, to the organization, turn off the **Add all available supervisors** toggle and select one or more Supervisors.
   b) Add, edit or remove one or more zones and assign CPU and memory limits and reservations.
   c) Click **Save**.
   d) Select or deselect one or more VM classes, storage classes, and infrastructure policies applicable to the organization.

      The list shows all VM and storage classes available in the selected Supervisor. You cannot assign a guaranteed VM class to a region quota that has 0 reservation. When adding storage classes, you can also edit the storage limit for the storage class.
   e) Click **Save**.
   f) If your environment has reserved VM classes, select or deselect one or more reserved VM classes.
10. Click **Save**.
11. If applicable, add the reserved VM classes that limit vGPU use for the organization and click **Save**.

## Edit the Zones of a Region Quota

After you create a full allocation region quota, you can use the VCF Automation Provider Management UI to change the zones and their resources which are allocated to an organization.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of an organization for all applications.

   The **Organizations for All Apps** do not have the `VM Apps` label. For more information about the different types of organizations, see Organization Management.
4. In the secondary left pane, click **Region Quota**.
5. Expand the region quota view by clicking the button to the left of the region name.
6. Under the **Capacity** section, click **Edit Reservation** or **Add Capacity**.

   If the entire capacity of the region is currently allocated, the button appears as **Edit Reservation**. If there is only partial allocation of the resources, the button appears as **Add Capacity**.
7. If you want to apply the same limits and reservations across all zones, select the **Add all zones and apply the same limits and reservations across all zones.** check box.
8. Add, edit, or delete zones and resources from the region quota.

   Any changes you make to the zones of the parent region also affect the zones for any child full allocation region quota.
9. Click **Save**.

## Edit the VM Classes, Storage Classes, Reserved VM Classes, and Infrastructure Policies of a Region Quota

You can use the VCF Automation Provider Management UI to change which VM classes are available for an organization in a specific region. Managing these classes ensures that organizations have the appropriate compute and storage resources for their workloads.

If you want to add infrastructure policies to the region quota, verify that you have at least one compatible infrastructure policy in your environment. See Create an Infrastructure Policy in VCF Automation.

> **Important:**
>
> For full allocation region quotas, the VM and storage classes are identical to the parent region and are given in full. You cannot edit these values, and the **Edit** button does not appear in the UI. Any changes to the VM and

storage classes of the parent region propagate to the full allocation region quota. You must explicitly assign reserved VM class capacity for both full allocation and partial capacity region quotas.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of an organization for all applications.

   The **Organizations for All Apps** do not have the `VM Apps` label. For more information about the different types of organizations, see Organization Management.
4. In the secondary left pane, click **Region Quota**.
5. Expand the region quota view by clicking the button to the left of the region name.
6. In the **Capacity** section, click **Edit Reservation** if the region quota is fully allocated or **Add Capacity**.
7. Add one or more zones and assign CPU and memory limits and reservations.

   If you want to apply the same limits and reservations across all zones, select the **Add all zones and apply the same limits and reservations across all zones** check box.
8. (Optional) Edit the **VM Classes**.

   The **Edit** button is only visible for partial capacity region quotas.

   a) Under the **VM Classes** section, click **Edit**.
   b) Select or deselect VM classes to set the applicable VM classes for the region quota and click **Save**.
9. (Optional) Edit the **Storage Classes**.

   The **Edit** button is only visible for partial capacity region quotas.

   a) Under the **Storage Classes** section, click **Edit** or **Add**.
   b) Edit the storage limit for the storage class and click **Save**.
10. (Optional) Edit the **Infrastructure Policies**.
    a) Under the **Infrastructure Policies** section, click **Edit**.
    b) Change the infrastructure policies applicable to the region quota of an organization and click **Save**.

       The list shows only infrastructure policies that are compatible with the region. All of the infrastructure policies you add to the region quota must be compatible with at least one of the zones added to the region quota. You cannot add an infrastructure policy before adding a compatible zone to the region quota.

       You can assign both mandatory and optional infrastructure policies. Mandatory infrastructure policies are not automatically assigned to all region quotas.
11. (Optional) Edit the **Reserved VM Classes**.
    a) Under the **Reserved VM Classes** section, click **Edit**.
    b) Change the reserved VM classes applicable to the region quota that limit the vGPU use of an organization and click **Save**.

## Modify the Metadata for a Region Quota in VCF Automation

You can use the Provider Management UI to add, edit, and delete metadata for a region quota in VCF Automation.

If you want to edit or delete metadata, verify that you have the **Create, update, and delete Virtual Datacenters** right. For read access, you must have the **View Virtual Datacenters** right.

You can use metadata to tag, categorize, and automate the management of tenant and provider resources, enabling policy-driven operations such as billing, compliance, workload placement, and service customization across multi-tenant environments.

By using metadata, you can associate user-defined `key=value` pairs with a region quota. You can use metadata in VCF Automation API query filter expressions.

You can edit only the value of an existing metadata item.

**Table 1807: Provider Metadata**

| Parameter | VCF Automation Metadata Limits |
|---|---|
| Key length | Up to 256 UTF-8 characters |
| Metadata entries per resource | Up to 200 entries, with a maximum limit of 300. |
| Value length | Up to 1,000 characters. The system discards entries that exceed this limit. |
| Supported types | `String`, `Bool`, and `Integer` |
| Search patterns | • `?metadata=org:vmware\|test:key:8=='string'` – searches for a string entry. When searching for a string, always enclose the value in single quotes (').<br>• `?metadata=test.key.3==42` – searches for a number entry.<br>• `?metadata=test.key.3==*` - searches for a key.<br>• `?metadata=test.key.*==*` - searches for a key starting with `test.key`.<br>• `?metadata=test.key.3=lt=43`<br>• `?metadata=test.key.4=='str*'` – searches for a string entry with a value starting with `str`. There are no limitations on the size of the FIQL expression or the number of different key-values in the expression. |
| Bulk creation | Supports bulk metadata creation. |

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of an organization for all applications.

   The **Organizations for All Apps** do not have the `VM Apps` label. For more information about the different types of organizations, see Organization Management.
4. In the secondary left pane, click **Region Quota**.
5. Expand the region quota view by clicking the button to the left of the region name.
6. Under the **Metadata** section, click **Edit**.
7. If you want to add a key-value pair, click **Add**, enter a key and a value, and select a type for the new key-value pair.

   The available metadata types are **Text**, **Yes or No**, and **Number**.
8. If you want to remove a key-value pair, in the right end of the row, click the **Delete** icon.
9. Click **Save**.

## Export the Organization Data in VCF Automation

You can use the VCF Automation Provider Management UI to export a CDV file with information about the IDs, names, and states of the organizations in VCF Automation.

Verify that your VCF Automation instance has at least one organization. See Create a VCF Automation Organization for All Applications or Create a VCF Automation Organization for VM Applications.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click **Export Orgs**.

## Delete a VCF Automation Region Quota

You can use the VCF Automation Provider Management UI to delete a region quota.

- Using the Organization UI or the VCF Automation API, delete all the namespaces created in this organization that use this region quota.
- Delete the content libraries built within the organization using resources from the region quota.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of the organization for which you want to delete the region quota.

   You must select an organization for all applications.
4. In the secondary left pane, click **Region Quota**.
5. Click the vertical ellipsis ( ⋮ ) next to the name of the region to which the region quota is applicable, and click **Delete**.
6. To confirm, click **Delete**.

## Modify the Metadata for a VCF Automation Organization

You can use the VCF Automation Provider Management UI to add, edit, and delete metadata for an organization.

If you want to edit or delete metadata, verify that you have the **Create Organization** right. For read access, you must have the **View all Organizations** right.

You can use metadata to tag, categorize, and automate the management of tenant and provider resources, enabling policy-driven operations such as billing, compliance, workload placement, and service customization across multi-tenant environments. See Metadata Management for querying and configuring metadata from the VCF Automation API.

By using metadata, you can associate user-defined `key=value` pairs with an organization. You can use metadata in VCF Automation API query filter expressions.

You can edit only the value of an existing metadata item.

You can also use metadata to configure organization-specific values for certain VCF Automation features. VCF Automation enforces limits on the number of IaaS policies that an organization can have. To learn how IaaS resource policies and Supervisor-level limits work, see Configure IaaS Resource Policies in VCF Automation. The default policy limit per organization is 10. You can use the metadata of an organization to override the default by setting a custom limit. Setting the limit to `0` deactivates the creation of new policies. Modifying a policy limit does not affect or deactivate policies created in the organization before the change.

**Table 1808: Provider Metadata**

| Parameter | VCF Automation Metadata Limits |
|---|---|
| Key length | Up to 256 UTF-8 characters |
| Metadata entries per resource | Up to 200 entries, with a maximum limit of 300. |
| Value length | Up to 1,000 characters. The system discards entries that exceed this limit. |
| Supported types | `String`, `Bool`, and `Integer` |
| Search patterns | • `?metadata=org:vmware\|test:key:8=='string '` – searches for a string entry. When searching for a string, always enclose the value in single quotes (').<br>• `?metadata=test.key.3==42` – searches for a number entry.<br>• `?metadata=test.key.3==*` - searches for a key.<br>• `?metadata=test.key.*==*` - searches for a key starting with `test.key`.<br>• `?metadata=test.key.3=lt=43`<br>• `?metadata=test.key.4=='str*'` – searches for a string entry with a value starting with `str`. There are no limitations on the size of the FIQL expression or the number of different key-values in the expression. |

| Parameter | VCF Automation Metadata Limits |
|---|---|
| Bulk creation | Supports bulk metadata creation. |

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of the organization.
4. From the secondary left pane, click **Metadata**.
5. Click **Edit**.
6. If you want to add a key-value pair, click **Add**, enter a key and a value, and select a type for the new key-value pair.

   The available metadata types are **Text**, **Yes or No**, and **Number**.

   To apply a custom policy limit to the organization, add a metadata entry with the following details:

   - Key: `vcfa.iaas.policy.limit`
   - Type: `Number`
   - Value: *`[New_limit]`*
7. If you want to remove a key-value pair, in the right end of the row, click the **Delete** icon.
8. Click **Save**.

## Edit the First User Details for a VCF Automation Organization

After you create an organization, you can use the VCF Automation Provider Management UI to change the first user details.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Infrastructure**, click **Organizations**.
3. Click the name of the organization.
4. In the secondary left pane, click **First User**.
5. Click **Edit**.
6. Change the user details.

   You can change the user name, password, and role.
7. Click **Save**.

# Managing VCF Automation System Settings

A VCF Automation system administrator can control system-wide settings that are related to LDAP, email notification, licensing, and general system preferences.

By default, VCF Automation uses FIPS 140-3 validated cryptographic modules and runs in FIPS-compliant mode. In VCF Automation, FIPS is deactivated only if you upgrade from a VMware Aria Automation installation with deactivated FIPS mode. After installing or upgrading to VCF Automation 9.1, you cannot activate or deactivate FIPS mode.

The Federal Information Processing Standard (FIPS) 140-3 is a U.S. and Canadian government standard that specifies security requirements for cryptographic modules. The NIST Cryptographic Module Validation Program (CMVP) validates the cryptographic modules compliant with the FIPS 140-3 standards.

VCF Automation uses the BouncyCastle FIPS 2.0.0 cryptographic module compliant with FIPS 140-3 (Certificate #4743).

## Length Limits on Names and Descriptions in Your VCF Automation Provider Management UI

Follow these guidelines when entering values in VCF Automation.

String values for the `name` attribute and the `Description` and `ComputerName` elements have length limitations that depend on the object to which they are attached.

**Table 1809: Length Limits on Object Properties**

| Object | Property | Maximum Length in Characters |
|---|---|---|
| `EdgeGateway` | `Name` | 35 |
| `Organization` | `Name` | 128<br>Must contain only the following characters: letters from A through Z, digits from 0 through 9, and the following special characters: `- _ . ( ) , $ ' !"` |
| `Organization` | `Description` | 256 |
| `Region` | `Name` | 63<br>Must follow the requirements as defined in RFC 1123. |
| `Region quota` | `Name` | 128 |
| `Region quota` | `Description` | 256 |

# Modify VCF Automation General System Settings

VCF Automation includes general system settings that are related to activity logs, localization, timeouts, organization limits, and operation limits. The default settings are appropriate for many environments, but you can modify the settings to meet your needs.

> **Note:**
>
> For information about changing the date, time, or time zone of the VCF Automation appliance, see Broadcom knowledge base article 59674.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Administration**, select **General Settings**.
3. Click **Edit** for the section you want to modify, edit the properties, and click **Save**.

| Name | Category | Description |
|---|---|---|
| `Log history to keep` | Activity Log | Number of days of the log history to keep before deleting it. Enter `0` never to delete logs. |
| `Log history shown` | Activity Log | Number of days of the log history to display.<br>To show all activity, enter `0`. |
| `Display debug information` | Activity Log | Enable this setting to display the debug information in the VCF Automation task log. |
| `API token lifetime` | Timeouts | The maximum amount of time, in minutes, a VCF Automation API token remains active. The value must be a whole number from 0 through 3600, where 0 represents an unlimited lifespan. |
| `Host refresh frequency` | Timeouts | How often VCF Automation checks whether its ESX hosts are accessible or inaccessible. |
| `Host hung timeout` | Timeouts | Select the amount of time to wait before marking a host as hung. |
| `Transfer session timeout` | Timeouts | Amount of time to wait before failing a paused or canceled upload task, for example, the upload of media. This timeout does not affect upload tasks that are in progress. |
| `Maximum number of Region Quotas` | Organization Limits | Enter the maximum number of region quotas per organization, or click **Unlimited**. |

| Name | Category | Description |
|---|---|---|
| `Number of resource-intensive operations per user` | Operation Limits | Enter the maximum number of simultaneous resource-intensive operations per user, or click **Unlimited**. |
| `Number of resource-intensive operations to be queued per user` | Operation Limits | Enter the maximum number of queued resource-intensive operations per user, or click **Unlimited**. |
| `Number of resource-intensive operations per organization` | Operation Limits | Enter the maximum number of simultaneous resource-intensive operations per organization, or click **Unlimited**. |
| `Number of resource-intensive operations to be queued per organization` | Operation Limits | Enter the maximum number of queued resource-intensive operations per organization, or click **Unlimited**. |

## Configure Blocking Task Settings On Your VCF Automation

You can configure certain operations as blocking tasks. These operations are suspended until a **system administrator** acts on them or a preconfigured timer expires. You can specify the timeout settings and default actions for blocking tasks. The settings apply to all organizations in the installation.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation bar, click **Events and Tasks**, and click the **Blocking tasks** tab.
3. To edit the default extension timeout and default timeout action, click **Configure**.
   a) Edit the **Default blocking task timeout**.
   b) Edit the **Default Action**.

      The **Default Action** is the action after a **Default blocking task timeout** expires.
   c) Click **Save**.
4. To edit the list of operations, considered as blocking tasks, click the **Edit** button under the **Operations** section.
   a) Select or deselect operations from the list of blocking tasks.

      Blocking tasks can be pre-blocking and post-blocking and occur before and after activities like deployments, system upgrades, and similar major changes.

      **Important:**

      If you deselect the organization-related `orgCreate` post-blocking task or `orgDeleteOrg` pre-blocking task, when you create a new organization, VCF Automation cannot create the default objects for the organization and the default VMware Cloud Foundation® Operations orchestrator integration for the provider consumption organization.

      If you change the organization-related `orgCreate` post-blocking task or `orgDeleteOrg` pre-blocking task selections, the organization creation and deletion might get stuck.
   b) Click **Save**.

## Monitor Blocked Tasks in VCF Automation

You can monitor the current blocked tasks or manually cancel, fail, or resume the tasks before the preconfigured timer expires.

Configure Blocking Task Settings On Your VCF Automation

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation bar, click **Events and Tasks**, and click the **Blocking tasks** tab.

   The tab displays a list of the current blocked tasks.

3. Select the task that you want to edit manually.
4. Decide between canceling, failing, or resuming the task and click the corresponding button.
5. Enter a message and click **Save**.
   The message appears in the task details.

## Subscribe to VCF Automation Provider Management Events, Tasks, and Metrics by Using an MQTT Client

As a provider administrator, you can use an MQTT client to subscribe to messages about Provider Management events and tasks.

- Verify that you have an MQTT client that supports WebSocket.
- Verify that you can add headers to a WebSocket-upgraded request.

MQTT is a lightweight, binary, messaging transport protocol. VCF Automation Provider Management UI uses MQTT to publish information about events and tasks to which you can subscribe by using an MQTT client. MQTT messages pass through an MQTT broker which can also store messages in case the clients are not online.

You can use an MQTT client to subscribe to metrics.

Organization administrators can use VCF Operations orchestrator workflows with event subscriptions. See Managing Event Subscriptions in VCF Automation.

1. Log in to VCF Automation Provider Management by using the OpenAPI endpoint.
   a) Make an API versions request to VCF Automation to obtain the VCF Automation API login URL.

      This request has the following form:

      ```
      GET http://vcfa.example.com/api/versions
      ```

   b) POST a request to the login URL to create a login session.

      This request has the following form:

      ```
      POST https://vcfa.example.com/cloudapi/1.0.0/sessions/provider
      ```

2. To establish a WebSocket connection, set the Sec-WebSocket-Protocol property to `mqtt`, set the client to connect to the `/messaging/mqtt` path, add an authorization header, and follow the standard MQTT connect flow.

   You receive the JWT token from the standard login request to VCF Automation.

   Enter your account name in the username field and the JWT token as the password in your MQTT client.

   ```
   Sec-WebSocket-Protocol: mqtt
   Authorization: Bearer {JWT_token}
   ```

3. Once the connection is established successfully, subscribe to topics through the MQTT client.

   ```
   publish/{user_org_id}/{user_id}
   ```

   ```
   publish/debd63a0-6eae-11ea-8c7b-0050561776be/d19fd8ff-6eae-11ea-bb42-0050561776c8
   ```

   **Organization administrators** can use wildcards to access all organization topics.

   ```
   publish/{user_org_id}/+
   ```

   **System administrators** can use wildcards to access all topics.

   ```
   publish/#
   ```

## Change the Language of the VCF Automation UI by Using the Provider Management UI

The VCF Automation user interface is available in English, French, Japanese, and Spanish. You can change the language of the UI and the messages you receive.

When you change the language of the VCF Automation UI, the language change applies only to your user and the change affects also the VCF Automation messages sent to you.

1. Log in to the VCF Automation Provider Management UI.
2. On the right side of the top navigation bar, click your user name.
3. Click **My Account**.
4. Click the **Preferences** tab, and click **Edit**.
5. From the drop-down menu, select your preferred language, and click **Save**.
6. Apply the language changes.

   To apply the changes, you must refresh the page. However, refreshing might interrupt tasks running in the background, such as content uploads.

   - To refresh the page, click **Refresh**.
   - If you want to postpone the refresh to a later time, click **Close**.

## Change Your Account Information

You can change your **service administrator** user details from the Provider Management UI.

1. On the right side of the top navigation bar, click your user name.
2. Click **My Account**.
3. On the **Profile** tab, click **Edit**.
4. Edit your user details, and click **Save**.

   You can change your username, display name, first and last name, description of the account, email, and phone number. You cannot edit the role.

## Changing the VCF Automation Portal Branding

To match your corporate branding standards and to create a fully custom cloud experience, you can customize the VCF Automation portals. In addition, you can modify and add custom links to the login and logout pages.

The VCF Automation UI provides a live preview of all changes so that you can see the modifications instantly without the need to save the theme.

> **Note:**
>
> By default, the individual organization branding is not shown outside of a logged-in session. The individual organization branding does not appear on the login and logout pages, so that organizations cannot discover the existence of other organizations. You can enable branding outside of logged in sessions by clicking **Settings** and turning on the **Enable Login and Logout Page Branding** toggle.

### Portal Branding

As part of the installation, VCF Automation contains two themes - light and dark. You can create, manage, and apply custom themes. In addition, you can change the portal name, the logo, the image on the login screen, and the browser icon. The browser title adopts the portal name that you set and the login background.

When you create a theme, you can assign it to one or more organizations. You can specify a theme to be the default theme, applicable to all organizations apart from the ones with specifically assigned themes through the **Assign** option.

For a particular theme, you can selectively override any combination of the portal name, colors, logo, icon, custom links, and the login background. Any color value that you do not set uses the corresponding theme base color.

Certain operations are not applicable to some of the theme types.

**Table 1810: Theme operations**

| Theme type | Edit | Delete | Assign | Make default | Export | Clone |
|---|---|---|---|---|---|---|
| Built-in themes | | | X | X | X | X |
| Custom theme | X | X | X | X | X | X |
| Assigned theme | X | | X | X | X | X |
| Default theme | X | | | | X | X |

## Create a Custom VCF Automation Portal Theme

You can change the branding for the VCF Automation Provider Management UI .

When changing the theme, you can instantly see how the selection would change the appearance of the theme.

1. Log in to the VCF Automation Provider Management UI.
2. In the left pane, click **Branding** and click **Create**.
3. Enter a name for the theme.
4. Change the platform branding from the option in the **Branding** tab.
   a) Enter a platform name.

      The platform name replaces the VCF Automation name on the top navigation bar, the login page, and the browser title.
   b) Upload a platform logo.

      The platform logo replaces the default logo on the top navigation bar and the login page. You can also select to include an additional dark theme logo.
   c) Upload a browser favicon.

      The favicon, or browser tab icon, is the icon that appears next to the page title in a browser tab.
   d) Upload a background image for the login page.

      The layout adapts to screen sizes at 768 px and 544 px. The login background image might fill the full size of the screen and part of it might be hidden below the login form. Depending on the browser window size and screen resolution settings, some parts of the background image might not appear. Consider using a background with a pattern or an image that looks complete even if parts of it do not appear.
5. In the left pane, click **Light Colors**, and select a color for each element of the light theme.
6. In the left pane, click **Dark Colors**, and select a color for each element of the dark theme.
7. Optional: Click the **Links** tab and customize the platform links.
   a) Enter new URLs for the product communities forum and for support requests.
   b) Add a link to the login page.

      For example, you can add links to the terms and conditions on your platform, privacy and accessibility information, trademarks, and so on. The custom link appears on the login page below the login button.

      **Note:**

      If you define custom links, the default links do not appear.
   c) Add a link to the logout page.

      For example, you can add links to the terms and conditions on your platform, privacy information, accessibility information, trademarks, and so on.

      **Note:**

      If you define custom links, the default links do not appear.

8. Click **Save**.

From the **Actions** menu on the theme, you can make the theme a default one, assign the theme to an organization, export, clone, and delete the theme. You can assign to VCF Automation a theme different from the default theme.

The default theme changes affect all organizations apart from the ones with specifically assigned themes through the **Assign** option.

If an assigned theme becomes the default, it stops being explicitly assigned to organizations. When you change the default theme, VCF Automation removes it from all organizations to which the theme was assigned to and replaces it with the new default.

### Import a Custom VCF Automation Portal Theme

You can export a custom portal theme from one VCF Automation site and import it to another site.

- Verify that you have an exported theme ZIP file. You can export an existing theme from its **Actions** menu.
- If you want to edit manually a ZIP file and use custom CSS for a theme, export the theme ZIP file and extract the file, make CSS changes to the theme, and compress the files to ZIP format again.

1. Log in to the VCF Automation Provider Management UI.
2. From the left panel, select **Branding** and click **Import**.
3. Select a name for the theme and a platform name.
4. Select a theme ZIP file, and click **Import**.

The new theme appears on the **Branding** page.
From the **Actions** menu on the theme, you can make the theme a default one, assign the theme to an organization, export, clone, and delete the theme.

The default theme changes affect all organizations apart from the ones with specifically assigned themes through the **Assign** option.

If an assigned theme becomes the default, it stops being explicitly assigned to organizations. When you change the default theme, VCF Automation removes it from all organizations to which the theme was assigned to and replaces it with the new default.

## Manage Your Customer Experience Improvement Program Participation in VCF Automation Provider Management

By default, the Customer Experience Improvement Program (CEIP) is active in VCF Automation. You can activate or deactivate your CEIP participation for a specific cluster in a workload domain by using the VCF Fleet LCM Service API.

- Identify the FQDN of your fleet lifecycle component.
- Verify that you have provider administrator credentials.
- If necessary, schedule a maintenance window. VCF Automation is temporarily unavailable while the pods restart during the CEIP update.

For details regarding the data collected through CEIP, and the purposes for data collection, see the Customer Experience Improvement Program page.

1. Generate a VCF services runtime access token by running the following `curl` command to request a token.

   To authenticate your requests to the VCF Fleet LCM Service API, you must obtain a Bearer token from the Identity Service.

```
curl -ks --request POST \
  --url https://<platform_fqdn>/api/v1/identity/token \
  --header 'Content-Type: application/x-www-form-urlencoded' \
  --data grant_type=password \
  --data 'username=admin@vsp.local' \
  --data 'password=<admin_password>' | jq -r '.access_token'
```

**Note:**

Replace *<platform_fqdn>* with the FQDN of your management VCF services runtime.

2. Retrieve the component ID for the cluster.
   a) Run a GET request to the components endpoint.

```
curl -ks -X GET \
  -H "Authorization: Bearer <access_token>" \
  -H 'Accept: application/json' \
  https://<fleet-fqdn>/fleet-lcm/v1/components | jq
```

   b) Search the `VspCluster` objects in the response to find the `componentId` for your VCF Automation cluster.
   c) Make note of the `componentId`.
3. Update the CEIP setting by running the following `PATCH` request to update the cluster configuration.

```
curl -ks -X PATCH \
  -H "Authorization: Bearer <access_token>" \
  -H 'Content-Type: application/json' \
  --data '{"type": "VspClusterConfigSpec", "ceip": true}' \
  https://<fleet-fqdn>/fleet-lcm/v1/components/<componentId>/config
```

# Managing Certificates Using Your VCF Automation

You can import, download, edit, and delete certificates from VCF Automation. You can copy the certificate PEM data to the clipboard.

## Test the VCF Automation Connection to a Remote Server and Establish a Trust Relationship Using the Provider Management UI

You can test the connection of VCF Automation to a remote server, and establish a trust relationship with it.

Verify that your role includes the **SSL: Test Connection** and the **Truststore: Manage** rights.

You can test and secure the connection of VCF Automation to a remote server by entering the server URL.

When you test a remote connection to a server, if the connection involves SSL communication and VCF Automation has not already established a trust relationship with the server, you are prompted to review a certificate, or a chain of certificates, and to make a trust choice.

If the certificate chain is not complete and there are additional certificates available, you can choose to retrieve them to view more details before making a trust choice.

When testing remote connections, there is the potential to exploit the Server-Side Request Forgery (SSRF) vulnerability. An attacker can scan the common Kubernetes subnet ranges, such as `10.244.0.0/16` or `100.96.0.0/16`, and so on, and then enumerate the services by scanning for open ports. To test connections to remote servers and to verify the server identity as part of an SSL handshake, you can use the VCF Automation API. An attacker can scan the common Kubernetes subnet ranges such as 10.244.0.0/16 or 100.96.0.0/16, and so on, and then, enumerate the services by scanning for open ports. To protect the internal network in which a VCF Automation instance is deployed from malicious attacks, you can configure a denylist of internal hosts that are unreachable to organizations. This way, if a malicious attacker with organizational access attempts to use the connection testing API to map the network in which VCF Automation is installed, they would not be able to connect to the internal hosts on the denylist. Managing this denylist is possible through the `/cloudapi/1.0.0/testConnection/restrictions` API.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Certificate Management**.
3. Click **Trusted Certificates**, and click **Test Remote Connection**.
4. Enter a URL for the server with which you want to test the connection.
5. From the drop-down menu, select the hostname verification algorithm to use when testing the connection.
6. Click **Connect**.
7. If the connection involves SSL communication and VCF Automation hasn't already established a trust relationship with the server, review the certificate information and make a trust choice.

| Option | Description |
|---|---|
| **Trust** | Verify that you trust the certificate information and establish a trust relationship for future communication with the server. |
| **Retrieve** | If the certificate chain is not complete and there are additional intermediate and leaf certificates available from the same issuing certificate authority, you can retrieve them to view more details. Depending on your security considerations, choose one of the options.<br>• Select which certificates to trust and click **Trust selected**<br>• To cancel the attempt to establish a trust relationship, click **Cancel** |
| **Cancel** | Cancel the attempt to establish a trust relationship. |

## Import Trusted Certificates Using Your VCF Automation Provider Management UI

You can import certificates of servers that VCF Automation communicates with, such as vCenter, NSX Manager, and so on.

Verify that your role includes the **Truststore: Manage** right.

> **Note:**
>
> Instead of importing certificates manually, you can test the connection to the remote server and establish a trust relationship with it. See Test the VCF Automation Connection to a Remote Server and Establish a Trust Relationship Using the Provider Management UI.

You must use FIPS-compatible private keys. You can use pyOpenSSL to generate private keys in FIPS-compatible PKCS#8 format. If you generate PKCS#8 private keys by using OpenSSL, the private keys are not FIPS-compatible.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, click **Certificate Management**.
3. Click **Trusted Certificates**, and click **Import**.
4. Upload a PEM file containing the certificates that you want to import, and click **Import**.
5. Optional: (Optional) Edit the certificate name.
6. Click **Import**.

You can use the vertical ellipsis ( ⋮ ) menu to perform the following actions:

- Download a certificate.
- Edit a certificate name.
- Delete a certificate.
- Copy the PEM data to the clipboard.

## Import Certificates to the Certificates Library Using Your VCF Automation Provider Management UI

In the VCF Automation certificates library, you can import certificates used when creating entities that you must secure, such as servers, Edge gateways, and so on.

- Verify that your role includes the **Certificate Library: Manage** right.
- Verify that the private keys you want to use are in the PKCS#8 format. VCF Automation does not support private keys generated with the Digital Signature Algorithm (DSA).

The certificate library contains information about single certificates, certificate chains, private keys, certificate expiration dates, the entities that the certificates secure, and so on.

You must manage the certificate libraries separately for each site.

You must use FIPS-compatible self-signed certificates and private keys. You can generate self-signed unencrypted certificates and private keys by using OpenSSL. If you generate self-signed certificates and private keys by using OpenSSL, the certificates and private keys are not FIPS-compatible.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation pane, under **Administration**, click **Certificate Management**.
3. Select **Certificates Library**, and click **Import**.
4. Enter a name, and optionally, a description for this certificate in the certificate library and click **Next**.
5. Upload a PEM file containing the certificate chain that you want to import and click **Next**.
6. Optional: Upload a private key file.
   Your private key file might not be protected with a passphrase.
7. Click **Import**.

The imported certificate appears in the list of available certificates during the creation of entities that you must secure.

You can use the vertical ellipsis ( ⋮ ) menu to perform the following actions:

- Download a certificate.
- Edit the name and description of a certificate.
- Delete a certificate. You can delete only certificates that do not secure any entities.
- Copy the certificate PEM data to the clipboard.

# Terraform Configuration in VCF Automation Provider Management

You can automate the deployment and configuration of VCF Automation infrastructure resources using Terraform.

Terraform is an Infrastructure as Code (IaC) tool developed by HashiCorp. You can use the open-source tool to define and provision your cloud infrastructure using the HashiCorp Configuration Language (HCL). Terraform uses the configuration files with the descriptions of the desired state of your infrastructure to create, manage, and delete the necessary resources across various cloud providers and platforms. Terraform providers are plug-ins that Terraform uses to interact with cloud providers, SaaS providers, third-party tools, and other APIs. Each provider defines its resources and data sources that map directly to the functionalities of the specific service.

For end-to-end operation of Terraform in VCF Automation, you need three open-source Terraform providers.

- Terraform Provider for VCF Automation - You can use this provider to perform CRUD operations for the Provider Management UI and a subset of the Organization UI resources that belong to organizations for All Apps, such as Supervisor namespaces. This provider is available on GitHub and is distributed as part of the HashiCorp Terraform registry. The greenfield folder contains examples of configuration files for fresh VCF Automation installations that show how all resources interact with each other and provide guidance on how you can use the provider. The folder contains two parts:

  - Provider: Using this part, you can create the necessary organizations and configure the underlying infrastructure, such as vCenter instances, NSX instances, provider gateways, roles, and so on.
  - Tenant: This part includes tenant-side operations. You can use this part to configure the layout of the organization, such as content libraries.
- Terraform Provider for Kubernetes - You can use the Kubernetes (K8S) provider to perform CRUD operations against the VCF Automation Organization UI resources exposed through the VCF Automation Kubernetes API layer, such as projects.
  To see a usage example, navigate to `https://<FQDN>/automation/api-docs/#/terraform-provider`.
- Terraform Provider for VMware Aria Automation - You can use this provider to perform CRUD operations against the VCF Automation Organization UI resources that belong to VM Apps organizations and for VCF Automation Organization UI resources that belong to
  organizations for All Apps but are not yet exposed through the VCF Automation Kubernetes API layers, such as blueprints.

**Table 1811: Terraform providers depending on the type of organization**

| Role | Organization Type | Terraform Provider for VCF Automation | Terraform Provider for Kubernetes | Terraform Provider for VMware Aria Automation |
|---|---|---|---|---|
| Provider Administrator | VM Apps | X | | |
| | All Apps | X | | |
| Organization Administrator | VM Apps | | | X |
| | All Apps | X | X | X |

**Table 1812: Terraform providers by roles**

| Role | Terraform Provider | Resource |
|---|---|---|
| Provider Administrator | Terraform Provider for VCF Automation | • Organization<br>• Region<br>• Region quota<br>• Networking<br>• Content library<br>• Supervisor namespace |
| Organization Administrator | Terraform Provider for Kubernetes | • VCF services projects<br>• Content library<br>• Virtual Private Cloud<br>• Subnets |
| | Terraform Provider for VMware Aria Automation | • Blueprint<br>• Catalog |
| Organization User | Terraform Provider for Kubernetes | Provisioning of IaaS services |
| | Terraform Provider for VMware Aria Automation | Catalog deployment |

# Managing Events and Tasks in the VCF Automation Provider Management UI

You can view and manage your VCF Automation events and tasks.

# View Tasks in Your VCF Automation Provider Management UI

From the Provider Management UI, you can view recent tasks and their status.

You can use the recent tasks view to monitor the status of tasks in your Provider Management UI. This view can be a good first step for troubleshooting any issues in your environment.

Next to the **Recent Tasks** button, running and failed tasks appear in blue and red, respectively.

1. Log in to the VCF Automation Provider Management UI.
2. In the lower-left corner, click **Events and Tasks**.
3. Optional: Sort and filter the list of recent tasks.

   The Provider Management UI displays the last 50 tasks for the last 12 hours.

A list of recent tasks displays, along with the status of the task, the type, the initiator, and the start and completion time.

# Stop a Task in Progress in Your VCF Automation Provider Management UI

If you accidentally start a VMware Cloud Foundation® Automation operation before applying or reviewing all necessary settings, you can stop the task in progress.

The **Recent Tasks** panel must be open.

By default, the **Recent Tasks** panel is displayed at the bottom of the portal. When you start an operation, for example to create a virtual machine, the task is displayed in the panel.

1. Log in to the VCF Automation Provider Management UI.
2. Start a long-running operation.
3. In the **Recent Tasks** panel, click the **Cancel** icon.
4. In the **Cancel Task** dialog box, confirm that you want to cancel the task by clicking **OK**.

The operation stops.

# View Events in Your VCF Automation Provider Management UI

In the VCF Automation UI, you can view the list of all events, their details, and status.

The events view is a way to view the status of the events in your portal. The view shows when the events happened, and whether they were successful. The events view contains one-time occurrences, such as user logins and object creation, or deletion.

1. Log in to the VCF Automation Provider Management UI.
2. In the left navigation bar, click **Events and Tasks**, and click the **Events** tab.

   The list of all events appears, along with the time the event happened and the status of the event.
3. Click **Manage Columns** to change the details you want to view about the events.
4. Optional: Click an event to view the event details.

| Detail | Description |
|---|---|
| Status | Status of the event, such as Succeeded or Failed. |
| Type | The object on which the task was performed. |
| Target | Target object of the event. |
| Owner | User who triggered the event. |
| Occurred At | Date and time when the event occurred. |
| Service namespace | Service name, such as *com.vmware.vcloud*. |

# Managing Defined Entities in VCF Automation

To create extensions that provide additional VCF Automation capabilities to the organizations, provider administrators can use the VCF Automation API.

Defined entities represent external resources that VCF Automation can manage. Extension and UI plug-in developers can create runtime defined entities, enabling extensions and UI plug-ins to store and manipulate the extension-specific information in VCF Automation. If you create an extension and it needs to store a state or references to external resources, the extension can use runtime defined entities instead of a local database. For example, a Kubernetes extension can store information about the Kubernetes clusters it manages in runtime defined entities. The extension can then provide extension APIs for managing those clusters using the information from the runtime defined entities.

You can access runtime defined entities only through the VCF Automation API. Users with admin privileges can track and observe the way extensions operate by verifying the state of the defined entities that the extensions create. The states of the defined entities can also help you to troubleshoot problems with the extensions.

## Defined Entity Type

You define at runtime the schema of the information that you can store by using defined entities. The schema is defined in the entity type through a JSON document. When you create a defined entity type, you create a new custom VCF Automation type which you can manage by extensions and behaviors.

Once you create an instance of a defined entity type, you cannot change the type, schema, and behaviors anymore. To add new functionalities to the entity type, you must create a new version of the type. When a defined entity instance is based on an earlier version of the entity type, you can upgrade the defined entity to use a later version of the type by setting the type property of the entity to the ID of the new type. When you create a defined entity, in the API call, the contents of the entity property must match the JSON schema specified in the entity type. You can grant access to an entity type only to specific users and organizations. You can set different access restrictions to the different versions of an entity type.

The schema of the entity type can specify the access to the different content fields. You can mark properties as `public`, `protected`, or `private`.

**Table 1813: Entity Type Property Access Control**

| Property Status | Access Control |
|---|---|
| Public | All users can view and modify the field. |
| Protected | All users can view the field. Only users with Full Control access can create and modify the field. The users must have the Admin Full Control: TYPE right or both the Full Control: TYPE right and a Full Control ACL. |
| Private | Only users with Full Control access can view, create, and modify the field. The users must have the Admin Full Control: TYPE right or both the Full Control: TYPE right and a Full Control ACL |

For example, the JSON schema for the entity property `clusterState` can be the following.

```
"clusterState" : {
  "type" : "string",
  "x-vcloud-restricted" : "protected"
}
```

## Interfaces and Behaviors

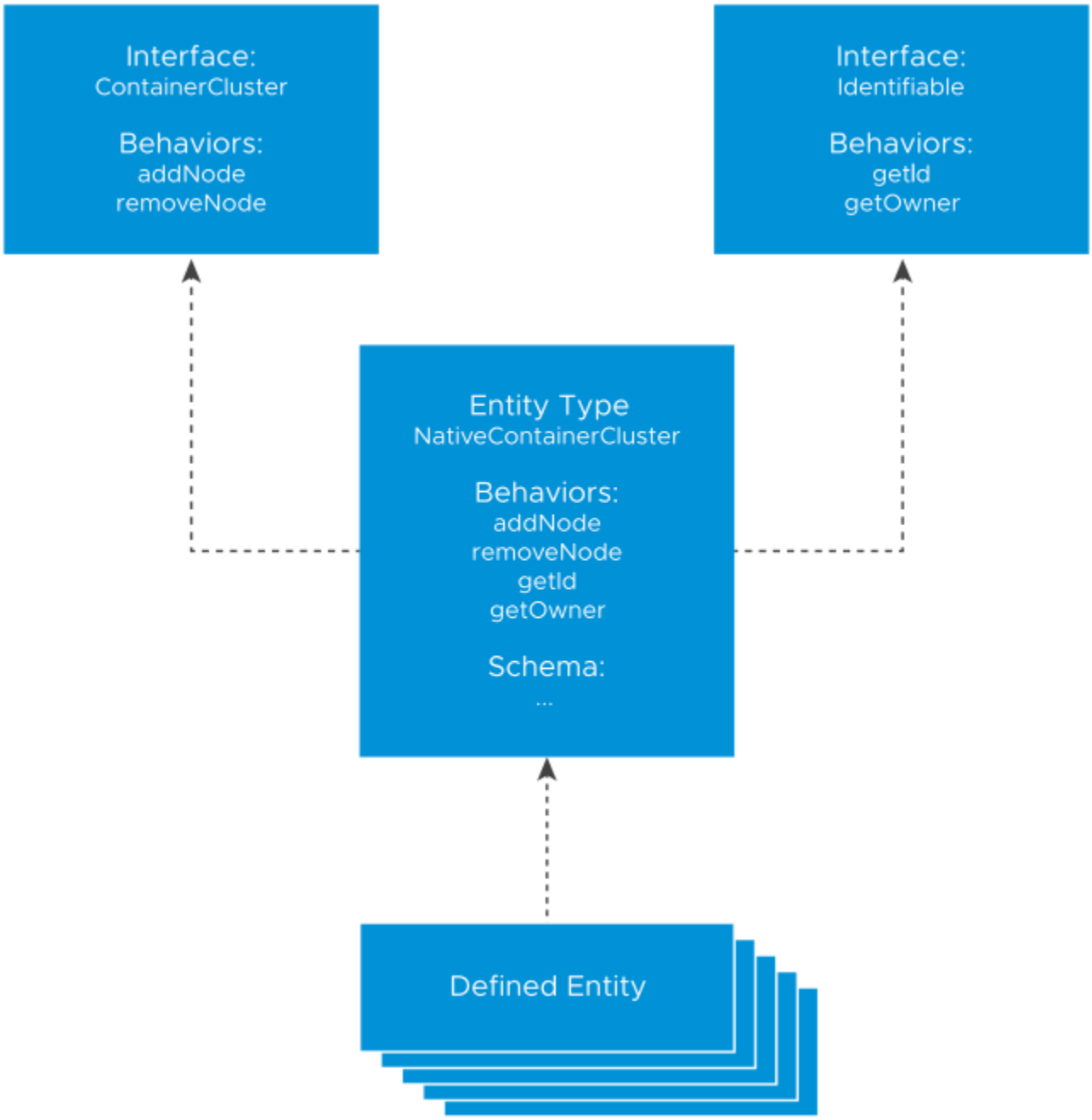
Defined entities also have interfaces that you can use to define behaviors. If we use the Kubernetes example, each cluster entity type has a different implementation of the behaviors defined in the container cluster interface. The Kubernetes cluster type behaviors communicate with vSphere to perform the necessary actions.

All users can see the interfaces of defined entities.

There are different ways to define behaviors.

- You can create and publish a VCF Operations orchestrator workflow and create ? behavior that invokes the workflow.
- You can link any functionality to a behavior by using a WebHook.
- With MQTT behaviors, you can call a functionality in an extension. By using MQTT you can communicate with the extension directly and the extension can provide a more detailed status.

**Figure 537: Runtime Defined Entity Interfaces**




Runtime defined entity interfaces, types, and behaviors have specific API IDs.

- Interface API IDs are in the form of `urn:vcloud:interface:vendor_name:interface_name:version`.
- Type API IDs are in the form of `urn:vcloud:type:vendor_name:type_name:version`.
- Behavior API IDs are in the form of `urn:vcloud:behavior-interface:behavior_name:vendor_name:interface_name:version` or `urn:vcloud:behavior-type:behavior_name:vendor_name:interface_name:version`.

### Creating a Runtime Defined Entity

To create a defined entity, you start by configuring the external resources and adding the necessary information to the defined entity. This process might require a few iterations. To resolve the defined entity, you must verify that the contents of the schema have all the necessary information. If the contents do not match the JSON schema specified in the type, the defined entity state changes to `Resolution_Error`. If you fill in the information correctly, the defined entity state changes to `Resolved` and is ready to use.

Because tasks track all long-running runtime defined entity operations, you can use the VCF Automation UI to monitor the creation of defined entities, behavior invocation, and so on. When the status of a defined entity changes to `Resolved`, the corresponding task in the UI appears as `Succeeded`.

When you create a defined entity in one organization, you cannot share the defined entity with other organizations. If you create a defined entity in the `System` organization, you can share it with other organizations and their users.

### Sample API Call to Create a Defined Entity

The example provides a sample API call to create a new entity of the `pksContainerCluster` type.

```
POST https://host/cloudapi/1.0.0/entityTypes/urn:vcloud:type:vendorA:pksContainerCluster:1.0.0
{
   "name": "testEntity",
   "entity": {
      "cluster": {
         "name": "testCluster",
         "nodes": ["node1"]
      }
   }
}
```

### Access to Defined Entity Instances

Two complementary mechanisms control the access to runtime defined entities.

- Rights - When you create a runtime defined entity type, you create a rights bundle for the type. To provide access to specific operations, you must assign rights from this bundle to other roles. Each bundle has five type-specific rights: **View: TYPE**, **Edit: TYPE**, **Full Control: TYPE**, **Administrator View: TYPE**, and **Administrator Full Control: TYPE**. The **View: TYPE**, **Edit: TYPE**, and **Full Control: TYPE** rights work only in combination with an ACL entry.
- Access Control List (ACL) - The ACL table contains entries defining the access users have to specific entities in the system. It provides an extra level of control over the entities. For example, while an **Edit: TYPE** right specifies that a user can modify entities to which they have access, the ACL table defines which entities the user has access to.
  **System administrators** with the **View General ACL** right can view the ACLs assigned to a specific defined entity by using the `accessControls` API.
  **System administrators** with the **Manage General ACL** right can create, modify, and remove specific ACLs by using the `accessControls` API.

**Table 1814: Rights and ACL Entries for RDE Operations**

| Entity Operation | Option | Description |
|---|---|---|
| Read | **Administrator View: TYPE** right | Users with this right can see all runtime defined entities of this type within an organization. |

| Entity Operation | Option | Description |
|---|---|---|
| | **View: TYPE** right and ACL entry **>= View** | Users with this right and a read-level ACL can view runtime defined entities of this type. |
| Modify | **Administrator Full Control: TYPE** right | Users with this right can create, view, modify, and delete runtime defined entities of this type in all organizations. |
| | **Edit: TYPE** right and ACL entry **>= Change** | Users with this right and modify-level ACL can create, view, and modify runtime defined entities of this type. |
| Delete | **Administrator Full Control: TYPE** right | Users with this right can create, view, modify, and delete runtime defined entities of this type in all organizations. |
| | **Full Control: TYPE** right and ACL entry **= Full Control** | Users with this right and full control-level ACL can create, view, modify, and delete runtime defined entities of this type. |

You can use the VCF Automation API or UI to publish the rights bundle to any organizations you want to manage the entities of this type. After publishing the rights bundle, you can assign rights from the bundle to roles within the organization.

You can use the VCF Automation API to edit the ACL table.

### Access to Defined Entity Interface and Type Operations

Defined entity interface and type operations require specific rights that certain service providers might not have. Before performing an operation, verify that you have the necessary rights.

**Table 1815: Rights for Defined Entity Interface and Type Operations**

| Operation | Requirement |
|---|---|
| View and query interface | All users have the necessary rights. |
| Create interface | You must have the **Create new custom entity definition** right. |
| Edit interface | You must have the **Edit custom entity definition** right. |
| Delete interface | You must have the **Delete custom entity definition** right. |
| Create type | You must have the **Create new custom entity definition** right. |
| View type | The minimum requirement is `ReadOnly` Type ACL. |
| Edit and delete type | You must have the **Edit/Delete custom entity definition** right and a `FullControl` Type ACL . |

### Working with Runtime Defined Entities

To invoke behaviors on a defined entity, it must be in the `Resolved` state. To delete a defined entity, it must be in the `Resolved` or `Resolution_Error` state.

You can modify entity types and interfaces only if they do not have instances. You can only modify the referenced entity type by changing its version.

To diagnose issues with the access to defined entities, you can use the entity ACL query APIs and verify the necessary rights. In addition, the rights bundle of an entity type must be published to the organizations that you want to use it.

# Sharing Defined Entities in VCF Automation

You can grant access to Runtime Defined Entities (RDEs) by sharing them with other VCF Automation system administrators or other organizations.

## Sharing Defined Entities with Another User

1. If you want to grant access to defined entities to other organizations, publish the rights bundle of the defined entity type to the organization of your choice. For example, for the creation and management of Kubernetes clusters, you must publish the **vmware:tkgcluster Entitlement** rights bundle. See Publish or Unpublish a Rights Bundle to VCF Automation.

   If you want to share the defined entity with a **system administrator**, skip this step.
2. Assign the **View: TYPE**, **Edit: TYPE**, or **Full Control: TYPE** right from the bundle to the user roles you want to have the specific level of access to the defined entity.

   For example, if you want the users with the **tkg_viewer** role to view Kubernetes clusters within the organization, you must add the **View: Tanzu Kubernetes Guest Cluster** right to the role. If you want the users with the **tkg_author** role to create, view, and modify Kubernetes clusters within this organization, add the **Edit: Tanzu Kubernetes Guest Cluster** to that role. If you want the users with the **tkg_admin** role to create, view, modify, and delete Kubernetes clusters within this organization, add the **Full Control: Tanzu Kubernetes Guest Cluster** right to the role.
3. Grant the specific user an Access Control List (ACL) by making the following REST API call.

   ```
   POST https://[address]/cloudapi/1.0.0/entities/urn:vcloud:entity:[vendor]:[type name]:[version]:[UUID]/
   accessControls
    {
      "grantType" : "MembershipAccessControlGrant",
      "accessLevelId" : "urn:vcloud:accessLevel:[Access_level]",
      "memberId" : "urn:vcloud:user:[User_ID]"
    }
   ```

   *Access_level* must be `ReadOnly`, `ReadWrite`, or `FullControl`. *User_ID* must be the ID of the user to which you want to grant the access to the defined entity.

   Users with the **tkg_viewer** role, described in the example, cannot grant ACL access. Users with the **tkg_author** or **tkg_admin** role can share access to a VMWARE:TKGCLUSTER entity with users who have the **tkg_viewer**, **tkg_author**, or **tkg_admin** role by granting them ACL access using the API request.

   You can also use REST API calls to revoke access or to view who has access to the entity. See the VCF Automation REST API documentation.

## Sharing Administrator Rights to Defined Entities

1. If you want to grant access to defined entities to organizations, publish the rights bundle of the defined entity type to an organization. For example, for the creation and management of Kubernetes clusters, you must publish the **vmware:tkgcluster Entitlement** rights bundle. See Publish or Unpublish a Rights Bundle to VCF Automation.

   If you want to share the defined entity with a **system administrator**, skip this step.
2. Assign the **Administrator View: TYPE** or **Administrator Full Control: TYPE** right from the bundle to the user roles you want to have the specific level of access to the defined entity.

   For example, if you want the users with this role to view all Kubernetes clusters within the organization, you must add the **Administrator View: Tanzu Kubernetes Guest Cluster** right to the role. If you want the users with this role to create, view, modify, and delete Kubernetes clusters in all organizations, add the **Administrator Full Control: Tanzu Kubernetes Guest Cluster** right to the user role.

   Users with the **Administrator Full Control: Tanzu Kubernetes Guest Cluster** right can grant ACL access to any VMWARE:TKGCLUSTER entity.

## Changing the Owner of a Defined Entity

The owner of a defined entity or a user with the **Administrator Full Control: TYPE** right can transfer the ownership to another user by updating the defined entity model and changing the owner field with the ID of the new owner.

# VCF Automation Webhook Behaviors

To connect two different applications, you can use webhooks. By using webhooks, you can set up event notifications.

A webhook is an HTTP callback. When an event happens, a web application implementing webhooks broadcasts data about that event and sends it to a webhook URL from the application that you want to receive the information. For more information, see https://github.com/vmware/.

In the context of VCF Automation, you can configure webhooks by using webhook behaviors. Webhook behaviors send out a POST request to a remote webhook server. The body of the request contains information about the behavior invocation and the entity that the behavior was invoked on. You can also customize the body by using a template. The webhook server response determines the result of the behavior invocation task. There are three different types of responses that the server can return.

The remote webhook server can be any server that VCF Automation can reach. However, before creating a webhook behavior, you must verify that the endpoint of the webhook is secure.

- Verify that the webhook endpoint is an HTTPS endpoint.
- Verify that the organization of the user invoking the behavior trusts the webhook server certificate. This means that even if the certificate is present in a VCF Automation trust store, if the organization of the user, who invokes the behavior does not trust the certificate, the webhook behavior will not run.
- You can add a certificate to the trust store by using the UI or the VCF Automation API. See Managing Certificates Using Your VCF Automation.

### Creating Webhook Behaviors

Request:

```
"POST https://host/cloudapi/1.0.0/interfaces/<interface_id>/behaviors{
   "name":"webhookBehavior",
   "execution":{
      "type":"WebHook",
      "id":"testWebHook",
      "href":"https://hooks.slack.com:443/services/T07UZFN0N/B01EW5NC42D/rfjhHCGIwzuzQFrpPZiuLkIX",
      "key":"secretKey",
      "execution_properties":{
         "template":{
            "content":"<template_content_string>"
         },
         "_secure_token":"secureToken",
         "invocation_timeout":7
      }
   }
}
```

| Field | Desciption |
|---|---|
| `type` | Required field. Must be `WebHook`. |
| `id` | Optional string field. |
| `href` | Required field for the URL that must receive requests from VCF Automation. |
| `_internal_key` | Required field for the shared secret used for signing the requests sent to the webhook endpoint. |
| `invocation_timeout` | Optional field specifying the timeout in seconds for the response from the webhook server |
| `template.content` | Optional field for customization of the payload sent to the webhook server. You can provide a template as a string. You can add other fields to `execution_properties`, accessible to the template. |

| Field | Desciption |
| --- | --- |
| _secure_ | Fields with prefix _secure_ are optional. The fields are write-only and cannot be obtained through a GET request on the behavior. The field value gets encrypted before it is saved to the VCF Automation database. The fields are accessible to the template. |

### Payload From VCF Automation to the Webhook Server

There is a default format of the payload sent to the webhook server. You can also customize the payload by using a template.

Default payload example:

```
{
   "entityId":"urn:vcloud:entity:vmware:test_entity_type:b95d96ec-c294-4a64-b5c4-9009abe6ab06",
   "typeId":"urn:vcloud:type:vmware:test_entity_type:1.0.0",
   "arguments":{
      "x":7
   },
   "_metadata":{
      "executionId":"testWebHook",
      "execution":{
         "href":"https://webhook.site/3ca4588f-c9f1-4c0f-911f-734ab598b2dc"
      },
      "invocation":{

      },
      "apiVersion":"36.0",
      "behaviorId":"urn:vcloud:behavior-interface:webhookBehavior:vmware:test_interface_3:1.0.0",
      "requestId":"8312df21-712c-4b85-86d4-c9b00669662f",
      "executionType":"WebHook",
      "invocationId":"115c5b99-09e8-44f7-8189-4b9d96e067b1",
      "taskId":"a9e0556b-7699-4ded-b56b-c51fee659008"
   },
   "entity":{
      "cluster":{
         "name":"testCluster0"
      },
      "clusterState":{
         "host":"testHost",
         "status":"valid"
      }
   }
}
```

For custom payloads with a template, you can specify a template for the payload sent to the webhook server. If you do not specify a template, VCF Automation sends the default request payload to the webhook server. The Apache FreeMarker template engine renders the payload from the provided template and the data model.

The data model represents all the data prepared for the template, in other words, all the data that you can include in the payload. The data model contains the invocation arguments of the webhook behavior, for example, arguments, metadata, execution properties, and entity information. The following example shows the tree-like structure of the data model.

```
+ - entityId
|
+ - typeId
|
+ - arguments
|
```

```
+ - arguments_string
|
+ - _execution_properties
|
+ - _metadata
|   |
|   + - executionId
|   + - behaviorId
|   + - executionType
|   + - taskId
|   + - execution
|   |   |
|   |   + - href
|   + - invocation
|   + - invocationId
|   + - requestId
|   + - apiVersion
|
+ - entity
|
+ - entity_string
```

**Table 1816: Invocation arguments**

| Argument | Description |
|---|---|
| `entityId` | ID of the invoked defined entity. |
| `typeId` | ID of the entity type of the invoked defined entity. |
| `arguments` | The arguments passed in the behavior invocation. |
| `arguments_string` | A JSON format string of arguments. You can use it if you want to add all the arguments to the payload. |
| `_execution_properties` | You can select all the values from the `execution_properties` invocation argument by key name. |
| `execution` | You can select all the values from the `execution` invocation argument by key name. |
| `invocation` | You can select all the values from the `invocation` argument by key name. |
| `entity` | A map of the entity content. You can select all values in the entity by key name. |
| `entity_string` | A JSON format string of entity. |

For more information on the Apache FreeMarker expression language for the template for the payload, see the Freemaker documentation.

Sample template:

```
<#assign header_Content\-Type= "application/json" />
{
"text": "Behavior with id ${_metadata.behaviorId} was executed on entity with id ${entityId}"
}
```

You can specify custom headers that you want to add to the POST request sent to the webhook servers. You can use the webhook template for adding custom headers which are variables in the template with the prefix `header_`. For example, you can specify the value for the `Authorization` header by adding the following line to the template:

```
<#assign header_Authorization = "${_execution_properties._secure_token}" />
```

**Payload From the Webhook Server to VCF Automation**

- Default response
  The default payload from the webhook server to VCF Automation must contain the following.
  - The response can either have no `Content-Type` header or a `Content-Type` header with `text/plain` value.
  - The response body must be a string.
  - The body of the response is used as the result of the behavior invocation. The status for the behavior invocation task is `success`. The task result is set to the body of the response.
- Single task update response
  Apart from the behavior result, the task update response can set some other behavior invocation task properties like `status`, `details`, `operation`, `error`, `progress`, and `result`. Because this is a one-time task update, the response must complete the task. If the response does not set the task status to `success`, `error`, or `aborted`, then the task fails with an error. The error message indicates that the status was not acceptable.
  - The response must have a `Content-Type` header with `application/vnd.vmware.vcloud.task+json` value.
  - The response body must contain a JSON representation of the `TaskType` class containing the task properties you want to modify.
  - Sample response body with status `success`:

```
{
   "status":"success",
   "details":"example details",
   "operation":"example operation",
   "progress":100,
   "result":{
      "resultContent":"example result"
   }
}
```

  - Sample response body with status `error`:

```
{
   "status":"error",
   "details":"example details",
   "operation":"example operation",
   "progress":50,
   "error":{
      "majorErrorCode":404,
      "minorErrorCode":"ERROR",
      "message":"example error message"
   }
}
```

- Continuous task update response
  The webhook server can send multiple task updates by using a multi-part HTTP response. Each update is a separate body part of the response body. The last body part of the webhook server response body must finish the task. If the response body does not finish the task, the task fails with an error.
  - The response must have a `Content-Type` header with `multipart/form-data; boundary=...` value.
  - The response body must start and end with a boundary string `--<boundary_string>`. Each part of the response body must end in a boundary string. You must specify the boundary in the `Content-Type` header of the HTTP response from the webhook server.

```
Headers:
...
Content-Type: multipart/form-data; boundary=<boundary_string>

Body:
--<boundary_string>
Content-Type: application/vnd.vmware.vcloud.task+json
```

```
{
"details": "example details",
"operation": "example operation",
"progress": 50
}
--<boundary_string>
Content-Type: application/vnd.vmware.vcloud.task+json
{
"status": "success",
"progress": 100,
"result": {
"resultContent": "example result"
}
}
--<boundary_string>
```

The body parts in the response body must be in the format for a task update or the format for a final update.

Task update example:

```
Content-Type: application/vnd.vmware.vcloud.task+json
{
"details": "example details",
"operation": "example operation",
"progress": 50
}
```

Final update example:

```
Content-Type: text/plain
<string>
```

## Authentication of the Webhook Behavior Requests

Hash-based message authentication code (HMAC) authentication secures webhook behavior requests. HMAC is a mechanism ensuring the authenticity and integrity of HTTP requests. To ensure the authenticity and integrity, HMAC includes two custom headers to each HTTP request. The custom headers are a signature header and a digest. For VCF Automation, the signature header is `x-vcloud-signature` and the digest is `x-vcloud-digest`.

The signature header `x-vcloud-signature` is a custom header sent with each webhook request. The signature header consists of an HMACSHA512 algorithm, headers, and a VCF Automation generated signature. The headers show what is in the base string which is signed to create the signature.

**Table 1817: Header Fields**

| Field | Description |
|---|---|
| `host` | The webhook server host |
| `date` | Date of the request |
| `(request-target)` | Linked lowercase HTTP method, ASCII space, and request path headers. For example, `post /webhook`. |
| `digest` | SHA-512 digest of the request body |

To generate the signature header, you need a shared secret. The shared secret is a string that only VCF Automation and the webhook server know.

The digest header `x-vcloud-digest` is a Base64-encoded SHA-512 digest of the request body.

Each webhook request from VCF Automation can be verified by generating the signature using the shared secret and comparing it to the signature in the signature header from VCF Automation.

## Invoking Webhook Behaviors

You can invoke webhook behaviors like any other behavior.

```
{
"arguments": {
"argument1": "data1",
"argument2": "data2",
...
 }
}
```